Exercise J-M

1. Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a vector such that $\vec{a}.\vec{c}=|\vec{c}|,|\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $(\vec{a}\times\vec{b})$ and $\vec{c}$ is $\frac{\pi}{6}$, then the value of $|(\vec{a}\times\vec{b})\times\vec{c})|$ is : **[JEE(Main)July2021]**

माना $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}$ है। यदि एक सदिश $\vec{c}$ के लिए $\vec{a}.\vec{c}=|\vec{c}|,|\vec{c}-\vec{a}|=2\sqrt{2}$ हैं तथा $(\vec{a}\times\vec{b})$ और $\vec{c}$ के बीच का कोण $\frac{\pi}{6}$ है, तो $|(\vec{a}\times\vec{b})\times\vec{c})|$ का मान है : **[JEE(Main)July2021]**

(A) $\frac{3}{2}$ (B) 4 (C) 3 (D) $\frac{2}{3}$

Ans. A

Sol. $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ $\quad\vec{b}=\hat{i}+\hat{j}$

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -2\\ 1 & 1 & 0\end{vmatrix}=2\hat{i}-2\hat{j}+\hat{k}$$

$|\vec{a}\times\vec{b}|=3$

$|\vec{c}-\vec{a}|=2\sqrt{2}$

$\vec{c}^{2}+a^{2}-2\vec{a}.\vec{c}=8$

$c^2 + a^2 - 2c = 8$

$[\because a^2 = 9]$

$c^2 - 2c + 1 = 0$

$(c - 1)^2 = 0$

$(c - 1)^2 = 0$

$c = 1$

$\Rightarrow|\vec{c}|=1$

$|(\vec{a}\times\vec{b})\times\vec{c}|=|\vec{a}\times\vec{b}||\vec{c}|\sin(\pi/6)$

$$=(3)(1)\left(\frac{1}{2}\right)$$

$$=\frac{3}{2}$$

2. In a triangle ABC, if $\left|\overrightarrow{BC}\right|=3,\left|\overrightarrow{CA}\right|=5$ and $\left|\overrightarrow{BA}\right|=7,$ then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to : **[JEE(Main)July2021]**

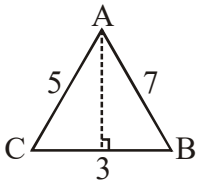
त्रिभुज ABC में यदि $\left|\overrightarrow{BC}\right|=3,\left|\overrightarrow{CA}\right|=5$ तथा $\left|\overrightarrow{BA}\right|=7$, तो सदिश $\overrightarrow{BA}$ का $\overrightarrow{BC}$ के अनुदिश प्रक्षेप होगा :

[JEE(Main)July2021]

(A) $\frac{11}{2}$ (B) $\frac{19}{2}$ (C) $\frac{13}{2}$ (D) $\frac{15}{2}$

Ans. A

Sol. $\left|\overrightarrow{BC}\right|=3,\left|\overrightarrow{CA}\right|=5,\left|\overrightarrow{BA}\right|=7$



Projection of $\overrightarrow{BA}$ on $\overrightarrow{BC}=\left|\overrightarrow{BA}\right|$ Cos$\angle$ABC

$=7\left(\frac{3^2+7^2-5^2}{2\times3\times7}\right)=\frac{11}{2}$

3. Let a vector $\vec{a}$ be coplanar with vectors $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d}=3\hat{i}+2\hat{j}+6\hat{k}$, and $|\vec{a}|=\sqrt{10}$. Then possible value of $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{a}\ \vec{b}\ \vec{d}\right]+\left[\vec{a}\ \vec{c}\ \vec{d}\right]$ is equal to :

[JEE(Main)July2021]

माना एक सदिश $\vec{a}$, सदिशों $\vec{b}=2\hat{i}+\hat{j}+\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ के सहतलीय है। यदि $\vec{a}$, सदिश $\vec{d}=3\hat{i}+2\hat{j}+6\hat{k}$ पर लम्बवत् है और $|\vec{a}|=\sqrt{10}$ है, तो $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{a}\ \vec{b}\ \vec{d}\right]+\left[\vec{a}\ \vec{c}\ \vec{d}\right]$ का एक संभावित मान है : **[JEE(Main)July2021]**

(A) – 40 (B) – 38 (C) – 42 (D) –29

Ans. C

Sol. $\vec{a}=\lambda\vec{b}+\mu\vec{c}$

$\vec{a}=\lambda\left(2i+j+k\right)+\mu\left(i-j+k\right)$

$\vec{a}=i\left(2\lambda+\mu\right)+j\left(\lambda-\mu\right)+k\left(\lambda+\mu\right)$

Now $\vec{a}.\vec{d}=0$

$3\left(2\lambda+\mu\right)+2\left(\lambda-\mu\right)+6\left(\lambda+\mu\right)=0$

$14\lambda+7\mu=0$

$2\lambda+\mu=0$ ....(1)

$|\vec{a}|=\sqrt{10}$

$\sqrt{\left(2\lambda+\mu\right)^2+\left(\lambda-\mu\right)^2+\left(\lambda+\mu\right)^2}=\sqrt{10}$

$4\lambda^2+4\lambda\mu+\mu^2+2\lambda^2+2\mu^2=10$

$6\lambda^2+3\mu^2+4\lambda\mu=10$ ....(2)

by (1) $\mu=-2\lambda$

$6\lambda^2+12\lambda^2-8\lambda^2=10$

$10\lambda^2=10$

$\lambda=\pm1$

if $\lambda=1$ then $\mu=-2$

if $\lambda=-1$ then $\mu=2$

Now $\vec{a}=3j-k$

or

$\vec{a}=-3j+k$

if we choose $\vec{a}=3j-k$

$\left[\vec{a}\,\vec{b}\,\vec{c}\right]+\left[\vec{a}\,\vec{b}\,\vec{d}\right]+\left[\vec{a}\,\vec{c}\,\vec{d}\right]=0-28-14=-42$

4. Let three vector $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{a}\times\vec{b}=\vec{c}$, $\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{a}|=2$. Then which one of the following is not true ? **[JEE(Main)July2021]**

(A) Projection of $\vec{a}$ on $\left(\vec{b}\times\vec{c}\right)$ is 2 (B) $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=51$

(C) $\vec{a}\times\left(\left(\vec{b}+\vec{c}\right)\times\left(\vec{b}-\vec{c}\right)\right)=\vec{0}$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{c}\ \ \vec{a}\ \ \vec{b}\right]=8$

माना तीन सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$, $\vec{a}\times\vec{b}=\vec{c}$, $\vec{b}\times\vec{c}=\vec{a}$ तथा $|\vec{a}|=2$ को संतुष्ट करते हैं। तो निम्न में से कौनसा कथन असत्य है? **[JEE(Main)July2021]**

(A) $\left(\vec{b}\times\vec{c}\right)$ पर $\vec{a}$ का प्रक्षेप 2 है (B) $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=51$

(C) $\vec{a}\times\left(\left(\vec{b}+\vec{c}\right)\times\left(\vec{b}-\vec{c}\right)\right)=\vec{0}$ (D) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]+\left[\vec{c}\ \ \vec{a}\ \ \vec{b}\right]=8$

Ans. B

Sol. $\vec{a}\times\vec{b}=\vec{c}$ .....(1)

$\vec{b}\times\vec{c}=\vec{a}$ .....(2)

and $|\vec{a}|=2$.

By (1) and (2)

$\vec{a}.\vec{c}=0$

$\vec{b}.\vec{c}=0$

$\vec{a}.\vec{b}=0$

Hence $\vec{a}$, $\vec{b}$ and $\vec{c}$ are mutually perpendicular

$\vec{a}\times\vec{b}=\vec{c}$

$|\vec{a}\times\vec{b}|=|\vec{c}|$

$a^2b^2\sin^2 90°=c^2$

$4b^2=c^2$ ....(3)

$\vec{b}\times\vec{c}=\vec{a}$

$|\vec{b}\times\vec{c}=\vec{a}|$

$b^2c^2\sin^2 90°=a^2$

$b^2c^2=4$ ....(4)

by (3) and (4)

$|\vec{b}|=1$

$|\vec{c}|=2$

Now

(A) option (A)

$$\left|\frac{\vec{a}.(\vec{b}\times\vec{c})}{\vec{b}\times\vec{c}}\right|=2$$

$$\frac{|\vec{a}||\vec{b}||\vec{c}|\sin\theta\cos\theta}{|\vec{b}||\vec{c}|\sin\theta}=2$$

2 = 2 which is correct.

(B) $\left|3\vec{a}+\vec{b}-2\vec{c}\right|^2=9a^2+b^2+4c^2+0+0+0$

$=36+1+16$

Which is wrong

hence option (2) is correct

5. Let the vectors

$(2+a+b)\hat{i}+(a+2b+c)\hat{j}-(b+c)\hat{k}$,

$(1+b)\hat{i}+2b\hat{j}-b\hat{k}$ and $(2+b)\hat{i}+2b\hat{j}+(1-b)\hat{k}$, $a, b, c \in R$

be co-planar. Then which of the following is true? **[JEE(Main)July2021]**

माना

$(2+a+b)\hat{i}+(a+2b+c)\hat{j}-(b+c)\hat{k}$,

$(1+b)\hat{i}+2b\hat{j}-b\hat{k}$ तथा $(2+b)\hat{i}+2b\hat{j}+(1-b)\hat{k}$, $a, b, c \in R$

सहतलीय सदिश हैं। तो निम्न में से कौन–सा कथन सही है ? **[JEE(Main)July2021]**

(A) $2b = a + c$ (B) $a = b + 2c$ (C) $3c = a + b$ (D) $2a = b + c$

Ans. A

Sol. vectors are co-planer then S.T.P = 0

$$\begin{vmatrix} 2+a+b & a+2b+c & -b-c \\ 1+b & 2b & -b \\ 2+b & 2b & 1-b \end{vmatrix} = 0$$

$R_3 \to R_3 - R_2$

$$\begin{vmatrix} 2+a+b & a+2b+c & -b-c \\ 1+b & 2b & -b \\ 1 & 0 & 1 \end{vmatrix} = 0$$

$C_1 \to C_1 - C_3$

$$\begin{vmatrix} 2+a+2b+c & a+2b+c & -b-c \\ 1+2b & 2b & -b \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$C_1 \to C_1 - C_2$

$$\begin{vmatrix} 2 & a+2b+c & -b-c \\ 1 & 2b & -b \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$4b - a - 2b - c = 0$

$2b = a + c$

6. Let a, b and c be distinct positive numbers. If the vectors $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ and $c\hat{i}+c\hat{j}+b\hat{k}$ are co-planar, then c is equal to : **[JEE(Main)July2021]**

माना a, b तथा c तीन भिन्न धनात्मक संख्याएँ हैं। यदि सदिश $a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k}$ तथा $c\hat{i}+c\hat{j}+b\hat{k}$ समतलीय हैं, तो c बराबर हैं: **[JEE(Main)July2021]**

(A) $\dfrac{2}{\frac{1}{a}+\frac{1}{b}}$ (B) $\sqrt{ab}$ (C) $\dfrac{a+b}{2}$ (D) $\dfrac{1}{a}+\dfrac{1}{b}$

Ans. B

Sol. $\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$

$a(-c) - a(b-c) + c(c) = 0$

$c^2 = ab$

$c = \sqrt{ab}$

7. If $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then $|\vec{a}.\vec{b}|$ is equal to : **[JEE(Main)July2021]**

यदि $|\vec{a}| = 2, |\vec{b}| = 5$ तथा $|\vec{a} \times \vec{b}| = 8$ हैं, तो $|\vec{a}.\vec{b}|$ बराबर है : **[JEE(Main)July2021]**

(A) 6 (B) 4 (C) 3 (D) 5

Ans. A

Sol. $\left|\vec{a} \times \vec{b}\right| = |\vec{a}||\vec{b}|\sin\theta$

$8 = 2 \times 5 \times \sin\theta$

$\sin\theta = 4/5 = \dfrac{P}{H}$ $\quad B = 3$

$\cos\theta = 3/5$

$\left|\vec{a} \cdot \vec{b}\right| = |\vec{a}||\vec{b}|\cos\theta$

$= 2 \times 5 \times \dfrac{3}{5} = 6$

8. Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$. Then the vector product $\left(\vec{a} + \vec{b}\right) \times \left(\left(a \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$ is equal to.

माना $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ तथा $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$ हैं। तो सदिश गुणनफल $\left(\vec{a} + \vec{b}\right) \times \left(\left(a \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$ बराबर है :

(A) $7\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$ (B) $7\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$ (C) $5\left(30\hat{i} - 5\hat{j} + 7\hat{k}\right)$ (D) $5\left(34\hat{i} - 5\hat{j} + 3\hat{k}\right)$

Ans. B

Sol. $\left(\vec{a} + \vec{b}\right) \times \left(\left(\vec{a} \times \left(\left(\vec{a} - \vec{b}\right) \times \vec{b}\right)\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(\left(\vec{a} \times \left(\vec{a} \times b\right)\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(\left(\left(\vec{a}.\vec{b}\right)\vec{a} - \left(\vec{a}.\vec{a}\right)\vec{b}\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(\left(7\vec{a} - 6\vec{b}\right) \times \vec{b}\right)$

$\Rightarrow \left(\vec{a} + \vec{b}\right) \times \left(7\left(\vec{a} \times \vec{b}\right)\right)$

$\Rightarrow 7\left\{\vec{a} \times \left(\vec{a} \times \vec{b}\right) + \vec{b} \times \left(\vec{a} \times \vec{b}\right)\right\}$

$\Rightarrow 7\left\{\left(\vec{a}.\vec{b}\right)\vec{a} - \left(\vec{a}.\vec{a}\right)\vec{b} + \left(\vec{b}.\vec{b}\right)\vec{a} - \left(\vec{b}.\vec{a}\right)\vec{b}\right\}$

$\Rightarrow 7\left\{7\vec{a}-6\vec{b}+14\vec{a}-7\vec{b}\right\}$

$\Rightarrow 7\left\{21\vec{a}-13\vec{b}\right\}$

$\Rightarrow 7\left\{21(i+j+2k)-13(-i+2j+3k)\right\}$

$\Rightarrow 7\ (34i-5j+3k)$

9. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a}=\vec{b}\times\left(\vec{b}\times\vec{c}\right)$. If magnitudes of the vectors $\vec{a},\vec{b}$ and $\vec{c}$ are $\sqrt{2},1$ and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is $\theta\left(0<\theta<\frac{\pi}{2}\right)$, then the value of $1+\tan\theta$ is equal to : **[JEE(Main)July2021]**

माना तीन सदिशों $\vec{a},\vec{b}$ तथा $\vec{c}$ के लिए $\vec{a}=\vec{b}\times\left(\vec{b}\times\vec{c}\right)$ है। यदि सदिशों $\vec{a},\vec{b}$ तथा $\vec{c}$ के परिमाण क्रमशः $\sqrt{2},1$ तथा 2 हैं और $\vec{b}$ तथा $\vec{c}$ के बीच कोण $\theta\left(0<\theta<\frac{\pi}{2}\right)$ है, तो $1+\tan\theta$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $\frac{\sqrt{3}+1}{\sqrt{3}}$ (B) $\sqrt{3}+1$ (C) 1 (D) 2

Ans. D

Sol. $\vec{a}=\vec{b}\times(\vec{b}\times\vec{c})$ ..... (i)

$\vec{a}=(\vec{b}.\vec{c})\vec{b}-(\vec{b}.\vec{b})\vec{c}$

Take dot producet with $\vec{b}$ in (i)

$\because \vec{a}.\vec{b}=0$ ..... (ii)

$\vec{a}=(2\cos\theta\vec{b}-\vec{c}$ ..... (iii)

Dot product with $\vec{a}$ in

$\vec{a}.\vec{b}=(2\cos\theta)\vec{b}.\vec{a}-\vec{c}.\vec{a}$

$\Rightarrow 2=0-\vec{c}.\vec{a}$

$\therefore \vec{c}\ .\vec{a}=-2$ .... (iv)

Take dot product with $\vec{c}$ in (iii)

$\vec{a}.\vec{c}=(2\cos\theta)\vec{b}.\vec{c}-\vec{c}.\vec{c}$

$2=4\cos^2\theta$

$\cos\theta=\pm\frac{1}{\sqrt{2}}$

$\cos^2\pm=\frac{1}{\sqrt{2}}$

$\theta \in \left(0, \frac{\pi}{2}\right)$

$\therefore \quad \cos\theta = \frac{1}{\sqrt{2}}$

$\theta = 45^\circ$

$1 + \tan\theta = 1 + \tan 45^\circ = 2$

10. If the shortest distance between the lines $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k}), \lambda \in R$, $\alpha > 0$ and $\vec{r_2} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k}), \mu \in R$ is 9, then α is equal to _____. **[JEE(Main)July2021]**

यदि रेखाओं $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k}), \lambda \in R$, $\alpha > 0$ तथा $\vec{r_2} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k}), \mu \in R$ के मध्य न्यूनतम दूरी 9 है, तो α बराबर है _____ **[JEE(Main)July2021]**

Ans. 6

Sol. $L_1 \ \vec{r} = \vec{a} + \lambda\vec{p} \qquad \vec{a} = (\alpha, 2, 2)$

$L_2 \ \vec{r} = \vec{b} + \mu\vec{q} \qquad \vec{b} = (-4, 0, -1)$

$\vec{p} \times \vec{q} = 8\hat{i} + 8\hat{j} + 4\hat{k}$

Shortest distance $= \left|\frac{(\vec{a} - \vec{b}).(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right| = 9$

$$\left|\frac{((\alpha + 4)\hat{i} + 2\hat{j} + 3\hat{k}).(2\hat{i} + 2\hat{j} + \hat{k})}{3}\right| = 9$$

$|2\alpha + 15| = 27$

$2\alpha + 15 = 27$ or $2\alpha + 15 = -27$

$\alpha = 6$ or $\alpha = -21$ (Rejected)

11. Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ, with the vector $\vec{a} + \vec{b} + \vec{c}$. Then $36\cos^2 2\theta$ is equal to _____. **[JEE(Main)July2021]**

माना बराबर परिमाण के तीन सदिश $\vec{a}, \vec{b}, \vec{c}$ परस्पर लंबवत है तथा सदिश $\vec{a} + \vec{b} + \vec{c}$ के साथ एक समान कोण θ बनाते हैं। तो $36\cos^2 2\theta$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 4

Sol. $\cos\theta = \frac{(\vec{a} + \vec{b} + \vec{c}).\vec{a}}{|\vec{a} + \vec{b} + \vec{c}||\vec{a}|}$

$= \frac{r^2}{(r\sqrt{3})(r)} = \frac{1}{\sqrt{3}}$

$$\cos(2\theta) = 2\cos^2\theta - 1 = \frac{-1}{3}$$

$$36\cos^2(2\theta) = 36\left(\frac{1}{9}\right) = 4$$

12. For p > 0, a vector $\overrightarrow{v_2} = 2\hat{i} + (p+1)\hat{j}$ is obtained by rotating the vector $\overrightarrow{v_1} = \sqrt{3}p\hat{i} + \hat{j}$ by an angle θ about origin in counter clockwise direction. If $\tan\theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$, then the value of α is equal to _____.

[JEE(Main)July2021]

p > 0 के लिए, सदिश $\overrightarrow{v_1} = \sqrt{3}p\hat{i} + \hat{j}$ को केन्द्रबिन्दु के गिर्द वामावृत्त दिशा में कोण θ से आवर्तित करने पर सदिश $\overrightarrow{v_2} = 2\hat{i} + (p+1)\hat{j}$ प्राप्त होता है। यदि $\tan\theta = \frac{(\alpha\sqrt{3}-2)}{4\sqrt{3}+3}$ है, तो α का मान बराबर है _____

[JEE(Main)July2021]

Ans. 6

Sol. $\left|\vec{V}_1\right| = \left|\vec{V}_2\right|$

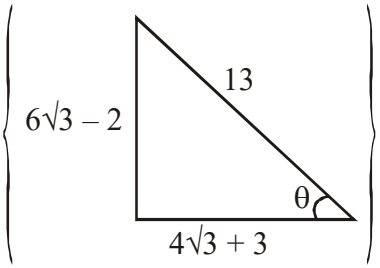
$3P^2 + 1 = 4 + (P+1)^2$

$P = 2$ or $P = -1$ (rejected)

$\vec{V}_1 = 2\sqrt{3}\,\hat{i} + \hat{j}$

$\vec{V}_2 = 2\hat{i} + 3\hat{j}$

$$\cos\theta = \frac{\vec{V}_1.\vec{V}_2}{\left|\vec{V}_1\right|\left|\vec{V}_2\right|} = \frac{4\sqrt{3}+3}{13}$$



$$\tan\theta = \frac{6\sqrt{3}-2}{4\sqrt{3}+3}$$

α=6

13. Let $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. If a vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p}+\vec{q})$ and $(\vec{p}-\vec{q})$, and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.

[JEE(Main)July2021]

माना $\vec{p}=2\hat{i}+3\hat{j}+\hat{k}$ तथा $\vec{q}=\hat{i}+2\hat{j}+\hat{k}$ दो सदिश हैं। यदि सदिश $\vec{r}=(\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k})$, दोनों सदिशों $(\vec{p}+\vec{q})$ तथा $(\vec{p}-\vec{q})$ के लम्बवत् है तथा $|\vec{r}|=\sqrt{3}$ है, तो $|\alpha|+|\beta|+|\gamma|$ बराबर है _____। **[JEE(Main)July2021]**

Ans. 3

Sol. $\vec{p}+\vec{q}=(3,5,2)$

$\vec{p}-\vec{q}=(1,1,0)$

$\vec{r}=(\alpha,\beta,\gamma)$

$\vec{r}.(\vec{p}+\vec{q})=0$

$3\alpha+5\beta+2\gamma=0$

$\vec{r}.(\vec{p}-\vec{q})=0$

$\alpha+\beta=0$

$\beta=-\alpha$

$\gamma=\alpha$

$\vec{r}=\alpha\hat{i}-\alpha\hat{j}+\alpha\hat{k}$

$|\vec{r}|=\sqrt{3\alpha^2}=\sqrt{3}$

$\alpha^2=1\Rightarrow\alpha=\pm1$

$|\alpha|+|\beta|+|\gamma|=3$

14. If $\left(\vec{a}+3\vec{b}\right)$ is perpendicular to $\left(7\vec{a}-5\vec{b}\right)$ and $\left(\vec{a}-4\vec{b}\right)$ is perpendicular to $\left(7\vec{a}-2\vec{b}\right)$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is _____. **[JEE(Main)July2021]**

यदि $\left(\vec{a}+3\vec{b}\right)$, $\left(7\vec{a}-5\vec{b}\right)$ के लंबवत है तथा $\left(\vec{a}-4\vec{b}\right)$, $\left(7\vec{a}-2\vec{b}\right)$ के लंबवत् है, तो $\vec{a}$ तथा $\vec{b}$ के बीच का कोण (डिग्री में) है _____ **[JEE(Main)July2021]**

Ans. 60

Sol. $\left(\vec{a}+3\vec{b}\right).\left(7\vec{a}-5\vec{b}\right)=0$

$7|\vec{a}|^2-5\vec{a}\cdot\vec{b}+21\vec{a}\cdot\vec{b}-15|\vec{b}|^2=0$

$7|\vec{a}|^2+16\vec{a}\cdot\vec{b}-15|\vec{b}|^2=0---\{1\}$

$(\vec{a}-4\vec{b}).(7\vec{a}-2\vec{b})=0$

$7|\vec{a}|^2-2\vec{a}\cdot\vec{b}-28\,\vec{a}\cdot\vec{b}+8|\vec{b}|^2=0$

$7|\vec{a}|^2-30\,\vec{a}\cdot\vec{b}+8|\vec{b}|^2=0----(2)$

$46\vec{a}\cdot\vec{b}-23|\vec{b}|^2=0$

$$\Rightarrow \vec{a}\cdot\vec{b}=\frac{|\vec{b}|^2}{2}$$

$$7|\vec{a}|^2+16\vec{a}\cdot\vec{b}-30\ \vec{a}\cdot\vec{b}=0$$

$$7|\vec{a}|^2=14\vec{a}\cdot\vec{b}$$

$$|\vec{a}|^2=2\vec{a}\cdot\vec{b} \quad \Rightarrow \quad \vec{a}\cdot\vec{b}=\frac{|\vec{a}|^2}{2}$$

$$\frac{|\vec{b}|^2}{2}=\frac{|\vec{a}|^2}{2}\Rightarrow|\vec{a}|=|\vec{b}|$$

$$2\vec{a}\cdot\vec{b}=|\vec{b}|^2$$

$$2|\vec{a}||\vec{b}|\cos\theta=|\vec{b}|^2$$

$$\cos\theta=1/2\Rightarrow\ \theta=\frac{\pi}{3}$$

15. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}$ and $\vec{c}=\hat{j}-\hat{k}$ be three vectors such that $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{a}\,.\,\vec{b}=1$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a}\times\vec{c}$ is l, then the value of $3l^2$ is equal to _____.

[JEE(Main)July2021]

माना $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}$ तथा $\vec{c}=\hat{j}-\hat{k}$ तीन सदिश हैं जिनके लिए $\vec{a}\times\vec{b}=\vec{c}$ तथा $\vec{a}\,.\,\vec{b}=1$ हैं। यदि सदिश $\vec{b}$ के, सदिश $\vec{a}\times\vec{c}$ पर प्रक्षेप सदिश की लम्बाई l है, तो $3l^2$ का मान बराबर है _____ **[JEE(Main)July2021]**

Ans. 2

Sol. $$l=\left|\frac{\vec{b}.(\vec{a}\times\vec{c})}{|\vec{a}\times\vec{c}|}\right|=\left|\frac{\left[\vec{b}\,\vec{a}\,\vec{c}\right]}{|\vec{a}\times\vec{c}|}\right|=\left|\frac{\left[\vec{a}\ \vec{b}\ \vec{c}\right]}{|\vec{a}\times\vec{c}|}\right|$$

$$\vec{a}\times\vec{c}=\begin{vmatrix}i & j & k\\ 1 & 1 & 1\\ 0 & 1 & -1\end{vmatrix}=-2i+j+k$$

$$\vec{a}\times\vec{b}=\vec{c}$$

$$\left(\vec{a}\times\vec{b}\right).\vec{c}=\vec{c}.\vec{c}\ \Rightarrow\ \left[\vec{a}\ \ \vec{b}\ \vec{c}\right]=2$$

$$l=\frac{2}{\sqrt{6}}=\sqrt{\frac{2}{3}}$$

$$3l^2=2$$

16. Let $\vec{a}=\hat{i}-\alpha\hat{j}+\beta\hat{k}$, $\vec{b}=3\hat{i}+\beta\hat{j}-\alpha\hat{k}$ and $\vec{c}=-\alpha\hat{i}-2\hat{j}+\hat{k}$, where α and β are integers. If $\vec{a}\cdot\vec{b}=-1$ and $\vec{b}\cdot\vec{c}=10$ then $\left(\vec{a}\times\vec{b}\right).\vec{c}$ is equal to _____. **[JEE(Main)July2021]**

माना $\vec{a}=\hat{i}-\alpha\hat{j}+\beta\hat{k}$, $\vec{b}=3\hat{i}+\beta\hat{j}-\alpha\hat{k}$ तथा $\vec{c}=-\alpha\hat{i}-2\hat{j}+\hat{k}$ हैं, जहाँ α तथा β पूर्णांक हैं। यदि $\vec{a}\cdot\vec{b}=-1$ तथा $\vec{b}\cdot\vec{c}=10$ हैं, तो $\left(\vec{a}\times\vec{b}\right).\vec{c}$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 9

Sol. $\vec{a}=\hat{i}-\alpha\hat{j}+\beta\hat{k}$

$\vec{b}=3\hat{i}+\beta\hat{j}-\alpha\hat{k}$

$\vec{c}=-\alpha j-2\hat{j}+\hat{k}$

$\therefore\ \vec{a}.\vec{b}=1$

$\Rightarrow\ 3-\alpha\beta-\alpha\beta=-1 \qquad \therefore\ \ \alpha\beta=2 \qquad$ .....(1)

$\therefore\ \vec{b}.\vec{c}=10$

$\therefore\ 3\alpha-2\beta-\alpha=10$

$4\alpha+2\beta=-10$

$\therefore\ 2\alpha+\beta=-5 \qquad$ .....(2)

$\therefore\ \alpha, \beta \in$ integers and satisfying above conditions

$\therefore\ \alpha=-2$ and $\beta=-1$

$$\left(\vec{a}\times\vec{b}\right).\vec{c}=\begin{vmatrix}1 & -\alpha & \beta\\ 3 & \beta & -\alpha\\ -\alpha & -2 & 1\end{vmatrix}$$

$$=\begin{vmatrix}1 & 2 & -1\\ 3 & -1 & 2\\ 2 & -2 & 1\end{vmatrix}$$

$= 1(-1+4)-2(3-4)-1(-6+2)$

$= 3+2+4=9$

3D Geometry

17. The lines $x=ay-1=z-2$ and $x=3y-2=bz-2$, $(ab\neq 0)$ are coplanar, if : **[JEE(Main)July2021]**

रेखाएँ $x=ay-1=z-2$ तथा $x=3y-2=bz-2$, $(ab\neq 0)$ समतलीय हैं, यदि : **[JEE(Main)July2021]**

(A) $b=1, a\in R-\{0\}$ (B) $a=2, b=2$ (C) $a=2, b=3$ (D) $a=1, b\in R-\{0\}$

Ans. A

Sol. $\dfrac{x-0}{1}=\dfrac{y-\frac{1}{a}}{1/a}=\dfrac{z-2}{1}$

$$\frac{x}{1}=\frac{y-\frac{2}{3}}{\frac{1}{3}}=\frac{z-\frac{2}{b}}{\frac{1}{b}}$$

if lines are coplaner then

$$\begin{vmatrix} 0 & \frac{2}{3}-\frac{1}{a} & \frac{2}{b}-2 \\ 1 & \frac{1}{a} & 1 \\ 1 & \frac{1}{3} & \frac{1}{b} \end{vmatrix}=0$$

$3b - 4a + 4ab = 3$

from option (1)

$b = 1\ a \in R - \{0\}$

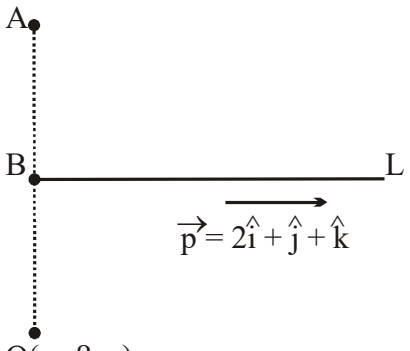
18. Consider the line L given by the equation $\frac{x-3}{2}=\frac{y-1}{1}=\frac{z-2}{1}$. Let Q be the mirror image of the point (2,3, –1) with respect to L. Let a plane P be such that it passes through Q, and the line L is perpendicular to P. Then which of the following points is on the plane P ? **[JEE(Main)July2021]**

माना Q, बिन्दु (2, 3, – 1) का रेखा L : $\frac{x-3}{2}=\frac{y-1}{1}=\frac{z-2}{1}$ में दपर्ण प्रतिबिम्ब है। माना एक समतल P बिन्दु Q से होकर जाता है तथा रेखा L, P पर लम्बवत् है। तो निम्न में से कौनसा बिन्दु समतल P है ? **[JEE(Main)July2021]**

(A) (1, 1, 2) (B) (1, 2, 2) (C) (1, 1, 2) (D) (–1, 1, 2)

Ans. B

Sol.



$Q(\alpha, \beta, \gamma)$

$B\ (2\lambda + 3, \lambda + 1, \lambda + 2)$

$\overrightarrow{AB}=(2\lambda+1)\hat{i}+(\lambda-2)\hat{j}+(\lambda+3)\hat{k}$

$\overrightarrow{AB}\ .\ \vec{P}=0$

$2(2\lambda + 1) + \lambda - 2 + \lambda + 3 = 0$

$6\lambda = -3$

$\lambda = \frac{-1}{2}$

$B\left(2, \frac{1}{2}, \frac{3}{2}\right)$

$\frac{\alpha+2}{2} = 2, \qquad \frac{\beta+3}{2} = \frac{1}{2}, \qquad \frac{\gamma-1}{2} = \frac{3}{2}$

$\alpha=2, \qquad \beta = 2, \qquad \gamma = 2$

equation of plane $2x + y + z =$ of passing through $(2, -2, 4)$

$d = 6$

$2x + y + z = 6$

19. Let L be the line of intersection of planes $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ and $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$. If P (α, β, γ) is the foot of perpendicular on L from the point (1,2, 0), then the value of $35(\alpha + \beta + \gamma)$ is equal to :

[JEE(Main)July2021]

माना L, समतलों $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ तथा $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$ की प्रतिच्छेदन रेखा है। यदि बिन्दु (1, 2, 0) से रेखा L पर डाले गए लम्ब का पाद P (α, β, γ) है, तो $35(\alpha + \beta + \gamma)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 134 (B) 143 (C) 119 (D) 101

Ans. C

Sol. Direction of line of inter section

$$\vec{P} = \vec{n}_2 \times \vec{n}_2$$

$$= \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$= i(1 - 2) - j(-1 - 4) + k(1 + 2)$

$= -1 + 5j + 3k$

Equation of plane in cartesicen form

$x - y + 2z = 2$ .....(1)

$2x + y - z = 2$ .....(2)

Now we will get one point on Line if intersection

then put $z = 0$ in (1) and (2)

$x = \frac{4}{3}, \quad y = -\frac{2}{3}$

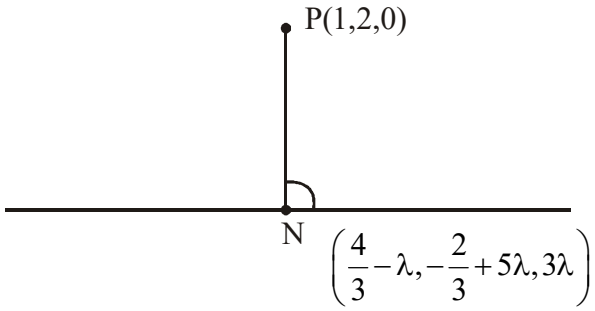
point $\left(\frac{4}{3}, -\frac{2}{3}, 0\right)$

Now equation of L

$\vec{r} = \vec{a} + \lambda\vec{P}$

$\vec{r} = \left(\frac{4}{3}i - \frac{2}{3}j + 0k\right) + \lambda\left(-i + 5j + 3k\right)$

We calcualte foot of perpendicular



$P\vec{N} = \left(\frac{1}{3} - \lambda\right)i + \left(5\lambda - \frac{8}{3}\right)j + \left(3\lambda\right)k$

$P\vec{N}.\vec{P} = 0$

$-1\left(\frac{1}{3} - \lambda\right) + 5\left(5\lambda - \frac{8}{3}\right) + 9\lambda = 0$

$\lambda = \frac{41}{105}$

Now $\alpha = \frac{4}{3} - \lambda$

$\beta = 5\lambda - \frac{2}{3}$

$\gamma = 3\lambda$

$\alpha + \beta + \gamma = 7\lambda + \frac{2}{3}$

$= 7\left(\frac{41}{105}\right) + \frac{2}{5} = \frac{51}{15}$

Now $35\left(\alpha + \beta + \gamma\right) = 35 \times \frac{51}{15} = 119$

20. If the shortest distance between the straight lines $3(x-1) = 6(y-2) = 2(z-1)$ and $4(x-2) = 2(y-\lambda) = (z-3)$, $\lambda \in R$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to : **[JEE(Main)July2021]**

यदि सरल रेखाओं $3(x-1)=6(y-2)=2(z-1)$ तथा $4(x-2)=2(y-\lambda)=(z-3)$, $\lambda \in R$ के बीच की न्यूनतम दूरी $\frac{1}{\sqrt{38}}$ है, तो λ का पूर्णांक मान बराबर है : **[JEE(Main)July2021]**

(A) 5 (B) – 1 (C) 2 (D) 3

Ans. D

Sol. Given lines

$3(x-1)=6(y-2)=2(z-1)$

$$\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-1}{3}---(1)$$

$4(x-2)=2(y-\lambda)=1(z-3)$

$$\frac{x-2}{1}=\frac{y-\lambda}{2}=\frac{z-3}{4}---(2)$$

Minimum distance between two skew lines

$$d=\left|\frac{\overrightarrow{AB}\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right| \quad \vec{p}=2i+j+3k \quad \vec{q}=i+2j+4k$$

$$\overrightarrow{AB}=i+(\lambda-2)j+(2)\hat{k}$$

$$\vec{p}\times\vec{q}=\begin{vmatrix} i & j & k \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix}=i(-2)-j(5)+k(3)$$

$$d=\left|\frac{-2+5(2-\lambda)+6}{\sqrt{4+25+9}}\right|$$

$$d=\left|\frac{-2+10-5\lambda+6}{\sqrt{38}}\right|=\frac{1}{\sqrt{38}}$$

$$d=\left|\frac{14-5\lambda}{\sqrt{38}}\right|=\frac{1}{\sqrt{38}} \quad =-2i-5j+3k$$

$|5\lambda-14|=1$

$5\lambda-14=\pm1$

$5\lambda=14\pm1$

$\lambda=3$ (integral value)

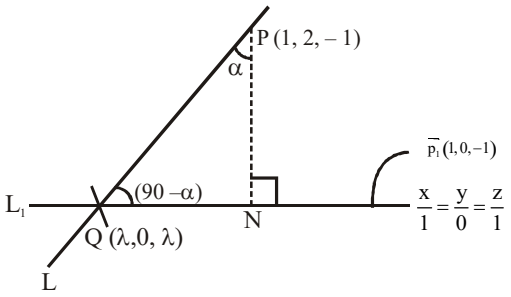
21. Let the foot of perpendicular from a point P(1, 2, –1) to the straight line L : $\frac{x}{1}=\frac{y}{0}=\frac{z}{-1}$ be N. Let a line be drawn from P parallel to the plane $x+y+2z=0$ which meets L at point Q. If α is the acute angle between the lines PN and PQ, then $\cos\alpha$ is equal to _____ . **[JEE(Main)July2021]**

माना सरल रेखा L : $\frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$ पर बिन्दु P(1, 2, – 1) से डाले गए लम्ब का पाद N है। माना P से समतल, x + y + 2z = 0 के समान्तर खींची गई एक रेखा, L को बिन्दु Q पर मिलती है। यदि रेखाओं PN तथा PQ के बीच का न्यूनकोण α है, तो cosα बराबर है: **[JEE(Main)July2021]**

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{5}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2\sqrt{3}}$

Ans. A

Sol.



$\overrightarrow{PQ} \parallel$ Plane

$\overrightarrow{PQ}(\lambda-1,-2,-\lambda+1)$

$\overrightarrow{PQ}.\vec{n} = 0 \quad [\vec{n} = (1,1,2)]$

$\lambda-1-2-2\lambda+2=0$

$-\lambda-1=0$

$Q(-1,0,-1)$

$\overrightarrow{PQ} = (-2,-2,2)$

$$\cos(90-\alpha) = \frac{\overrightarrow{PQ}.\overrightarrow{P_1}}{|\overrightarrow{PQ}|.|\overrightarrow{P_1}|}$$

$$\sin\alpha\left|\frac{-2+0-2}{\sqrt{4+4+4\sqrt{2}}}\right| = \frac{4}{2\sqrt{3}\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\cos\alpha = \frac{1}{\sqrt{3}}$$

22. Let the plane passing through the point (–1, 0, – 2) and perpendicular to each of the planes 2x + y – z = 2 and x – y – z = 3 be ax + by + cz + 8 = 0. Then the value of a + b + c is equal to : **[JEE(Main)July2021]**

माना बिन्दु (–1, 0, – 2) से होकर जाने वाले तथा समतलों 2x + y – z = 2 और x – y – z = 3 पर लम्बवत् समतल का समीकरण ax + by + cz + 8 = 0 है, तो a + b + c का मान बराबर है : **[JEE(Main)July2021]**

(A) 5 (B) 8 (C) 3 (D) 4

Ans. D

Sol. A (–1, 0, – 2)

P_1; $2x + y - z = 2$; $\vec{n}_1 = 2i + j - k$

P_2; $x - y - z = 3$; $\vec{n}_2 = i - j - k$

$\vec{n} = \vec{n}_1 \times \vec{n}_2$

$$\vec{n} = \begin{vmatrix} i & j & k \\ 2 & 1 & -1 \\ 1 & -1 & -1 \end{vmatrix} = -2i + j - 3k$$

DR's of normal = 2, –1, 3

Equation of plane $\Rightarrow 2(x + 1) - 1(y - 0) + 3(z + 2) = 0$

$\Rightarrow 2x - y + 3z + 8 = 0$

$a + b + c = 2 - 1 + 3 = 4$

23. For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines

$$\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3} \text{ and } \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3},$$

lies on the plane $x + 2y - z = 8$ then $\alpha - \beta$ is equal to : **[JEE(Main)July2021]**

वास्तविक संख्याओं α तथा $\beta \neq 0$ के लिए, यदि सरल रेखाओं

$$\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3} \text{ तथा } \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3},$$

का प्रतिच्छेदन बिन्दु, समतल $x + 2y - z = 8$ पर है, तो $\alpha - \beta$ बराबर है : **[JEE(Main)July2021]**

(A) 5 (B) 7 (C) 3 (D) 9

Ans. B

Sol. Given lines :

$$L_1 : \frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$$

$$L_2 : \frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$$

General point on the line L_1 :

$= (\alpha + t_1, 1 + 2t_1, 1 + 3t_1)$

General point on the line L_2 :

$= (4 + \beta + t_2, 6 + 3t_2, 7 + 3t_2)$

for point of intersection of 4 L_2

$\alpha + t_1 = 4 + \beta t_2$ .....(i)

$1 + 2t_2 = 6 + 3t_2$ .....(ii)

$1 + 3t_1 = 7 + 3t_2$ .....(iii)

On solving (ii) and (iii)

$t_1 = 1, t_2 = -1$

Using (i) $\alpha + 1 = 4 - \beta$

$\alpha + \beta = 3$ .....(iv)

Point of intersection of L_1 and L_2 in term of α

$= (\alpha + 1, 1+ 2, 1 +3)$

$= (1 +\alpha , 3, 4)$ lies on the plane $x + 2y - z = 8$

$1 + \alpha + 6 - 4 = 8 \quad \alpha = 5, \beta = 2$

$\alpha - \beta = 7$

24. Let P be a plane passing through the pionts (1,0,1), (1, –2, 1) and (0, 1, –2). Let a vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ be such that $\vec{a}$ is parallel to the plane P, perpendicular to $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $\vec{a}.(\hat{i} + \hat{j} + 2\hat{k}) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____. **[JEE(Main)July2021]**

माना समतल P बिंदुओं (1,0,1), (1, –2, 1) तथा (0, 1, –2) से होकर जाता है। माना एक सदिश $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$, समतल P के समांतर है, सदिश $(\hat{i} + 2\hat{j} + 3\hat{k})$ के लंबवत है तथा $\vec{a}.(\hat{i} + \hat{j} + 2\hat{k}) = 2$ को संतुष्ट करता है। तो $(\alpha - \beta + \gamma)^2$ बराबर है _____ **[JEE(Main)July2021]**

Ans. 81

Sol. Plane P A(1, 0, 1), B(1, –2, 1), C(0, 1, –2)

$\vec{P} = \overrightarrow{AB} = 2\hat{j}$

$\vec{q} = \overrightarrow{AC} = -\hat{i} + \hat{j} - 3\hat{k}$

let $\hat{i} + 2\hat{j} + 3\hat{k} = \vec{r}$

$\vec{a}$ is lying in plane of $\vec{p}$ & $\vec{q}$ and perpendicular or $\vec{r}$

$\vec{a} = \lambda(\vec{p} \times \vec{q}) \times \vec{r}$

$= \lambda((\vec{p}.\vec{r})\vec{q} - (\vec{q}.\vec{r})\vec{p})$

$= \lambda(4\vec{q} + 8\vec{p})$

$\vec{a} = \lambda(-4\hat{i} + 10\hat{j} - 12\hat{k})$

$\vec{a}.(\hat{i} + \hat{j} + 2\hat{k}) = 2$

$-4\lambda + 20\lambda - 24\lambda = 2$

$-8\lambda = 2$

$\lambda = -1/4$

$\alpha - \beta + \gamma = -4\lambda - 20\lambda - 12\lambda$

$= -36\lambda$

$\alpha - \beta + \gamma = 9$

$(\alpha - \beta + \gamma)^2 = 81$

25. If the line $\frac{x-k}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x+1}{3}=\frac{y+2}{2}=\frac{z+3}{1}$ are co-planar, then the value of k is _____.

[JEE(Main)July2021]

यदि रेखाएँ $\frac{x-k}{1}=\frac{y-2}{2}=\frac{z-3}{3}$ तथा $\frac{x+1}{3}=\frac{y+2}{2}=\frac{z+3}{1}$ समतलीय हैं, तो k का मान है _____

[JEE(Main)July2021]

Ans. 1

Sol. $\vec{r}=(k,2,3)+\lambda(1,2,3)$

$\frac{1}{2}$

$\vec{d}=(k+1)\hat{i}+4\hat{j}+6\hat{k}$

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ k+1 & 4 & 6 \end{vmatrix}=0$$

$1(8) - 2(18 - k - 1) + 3\,(12 - 2k - 2) = 0$

$2k + 10 - 6K - 6 = 0$

$4 = 4k$

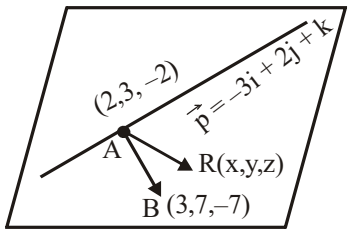
$k = 1$

26. Let a plane P pass through the point (3, 7, –7) and contain the line $\frac{x-2}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$. If distance of the plan P from the originis d, then d^2 is equal to. **[JEE(Main)July2021]**

माना समतल P बिन्दु (3, 7, –7) से होकर जाता है। तथा रेखा $\frac{x-2}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$ इसमें स्थित है। यदि समतल P की मूलबिन्दु से दूरी d है, तो d^2 बराबर है ––––––– **[JEE(Main)July2021]**

Ans. 3

Sol.



$\overrightarrow{AR}, \overrightarrow{AB}, \vec{P}$ are collinear vectors

$$\begin{vmatrix} x-2 & y-3 & z+2 \\ 1 & 4 & -5 \\ -3 & 2 & 1 \end{vmatrix}=0$$

$\Rightarrow (x-2)\,(14) - (y-3)\,(-14) + (z+2)\,(14) = 0$

$\Rightarrow x + y + 7 - 3 = 0$

$d = \left|\frac{0+0+0-3}{\sqrt{3}}\right| = \sqrt{3}$

$d^2 = 3$

27. The distance of the point P(3, 4, 4) from the point of intersection of the line joining the points Q(3, –4, –5) and R(2, –3, 1) and the plane 2x + y + z = 7, is equal to _____. **[JEE(Main)July2021]**

बिन्दुओं Q(3, – 4, – 5) तथा R(2, – 3, 1) को मिलाने वाली रेखा तथा समतल 2x + y + z = 7 के प्रतिच्छेदन बिन्दु से बिन्दु P (3,4, 4) की दूरी है _____ **[JEE(Main)July2021]**

Ans. 7

Sol. D.R. of the line QR

$= < -1, 1, 6 >$

General point on the line QR

$= (3 - t, -4 + t, -5 + 6t)$

Let It lies on the plane 2x + y + z = 7

$\Rightarrow 6 - 2t - 9 + 7t = 7$

$\Rightarrow 5t = 10$

$t = 2$

$\therefore$ Point of intersection of line QR and the plane = (3 – 2, – 4 + 2, –5 + 12)

$= (1, -2, 7)$

Required distance = $\sqrt{(3-1)^2 + (4+2)^2 + (4-7)^2}$

$= \sqrt{4+36+9}$

$= 7$

28. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{a} \cdot (\vec{b} \times \vec{c})$ is equal to : **[JEE(Main)Aug.2021]**

माना $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ तथा $\vec{b} = \hat{j} - \hat{k}$ है। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $\vec{a} \times \vec{c} = \vec{b}$ तथा $\vec{a} \cdot \vec{c} = 3$ है, तो $\vec{a} \cdot (\vec{b} \times \vec{c})$ बराबर है: **[JEE(Main)Aug.2021]**

(A) – 6 (B) 6 (C) 2 (D) – 2

Ans. D

Sol. $\vec{a} \times \vec{c} = \vec{b}$

cross with $\vec{a}$ $\qquad \vec{a} \times \vec{b} = -2\hat{i} + \hat{j} + \hat{k}$

$\vec{a} \times (\vec{a} \times \vec{c}) = \vec{a} \times \vec{b}$ $\qquad \left|\vec{a} \times \vec{b}\right| = \sqrt{6}$

$3\vec{a} - 3\vec{c} = \vec{a} \times \vec{b}$

$3\vec{c} = 3\vec{a} - \vec{a} \times \vec{b}$

dot with $\vec{a} \times \vec{b}$

$3\left(\vec{a}\times\vec{b}\right).\vec{c}=0-\left(\vec{a}\times\vec{b}\right)^2$

$3\left(\vec{a}\times\vec{b}\right).\vec{c}=-\ 6$

$\left(\vec{a}\times\vec{b}\right).\vec{c}=-2=\vec{a}.\left(\vec{b}\times\vec{c}\right)$

29. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|2\vec{a}+3\vec{b}\right| = \left|3\vec{a}+\vec{b}\right|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60º. If $\frac{1}{8}\ \vec{a}$ is a unit vector, then $|\vec{b}|$ is equal to : **[JEE(Main)Aug.2021]**

माना $\vec{a}$ तथा $\vec{b}$ दो सदिशों हैं जिनके लिए $\left|2\vec{a}+3\vec{b}\right| = \left|3\vec{a}+\vec{b}\right|$ हैं और $\vec{a}$ तथा $\vec{b}$ के बीच का कोण 60º है। यदि $\frac{1}{8}\ \vec{a}$ एक इकाई सदिश हैं, तो $|\vec{b}|$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 6 (B) 4 (C) 8 (D) 5

Ans. D

Sol. $\vec{a}\hat{}\vec{b}=60^o$

$|\vec{a}|=8$

$|2\vec{a}+3\vec{b}|^2=|3\vec{a}+\vec{b}|^2$

$4a^2+12ab\cos 60^o+9b^2=9a^2+6ab\cos 60^o+b^2$

$\Rightarrow b^2+3b-40=0$

$\Rightarrow b=5(b>0)$

30. Let $\vec{a},\vec{b},\vec{c}$ be three vectors mutually perpendicular to each other and have same magnitude. If a vector $\vec{r}$ satisfies $\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\}+\vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\}+\vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\}=\vec{0}$ then $\vec{r}$ is equal to : **[JEE(Main)Aug.2021]**

माना तीन सदिश $\vec{a},\vec{b},\vec{c}$ परस्पर लम्बवत हैं तथा इनके परिमाण बराबर है। यदि एक सदिश $\vec{r}$, $\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\}+\vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\}+\vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\}=\vec{0}$ को सन्तुष्ट करता है, तो $\vec{r}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{1}{2}(\vec{a}+\vec{b}+2\vec{c})$ (B) $\frac{1}{3}(2\vec{a}+\vec{b}-\vec{c})$ (C) $\frac{1}{3}(\vec{a}+\vec{b}+\vec{c})$ (D) $\frac{1}{2}(\vec{a}+\vec{b}+\vec{c})$

Ans. D

Sol. $\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$

$|\vec{a}|=|\vec{b}|=|\vec{c}|=k$

$\vec{a}\times\{(\vec{r}-\vec{b})\times\vec{a}\}+\vec{b}\times\{(\vec{r}-\vec{c})\times\vec{b}\}+\vec{c}\times\{(\vec{r}-\vec{a})\times\vec{c}\}=0$

$\vec{a}\times\left((\vec{r}-\vec{b})\times\vec{a}\right)$

$$=|\vec{a}|^2(\vec{r}-\vec{b})-\left(\vec{a}.(\vec{r}-\vec{b})\right)\vec{a}$$

$$=k^2(\vec{r}-\vec{b})-\left(\vec{a}.\vec{r}\right)\vec{a}$$

$$=k^2(\vec{r}-\vec{b})-\left(\vec{a}.(x\vec{a}+y\vec{b}+z\vec{c})\right)\vec{a}$$

$$=k^2(\vec{r}-\vec{b})-\left(k^2x\right)\vec{a}$$

$$=k^2(\vec{r}-\vec{b})-k^2x\vec{a}+k^2(\vec{r}-\vec{c})-k^2y\vec{b}+k^2(\vec{r}-\vec{a})-k^2z\vec{c}=0$$

$$\vec{r}-\vec{b}-x\vec{a}+\vec{r}-\vec{c}-y\vec{b}+\vec{r}-\vec{a}-z\vec{c}=0$$

$$3\vec{r}-\vec{a}-\vec{b}-\vec{c}-x\vec{a}-y\vec{b}-z\vec{c}=0$$

$$3\vec{r}-(\vec{a}+\vec{b}+\vec{c})-(x\vec{a}+y\vec{b}+z\vec{c})=0$$

$$2\vec{r}=\vec{a}+\vec{b}+\vec{c}$$

$$\vec{r}=\frac{\vec{a}+\vec{b}+\vec{c}}{2}$$

31. If the proejction of the vector $\hat{i}+2\hat{j}+\hat{k}$ on the sum of the two vectors $2\hat{i}+4\hat{j}-5\hat{k}$ and $-\lambda\hat{i}+2\hat{j}+3\hat{k}$ is 1, then λ is equal to _____. **[JEE(Main)Aug.2021]**

यदि दो सदिशों $2\hat{i}+4\hat{j}-5\hat{k}$ तथा $-\lambda\hat{i}+2\hat{j}+3\hat{k}$ के योग पर सदिश $\hat{i}+2\hat{j}+\hat{k}$ का प्रक्षेप 1 है, तो λ बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 5

Sol. $\overline{v}_1=\hat{i}+2\hat{j}+\hat{k},\overline{v}_2=2\hat{i}+4\hat{j}-5\hat{k},\overline{v}_3=-\lambda\hat{i}+2\hat{j}+3\hat{k}$

$$\overline{v}_2+\overline{v}_3=(2-\lambda)\hat{i}+6\hat{j}-2\hat{k}=\overline{v}_4$$

Projection of $\overline{v}_1$ on $\overline{v}_4=\overline{v}_1\cdot\frac{\overline{v}_4}{|\overline{v}_4|}$

$$\Rightarrow\frac{1\times(2-\lambda)+2\times6+1\times(-2)}{\sqrt{(2-\lambda)^2+6^2+(-2)^2}}=1$$

$\Rightarrow (12-\lambda)^2=(2-\lambda)^2+40$

On solving

$\lambda = 5$

32. Let $\vec{a}=\hat{i}+5\hat{j}+a\hat{k},\vec{b}=\hat{i}+3\hat{j}+\beta\hat{k}$ and $\vec{c}=-\hat{i}+2\hat{j}-3\hat{k}$ be three vectors such that, $|\vec{b}\times\vec{c}|=5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____. **[JEE(Main)Aug.2021]**

माना तीन सदिशों $\vec{a}=\hat{i}+5\hat{j}+a\hat{k},\vec{b}=\hat{i}+3\hat{j}+\beta\hat{k}$ तथा $\vec{c}=-\hat{i}+2\hat{j}-3\hat{k}$ के लिए $\left|\vec{b}\times\vec{c}\right|=5\sqrt{3}$ है तथा सदिश $\vec{a}$ सदिश $\vec{b}$ के लम्बवत् है तो $|\vec{a}|^2$ के मानों में अधिकतम मान है _____ **[JEE(Main)Aug.2021]**

Ans. 90

Sol. $\vec{a}=i+5j+\alpha k$

$\vec{b}=i+3j+\beta k$

$\vec{c}=-i+2j-3k$

$|\vec{b}\times\vec{c}|^2=75$

$\Rightarrow b^2c^2-(\vec{b}.\vec{c})^2=75$

$\Rightarrow (10+\beta^2).14-(5-3\beta)^2=75$

$\Rightarrow \beta=-2,-4$

$\vec{a}.\vec{b}=0$

$\Rightarrow 1+15+\alpha\beta=0$

$\Rightarrow \alpha=8,4$

$|\vec{a}|^2=1+25+\alpha^2$

$|\vec{a}|^2_{max}=1+25+64$

$=90$

33. Let $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$. Let a vector $\vec{\upsilon}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{\upsilon}$ is perpendicular to the vector $3\hat{i}+2\hat{j}-\hat{k}$ and its projection on $\vec{a}$ is 19 units, then $\left|2\vec{\upsilon}\right|^2$ is equal to ____ .

[JEE(Main)Aug.2021]

माना $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ तथा $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$ है। माना $\vec{a}$ तथा $\vec{b}$ को रखने वाले समतल में एक सदिश $\vec{\upsilon}$ है। यदि $\vec{\upsilon}$ सदिश $3\hat{i}+2\hat{j}-\hat{k}$ के अभिलम्ब है तथा $\vec{a}$ पर उसका प्रक्षप 19 इकाई है, तब $\left|2\vec{\upsilon}\right|^2$ बराबर है ____

[JEE(Main)Aug.2021]

Ans. 1494

Sol. $\vec{a}\times\vec{b}=\begin{vmatrix} i & j & k \\ 2 & -1 & 2 \\ 1 & 2 & -1 \end{vmatrix}$

$\vec{a}\times\vec{b}=-3\hat{i}+4\hat{j}+5\hat{k}$

$\therefore$ $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$ and it is perpendicular to $\vec{c}=(3\hat{i}+2\hat{j}-\hat{k})$

$\therefore \vec{v}=\lambda(\vec{c}\times(\vec{a}\times\vec{b}))$

$\vec{v} = \lambda((\vec{c}.\vec{b})\vec{a} - (\vec{c}.\vec{a})\vec{b})$

$\vec{v} = \lambda(8\vec{a} - 2\vec{b})$

$\vec{v} = \lambda(14\hat{i} - 12\hat{j} + 18\hat{k})$

Given,

$\vec{v}.\hat{a} = 19$

$\lambda(14\hat{i} - 12\hat{j} + 18\hat{k}).\frac{(2\hat{i} - \hat{j} + 2\hat{k})}{3} = 19$

$\Rightarrow \lambda\frac{(28 + 12 + 36)}{3} = 19$

$\lambda \times \frac{76}{3} = 19$

$\therefore \lambda = \frac{3}{4}$

$\vec{v} = \frac{3}{4}(14\hat{i} - 12\hat{j} + 18k) = \frac{3}{2}(7\hat{i} - 6\hat{j} + 9\hat{k})$

$2\vec{v} = 3(7\hat{i} - 6\hat{j} + 9\hat{k})$

$|2\vec{v}|^2 = \left(\sqrt{21^2 + 18^2 + 27^2}\right)^2 = 1494$

34. A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P ? **[JEE(Main)Aug.2021]**

एक समतल P में रेखा $x + 2y + 3z + 1 = 0 = x - y - z - 6$ स्थित है तथा P, समतल $-2x + y + z + 8 = 0$ के लम्बवत् है तो निम्न में से कौनसा बिन्दु समतल P पर है ? **[JEE(Main)Aug.2021]**

(A) (2, – 1, 1) (B) (1, 0, 1) (C) (–1, 1, 2) (D) (0, 1, 1)

Ans. D

Sol. Equation of required plane

$x + 2y + 3z + 1 + \lambda(x - y - z - 6) = 0$

It is perpendicular to $-2x + y + z + 8 = 0$

$(1 + \lambda, 2 - \lambda, 3 - \lambda).(-2, 1, 1) = 0$

$-2 - 2\lambda + 2 - \lambda + 3 - \lambda = 0$

$\lambda = \frac{3}{4}$

$x + 2y + 3z + 1 + \frac{3}{4}(x - y - z - 6) = 0$

$7x + 5y + 9z - 14 = 0$

(0, 1, 1) lies in this plane

35. Let P be the plane passing through the point (1, 2, 3) and the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=$ 16 and $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$. Then which of the following points does NOT lie on P ?

[JEE(Main)Aug.2021]

माना P एक समतल है जो बिन्दु (1, 2, 3) तथा समतलों $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=16$ और $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$ की प्रतिच्छेदन रेखा से होकर जाता है तो निम्न में से कौनसा बिन्दु P पर स्थित नहीं है? **[JEE(Main)Aug.2021]**

(A) (3, 3, 2) (B) (6, – 6, 2) (C) (– 8, 8, 6) (D) (4, 2, 2)

Ans. D

Sol. Equation of plane

$$(x+y+4x-16)+\lambda(-x+y+z-6)=0$$

$$(1,2,3)$$

$$\lambda=\frac{-1}{2}$$

equation of Req. plane $3x+y+7z=26$

36. Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes $x-y-z-1=0$ and $2x+y-3z+4=0$, is : **[JEE(Main)Aug.2021]**

मूलबिन्दु से $\sqrt{\frac{2}{21}}$ की दूरी पर एक समतल, जिसमें समतलों $x-y-z-1=0$ तथा $2x+y-3z+4=0$ की प्रतिच्छेदन रेखा स्थित है, का समीकरण है : **[JEE(Main)Aug.2021]**

(A) $3x-4z+3=0$ (B) $-x+2y+2z-3=0$

(C) $4x-y-5z+2=0$ (D) $3x-y-5z+2=0$

Ans. C

Sol. $P_1: x-y-z-1=0$

$P_2: 2x-y-3z+4=0$

$P: P_1+\lambda P_2=0$

$(x-y-z-1)+\lambda(2x+y-3z+4)=0$

$\Rightarrow (1+2\lambda)x+(-1+\lambda)y+(-1-3\lambda)z-1+4\lambda=0$

Distance from $(0,0)=\sqrt{\frac{2}{21}}$

$$\left|\frac{0+0+0-1+4\lambda}{\sqrt{(1+2\lambda)^2+(-1+\lambda)^2+(-1-3\lambda)^2}}\right|=\sqrt{\frac{2}{21}}$$

$\Rightarrow 308\lambda^2-184\lambda+15=0$

$(2\lambda-1)(154\lambda-15)=0$

$\lambda=\frac{1}{2},\frac{15}{154}$

for $\lambda=\frac{1}{2}$

$P:(x-y-z-1)+\frac{1}{2}(2x+y-3z+4)=0$

$P:4x-y-5z+2=0$

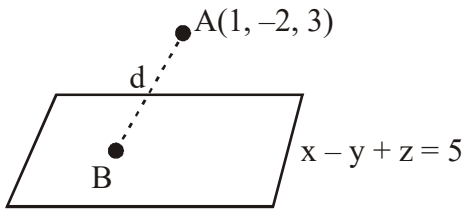
37. The distance of the point (1, –2, 3) from the plane x – y + z = 5 measured parallel to a line, whose direction ratios are 2, 3, –6 is : **[JEE(Main)Aug.2021]**

बिन्दु (1, –2, 3) की, एक रेखा जिसके दिक् अनुपात 2, 3, –6 हैं, के समान्तर समतल x – y + z = 5 से दूरी है : **[JEE(Main)Aug.2021]**

(A) 3 (B) 2 (C) 5 (D) 1

Ans. D

Sol.



DR's of AB = 2, 3, – 6

Equation of AB $=\frac{x-1}{2}=\frac{y+2}{3}=\frac{z-3}{-6}=\lambda$

$B(1+2\lambda,-2+3\lambda,3-6\lambda)$

Lies on given plane

$1+2\lambda+2-3\lambda+3-6\lambda=5$

$\lambda=\frac{1}{7}$

$\overrightarrow{AB}=2\lambda i+3\lambda j-6\lambda k$

$d=|\overrightarrow{AB}|=\lambda.7=\frac{1}{7}.7=1$

38. The equation of the plane passing through the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and parallel to the x-axis is : **[JEE(Main)Aug.2021]**

समतलों $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ तथा $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ की प्रतिच्छेदन रेखा से होकर जाने वाले तथा x-अक्ष के समान्तर समतल का समीकरण है : **[JEE(Main)Aug.2021]**

(A) $\vec{r}\left(\hat{j}-3\hat{k}\right)+6=0$

(B) $\vec{r}\left(\hat{j}-3\hat{k}\right)-6=0$

(C) $\vec{r}\left(\hat{i}-3\hat{k}\right)+6=0$

(D) $\vec{r}\left(\hat{i}+3\hat{k}\right)+6=0$

Ans. A

Sol. Given plane in cartesian form

$x + y + z = 1$ _____(1)

$2x + 3y - z + 4 = 0$ _____(2)

Required plane

$(x + y + z - 1) + \lambda\ (2x + 3y - z + 4) = 0$

$(1 + 2\lambda)x + (1 + 3\lambda)y + (1 - \lambda)z + 4\lambda - 1 = 0$

Here, normal is perpendicular to x-axis.

Hence $(1+2\lambda)+(1+3\lambda)(0)+(1-\lambda)(0)=0$

$d = -\frac{1}{2}$

Now required plane

$y - 3z + 6 = 0$

39. The angle between teh straight lines, whose directioncosines are given by the equations $2l + 2m - n = 0$ and $mn + nl + lm = 0$, is : **[JEE(Main)Aug.2021]**

सरल रेखाओं, जिनके दिक्–कोसाइन समीकरणों $2l + 2m - n = 0$ तथा $mn + nl + lm = 0$ द्वारा दिए गए हैं, के बीच का कोण है : **[JEE(Main)Aug.2021]**

(A) $\frac{\pi}{3}$ (B) $\cos^{-1}\left(\frac{8}{9}\right)$ (C) $\frac{\pi}{2}$ (D) $\pi-\cos^{-1}\left(\frac{4}{9}\right)$

Ans. C

Sol. $n = 2(l + m)$ _____ (1)

$n(m + l) + lm = 0$ _____ (2)

$2(l + m)^2 + lm = 0$

$2l^2 + 5lm + 2m^2 = 0$

$(m + 2l)(2m + l) = 0$

$l = -2m$	$l = -\frac{m}{2}$
Case - I	
$<-2m, m, -2m,>$	$<-\frac{m}{2}, m, m>$
$<-2, 1, -2>$	$<-1, 2, 2>$

Hence, angle between these lines

$$\cos\theta = \frac{2+2-4}{\sqrt{9}\sqrt{9}}$$

$$\cos\theta = 0$$

$$\theta = \frac{\pi}{2}$$

40. Let the equation of the plane, that passes through the point (1, 4, –3) and contains the line of intersection of the planes $3x - 2y + 4z - 7 = 0$ and $x + 5y - 2z + 9 = 0$, be $\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is equal to : **[JEE(Main)Aug.2021]**

माना समतल, जो बिन्दु (1, 4, –3) से होकर जाता है तथा जिसमें समतलों $3x - 2y + 4z - 7 = 0$ तथा $x + 5y - 2z + 9 = 0$ की प्रतिच्छेदन रेखा स्थित है, का समीकरण $\alpha x + \beta y + \gamma z + 3 = 0$ है, तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 15 (B) 23 (C) – 23 (D) – 15

Ans. C

Sol. $P_1 : 3x - 2y + 4z - 7 = 0$

$P_2 : x + 5y - 2z + 9 = 0$

$P : P_1 + \lambda P_2 = 0$

$(3x - 2y + 4z - 7) + \lambda(x + 5y - 2z + 9) = 0$

$\alpha + \beta + \gamma = -11 - 4 - 8 = -23$

$P : (3x - 2y + 4z - 7) + \frac{2}{3}(x + 5y - 2z + 9) = 0$

$P : -11x - 4y - 8z + 3 = 0$

$\alpha + \beta + \gamma = -11 - 4 - 8 = -23$

41. The distance of the point (–1, 2, –2) from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is : **[JEE(Main)Aug.2021]**

बिन्दु (–1, 2, –2) की समतलों $2x + 3y + 2z = 0$ और $x - 2y + z = 0$ की प्रतिच्छेदन रेखा से दूरी है : **[JEE(Main)Aug.2021]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{34}}{2}$ (C) $\frac{5}{2}$ (D) $\frac{\sqrt{42}}{2}$

Ans. B

Sol. $P_1 : 2x + 3y + 2z = 0$

$\vec{n}_1 = 2\hat{i} + 3\hat{j} + 2\hat{k}$

$P_2 : x - 2y + z = 0$

$\vec{n}_2 = \hat{i} - 2\hat{j} + \hat{k}$

$\vec{b} = \vec{n}_1 \times \vec{n}_2$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 1 & -2 & 1 \end{vmatrix}$$

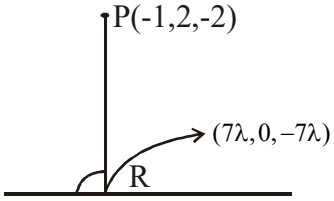
$= \hat{i}(7) - \hat{j}(0) + \hat{k}(-7)$

$\vec{b} = 7\hat{i} - 7\hat{k}$

$\vec{r} = \vec{a} + \lambda\vec{b}$

$= (0,0,0) + \lambda(7\hat{i} - 7\hat{k})$

$= 7\lambda(\hat{i} - \hat{k})$



$\overrightarrow{PR} = (7\lambda + 1)\hat{i} - 2\hat{j} + (2 - 7\lambda)\hat{k}$

$\overrightarrow{PR}.\vec{b} = 0$

$7(7\lambda + 1) - 7(2 - 7\lambda) = 0$

$\lambda = \frac{1}{14}$

$R = \left(-\frac{1}{2}, 0, -\frac{1}{2}\right)$

$PR = \sqrt{\left(\frac{3}{2}\right)^2 + 4 + \left(\frac{3}{2}\right)^2}$

$= \sqrt{\frac{9}{2} + 4} = \sqrt{\frac{17}{2}}$

42. The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point (2, –1, 6) is :

[JEE(Main)Aug.2021]

रेखा $3y - 2z - 1 = 0 = 3x - z + 4$ की बिन्दु (2, –1, 6) से दूरी है : **[JEE(Main)Aug.2021]**

(A) $\sqrt{26}$ (B) $4\sqrt{2}$ (C) $2\sqrt{5}$ (D) $2\sqrt{6}$

Ans. D

Sol. Let direction ratio of the line :

< a, b, c >

Hence

$0.a+3b-2c=0$

$3.a+0.b-1.c=0$

Using cross multiplication rule

$$\frac{a}{-3-0}=\frac{b}{-6-0}=\frac{c}{0-9}$$

$$\Rightarrow \frac{a}{-3}=\frac{b}{-6}=\frac{c}{-9}$$

$$\frac{a}{1}=\frac{b}{2}=\frac{c}{3}$$

for point of the line

when x = 0, z = 4, y = 3

Equation of the line :

$$\frac{x-0}{1}=\frac{y-3}{2}=\frac{z-4}{3} \quad \text{.....(1)}$$

General point on the line

$P=(t,3+2t,4+3t)$

Let given point be $Q=(2,-1,6)$

D.R, of PQ = $(2-t,\ -4-2t,\ 2-3t)$

PQ $\perp^r$ to the line L

$1(2-t)+2(-4-2t)+3(2-3t)=0$

$\Rightarrow 2-t-8-4t+6-9t=0$

$\Rightarrow\ -14t=0$

$t=0$

$\therefore P\equiv(0,3,4)$

$PQ=\sqrt{2^2+4^2+2^2}$

$=\sqrt{4+16+4}=\sqrt{24}$

$=2\sqrt{6}$

43. Let the acute angle bisector of the two planes x – 2y – 2z + 1 = 0 and 2x – 3y – 6z + 1 = 0 be the plane P. Then which of the following points lies on P? **[JEE(Main)Aug.2021]**

माना दो समतलों x – 2y – 2z + 1 = 0 तथा 2x – 3y – 6z + 1 = 0 के न्यून कोण का समद्विभाजक समतल P है। तब इनमें से कौनसा बिन्दु P पर स्थित है? **[JEE(Main)Aug.2021]**

(A) $\left(-2,0,-\frac{1}{2}\right)$ (B) (0, 2, –4) (C) $\left(3,1,-\frac{1}{2}\right)$ (D) (4, 0, –2)

Ans. A

Sol. $P_1 : x - 2y - 2z + 1 = 0$

$P_2 : 2x - 3y - 6z + 1 = 0$

Equation of angle bisectors

$$\frac{x-2y+2z+1}{3} = \pm\frac{2x-3y+6z+1}{7}$$

since $a_1a_2 + b_1b_2 + c_1c_2 > 0$

$\therefore$ Negative sign will give acute angle bisector

$$\Rightarrow 7x-14y-14z+7 = -6x+9y+18z-3$$

$$13x-23y-32z+10 = 0$$

$\left(-2,0,-\frac{1}{2}\right)$ satisfy it.

44. Suppose the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. The $(\alpha + \beta)$ is equal to_____.

[JEE(Main)Aug.2021]

माना रेखा $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ समतल $x + 3y - 2z + \beta = 0$ में स्थित है तो $(\alpha + \beta)$ बराबर है _____

[JEE(Main)Aug.2021]

Ans. 7

Sol. $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$

$\vec{b} = \alpha\hat{i} - 5\hat{j} + 2\hat{k}$

$\vec{n} = \hat{i} + 3\hat{j} - 2\hat{k}$

$\vec{b}.\vec{n} = 0$

$\alpha - 15 - 4 = 0$

$\alpha = 19$

$2 + 6 + 4 + \beta = 0$

$\beta = -12$

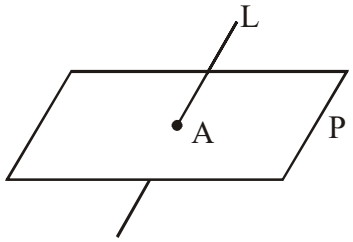
$\alpha + \beta = 19 - 12 = 7$

45. The square of the distance of the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point (–1, – 1, 2) is _____. **[JEE(Main)Aug.2021]**

बिन्दु (–1, – 1, 2) से रेखा $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+1}{6}$ तथा समतल 2x – y + z = 6 के प्रतिच्छेदन बिन्दु की दूरी का वर्ग बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 61

Sol.



$P: 2x-y+z=6$

$L: \frac{x-1}{2}=\frac{y-2}{3}=\frac{z+1}{6}=\lambda$

$A(1+2\lambda, 2+3\lambda, -1+6\lambda)$

$\Rightarrow 2+4\lambda-2-3\lambda-1+6\lambda=6$

$\Rightarrow \lambda=1$

$A(3,5,5)$

$B(-1,-1,2)$

$AB^2=61$

46. Let S be the mirror image of the point Q(1, 3, 4) with respect to the plane 2x–y+z+3=0 and let R (3, 5, λ) be a point of this plane. Then the square of the length of the line segment SR is _____. **[JEE(Main)Aug.2021]**

माना समतल 2x–y+z+3=0 के सापेक्ष बिन्दु Q(1, 3, 4) का दर्पण प्रतिबिम्ब S है तथा माना इस समतल पर एक बिन्दु R (3, 5, λ) है तो रेखाखण्ड SR की लम्बाई का वर्ग है_____ **[JEE(Main)Aug.2021]**

Ans. 72

Sol. R (3, 5, r) on the plane

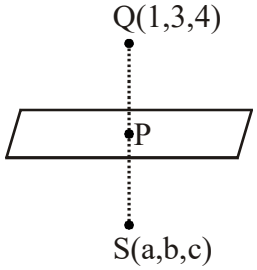
2x – y + z + 3 = 0

then 6 – 5 + r + 3 = 0

r = –4

R(3, 5, –4)

Equation of line QS



$\vec{r}=(1,3,4)+\lambda(2,-1,1)$

$P(1+2\lambda, 3-\lambda, 4+\lambda)$

P is one plane

$2(1+2\lambda)+\lambda-3+4+\lambda+3=0$

$2+4\lambda+\lambda-3+4+\lambda+3=0$

$6\lambda+6=0$

$\lambda = -1$

$(-1, 4, 3)$

P is mid-point of QS then $\frac{a+1}{2} = -1$ $\quad a = -3$

$\frac{b+3}{2} = 4$ $\quad b = 5$

$\frac{c+4}{2} = 3$ $\quad c = 2$

S(–3, 5, 2)

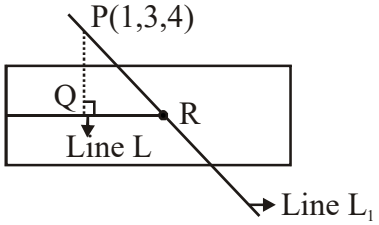
$RS^2 = (\sqrt{36+36})^2 = 72$

47. Let the line L be the projection of the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ in the plane $x - 2y - z = 3$. If d is the distance of the point (0, 0, 6) from L, then d^2 is equal to _____. **[JEE(Main)Aug.2021]**

माना रेखा $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ का समतल $x - 2y - z = 3$ में प्रक्षेप रेखा L है। यदि बिन्दु (0, 0, 6) की L से दूरी d है, तो d^2 बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 26

Sol.



R is POI of line L_1 and Plane

$R(1+2\gamma, 3+\gamma, 4+2\gamma)$

R will lie on $x - 2y - z = 3$

$1 + 2\gamma - 2(3+\gamma) - (4+2\gamma) = 3$

$2\gamma = 1 - 6 - 4 - 3 \Rightarrow \gamma = -6$

$R(-11, -3, -8)$

$Q \equiv$ Foot of perpendicular from P on P line

$Q(1+\mu, 3-2\mu, 4-\mu)$

$1 + \mu - 2(3-2\mu) - (4-\mu) = 3$

$6\mu = 6 + 4 + 3 - 1$

$\mu = 2$ $\quad \overrightarrow{QR} = (-14, -2, -10)$

$Q(3,-1,2)$

Equation of line L

$\vec{r} = (3,-1,2) + t(7,1,5)$

$S(0,0,6)$

distance of S from line $= \dfrac{|\overrightarrow{SQ} \times \overrightarrow{QR}|}{|\overrightarrow{QR}|}$

$= \dfrac{|\hat{i} + 43\hat{j} - 10\hat{k}|}{|7\hat{i} + \hat{j} + 5\hat{k}|}$

$d = \sqrt{26}$

$\Rightarrow d^2 = 26$

48. Let Q be the foot of the perpendicular from the point P(7, –2, 13) on the plane containing the lines $\dfrac{x+1}{6} = \dfrac{y-1}{7} = \dfrac{z-3}{8}$ and $\dfrac{x-1}{3} = \dfrac{y-2}{5} = \dfrac{z-3}{7}$ .Then $(PQ)^2$, is equal to _____.

[JEE(Main)Aug.2021]

माना बिन्दु P(7, –2, 13) से समतल, जिसमें रेखाएँ $\dfrac{x+1}{6} = \dfrac{y-1}{7} = \dfrac{z-3}{8}$ तथा $\dfrac{x-1}{3} = \dfrac{y-2}{5} = \dfrac{z-3}{7}$ स्थित है, पर डाले गये लम्ब का पाद Q है तो $(PQ)^2$ बराबर _____ है। **[JEE(Main)Aug.2021]**

Ans. 96

Sol. Equation of plane $\begin{vmatrix} x-1 & y-2 & z-3 \\ 6 & 7 & 8 \\ 3 & 5 & 7 \end{vmatrix} = 0$

$\Rightarrow x - 2y + z = 0$

$d^2(7,-2,13) = \dfrac{|7+4+13|^2}{6} = 96$

49. Let $\hat{a}$, $\hat{b}$ be unit vectors. If $\vec{c}$ be a vectors such that the angle between $\hat{a}$ and $\hat{c}$ is $\dfrac{\pi}{12}$, and $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$, then $|6\vec{c}|^2$ is equal to : **[JEE(Main)June2022]**

माना Let $\hat{a}$, $\hat{b}$ इकाई सदिश हैं। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $\hat{a}$ तथा $\hat{c}$ के बीच का कोण $\dfrac{\pi}{12}$ है तथा $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$ है, तो $|6\vec{c}|^2$ बराबर है : **[JEE(Main)June2022]**

(A) $6\left(3-\sqrt{3}\right)$ (B) $3 + \sqrt{3}$ (C) $6\left(3+\sqrt{3}\right)$ (D) $6\left(\sqrt{3}+1\right)$

Ans. C

Sol. Let $|\vec{c}| = x$

$\hat{b} = \vec{c} + (\vec{c} \times \hat{a})$

$|\hat{b}| = 1$

$|\vec{c} + 2(\vec{c} \times \hat{a})| = 1$

$|\vec{c}|^2 + 4(\vec{c} \times \hat{a})^2 + 4\vec{c}.(\vec{c} \times \hat{a}) = 1$

$x^2 + 4(x^2)(1)\sin^2(15°) + 0 = 1$

$x^2 + 2x^2\left(1 - \frac{\sqrt{3}}{2}\right) = 1$

$x^2 + x^2(2 - \sqrt{3}) = 1$

$x^2(3 - \sqrt{3}) = 1$

$x^2 = \frac{3 + \sqrt{3}}{6}$

Ans = $(6x)^2 = 36x^2 = 6(3 + \sqrt{3})$

50. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $\left|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})\right| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : The projection of $\hat{a}$ on $(\hat{a} + \hat{b})$ is $\frac{1}{2}$ **[JEE(Main)June2022]**

(A) only (S1) is true (B) only (S2) is true

(C) Both (S1) and (S2) are true (D) Both (S1) and (S2) are false

माना दो इकाई सदिशों $\hat{a}$ तथा $\hat{b}$ के लिए $\left|(\hat{a} + \hat{b}) + 2(\hat{a} \times \hat{b})\right| = 2$ है। यदि $\hat{a}$ तथा $\hat{b}$ के बीच का कोण $\theta \in (0, \pi)$ है, है, तो कथनों :

(S1) : $2|\hat{a} \times \hat{b}| = |\hat{a} - \hat{b}|$

(S2) : $\hat{a}$ का $(\hat{a} + \hat{b})$ पर प्रक्षेप $\frac{1}{2}$ है **[JEE(Main)June2022]**

(A) केवल (S1) सत्य है (B) केवल (S2) सत्य है

(C) (S1) तथा (S2) दोनों सत्य हैं (D) (S1) तथा (S2) दोनों असत्य हैं

Ans. C

Sol. $\left|\left(\hat{a}+\hat{b}\right)+2\left(\hat{a}\times\hat{b}\right)\right|=2$

$\left(\hat{a}+\hat{b}\right)^2+4\left(\hat{a}+\hat{b}\right)^2+4\left(\vec{a}+\vec{b}\right)\left(\vec{a}\times\vec{b}\right)=4$

$1+1+2\cos\theta+4\sin^2\theta=4$

$1+\cos\theta+2\left(1-\cos^2\theta\right)=2$

$\cos\theta=\dfrac{-1}{2}$ is correct in $\theta\in(0,\pi)$

$\theta=\dfrac{2\pi}{3}$

(S1) $2\left|\hat{a}\times\hat{b}\right|=\left|\hat{a}-\hat{b}\right|$

$2\times\dfrac{\sqrt{3}}{2}=\sqrt{1+1+2\times\dfrac{1}{2}}$

$\sqrt{3}=\sqrt{3}$

which is correct

(S2) $\left|\dfrac{\hat{a}.\left(\hat{a}+\hat{b}\right)}{\left|\hat{a}+\hat{b}\right|}\right|=\dfrac{1}{2}$

$\left|\dfrac{1-\dfrac{1}{2}}{1}\right|=\dfrac{1}{2}$

which is also correct

51. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ $a_i>0$, i = 1, 2, 3 be vector which makes equal angles with the coordinate axes OX, OY and OZ. Also, Let the projection of $\vec{a}$ on the vector $3\hat{i}+4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a},\vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i}+4\hat{j}$ is equal to :

[JEE(Main)June2022]

माना सदिश $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ $a_i>0$, i = 1, 2, 3 निर्देशांक अक्षों OX, OY तथा OZ से बराबर कोण बनाता है। यह भी मान लीजिए कि $\vec{a}$ का सदिश $3\hat{i}+4\hat{j}$ पर प्रक्षेप 7 है। माना सदिश $\vec{a}$ को 90° घुमाने पर सदिश $\vec{b}$ प्राप्त होता है। यदि $\vec{a},\vec{b}$ तथा x-अक्ष सह–तलीय हैं, तो एक सदिश $\vec{b}$ का $3\hat{i}+4\hat{j}$ पर प्रक्षेप बराबर है : **[JEE(Main)June2022]**

(A) $\sqrt{7}$ (B) $\sqrt{2}$ (C) 2 (D) 7

Ans. B

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$$\vec{a} = \lambda\left(\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}\right) = \frac{\lambda}{\sqrt{3}}\left(\hat{i} + \hat{j} + \hat{k}\right)$$

Now projection of $\vec{a}$ on $\vec{c} \equiv 3\hat{i} + 4\hat{j}$ is 7

$$\Rightarrow \frac{\vec{a}\cdot\vec{c}}{|\vec{c}|} = 7$$

$$\frac{\lambda}{\sqrt{3}}\frac{\left(\hat{i} + \hat{j} + \hat{k}\right)\cdot\left(3\hat{i} + 4\hat{j}\right)}{5} = 7$$

$$\lambda = 5\sqrt{3}$$

$$\vec{a} = 5(\hat{i} + \hat{j} + \hat{k})$$

now $\vec{b} = 5\alpha\left(\hat{i} + \hat{j} + \hat{k}\right) + \beta\left(\hat{i}\right)$

$$\vec{a}\cdot\vec{b} = 0$$

$$\Rightarrow 25\alpha(3) + 5\beta = 0$$

$$\Rightarrow 15\alpha + \beta = 0 \Rightarrow \beta = -15\alpha$$

$$\vec{b} = 5\alpha(-2\hat{i} + \hat{j} + \hat{k})$$

$$|\vec{b}| = 5\sqrt{3} = |\vec{a}|$$

$$\Rightarrow \alpha = \pm\frac{1}{\sqrt{2}}$$

$$\vec{b} = \pm\frac{5}{\sqrt{2}}(-2\hat{i} + \hat{j} + \hat{k})$$

Projection of $\vec{b}$ on $3\hat{i} + 4\hat{j}$ is

$$\frac{\vec{b}\cdot(3\hat{i} + 4\hat{j})}{5} = \pm\frac{5}{\sqrt{2}}\left(\frac{-6+4}{5}\right) = \pm\sqrt{2}$$

52. If $\vec{a}\,.\,\vec{b} = 1, \vec{b}\,.\,\vec{c} = 2$ and $\vec{c}\,.\,\vec{a} = 3$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right), \vec{b}\times\left(\vec{c}\times\vec{a}\right), \vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ is :

[JEE(Main)June2022]

यदि $\vec{a}\,.\,\vec{b} = 1, \vec{b}\,.\,\vec{c} = 2$ तथा $\vec{c}\,.\,\vec{a} = 3$ हैं, तो $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right), \vec{b}\times\left(\vec{c}\times\vec{a}\right), \vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ का मान बराबर है :

[JEE(Main)June2022]

(A) 0 (B) $-6\vec{a}.\left(\vec{b}\times\vec{c}\right)$ (C) $12\vec{c}.\left(\vec{a}\times\vec{b}\right)$ (D) $-12\vec{b}.\left(\vec{c}\times\vec{a}\right)$

Ans. A

Sol. $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\left(\vec{a}.\vec{c}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{c}=3\vec{b}-\vec{c}$

$\vec{b}\times\left(\vec{c}\times\vec{a}\right)=\left(\vec{b}.\vec{a}\right)\vec{c}-\left(\vec{b}.\vec{c}\right)\vec{a}=\vec{c}-2\vec{a}$

$\vec{c}\times\left(\vec{b}\times\vec{a}\right)=\left(\vec{c}.\vec{a}\right)\vec{b}-\left(\vec{c}.\vec{b}\right)\vec{a}=3\vec{b}-2\vec{a}$

$\left[3\vec{b}-\vec{c},\vec{c}-2\vec{a},3\vec{b}-2\vec{a}\right]$

$\left(3\vec{b}-\vec{c}\right).\left[\left(\vec{c}-2\vec{a}\right)\times\left(3\vec{b}-2\vec{a}\right)\right]$

$\left(3\vec{b}-\vec{c}\right).\left[3\left(\vec{c}\times\vec{b}\right)-2\left(\vec{c}\times\vec{a}\right)-6\left(\vec{a}\times\vec{b}\right)\right]$

$-6\left[\vec{b}\vec{c}\vec{a}\right]+6\left[\vec{c}\vec{a}\vec{b}\right]$

$=0$

53. If the lines $\vec{r}=\left(\hat{i}-\hat{j}+\hat{k}\right)+\lambda\left(3\hat{j}-\hat{k}\right)$ and $\vec{r}=\left(\alpha\,\hat{i}-\hat{j}\right)+\mu\left(2\hat{i}-3\hat{k}\right)$ are coplanar, then the distance of the plane containing these two lines from the point $(\alpha, 0,0)$ is : **[JEE(Main)June2022]**

यदि रेखाएँ $\vec{r}=\left(\hat{i}-\hat{j}+\hat{k}\right)+\lambda\left(3\hat{j}-\hat{k}\right)$ तथा $\vec{r}=\left(\alpha\,\hat{i}-\hat{j}\right)+\mu\left(2\hat{i}-3\hat{k}\right)$ सह–तलीय हैं, तो इन रेखाओं से होकर जाने वाले समतल की बिन्दु $(\alpha, 0,0)$ से दूरी है : **[JEE(Main)June2022]**

(A) $\frac{2}{9}$ (B) $\frac{2}{11}$ (C) $\frac{4}{11}$ (D) 2

Ans. B

Sol. $\vec{r}=(1,3\lambda-1,1-\lambda)$

$\vec{r}=(\alpha+2\mu,-1,-3\mu)$

$$\begin{bmatrix}\alpha-1 & 0 & -1\\ 0 & 3 & -1\\ 2 & 0 & -3\end{bmatrix}=0$$

$\Rightarrow\alpha=\frac{5}{3}$

$$\vec{n}=\begin{bmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 3 & -1\\ 2 & 0 & -3\end{bmatrix}=\hat{i}(-9)-\hat{j}(2)+\hat{k}(-6)$$

$=-9\hat{i}-2\hat{j}-6\hat{k}$

equation of plane

$9(x-1)+2(y+1)+6(z-1)=0$

$9x+2y+6z=13$

Perpendicular distance from (5/3, 0, 0)

$$=\left|\frac{9\frac{5}{3}+0+0-13}{\sqrt{81+4+36}}\right|=\left|\frac{2}{\sqrt{121}}\right|=\frac{2}{11}$$

54. Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k},\vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v}.\hat{j}=7$, then $\vec{v}.\left(\hat{i}+\hat{k}\right)$ is equal to : **[JEE(Main)June2022]**

माना तीन सदिश $\vec{a}=\hat{i}+\hat{j}+2\hat{k},\vec{b}=2\hat{i}-3\hat{j}+\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}+\hat{k}$ है। माना $\vec{a}$ तथा $\vec{b}$ के समतल में एक सदिश $\vec{v}$ का $\vec{c}$ पर प्रक्षेप $\frac{2}{\sqrt{3}}$ है। यदि $\vec{v}.\hat{j}=7$ है, तो $\vec{v}.\left(\hat{i}+\hat{k}\right)$ बराबर है : **[JEE(Main)June2022]**

(A) 6 (B) 7 (C) 8 (D) 9

Ans. D

Sol. $\vec{v}=x\vec{a}+y\vec{b}$

$\vec{v}=x\left(\hat{i}+\hat{j}+2\hat{k}\right)+y\left(2\hat{i}-3\hat{j}+\hat{k}\right)$

$\frac{\vec{v}.\vec{c}}{|\vec{c}|}=2\sqrt{3}$

$x+2y-(x-3y)+2x+y=2$

$2x+6y=2$

$x+3y=1$ ———(1)

$\vec{v}.\hat{j}=7$

$x-3y=7$ ———(2)

$2y=8 \Rightarrow x=4$

$3y=-3 \Rightarrow y=-1$

$\vec{v}=2\hat{i}+7\hat{j}+7\hat{k}$

$\vec{v}.\left(\hat{i}+k\right)$

$2+7=9$

55. Let $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$ .Then the number of vectors $\vec{b}$ such that $\vec{b}\times\vec{c}=\vec{a}$ and $|\vec{b}|\in\{1,2,.....,10\}$ is: **[JEE(Main)June2022]**

माना $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$ हैं। तो सदिशों $\vec{b}$ की संख्या, जिनके लिए $\vec{b}\times\vec{c}=\vec{a}$ तथा $|\vec{b}|\in\{1,2,.....,10\}$ हैं, है: **[JEE(Main)June2022]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\vec{a}=\hat{i}+\hat{j}-\hat{k}$

$\vec{c}=2\hat{i}-3\hat{j}+2\hat{k}$

$\Rightarrow \vec{b}\times\vec{c}\ ||\ \vec{a}$

$\Rightarrow c\perp\vec{a}$

$\Rightarrow \vec{a}.\vec{c}=0$

$\left(\hat{i}+\hat{j}-\hat{k}\right).\left(2\hat{i}-3\hat{j}+2\hat{k}\right)$

$2-3-2\neq 0$

which is coutradiction

So no of vectors $\vec{b}=0$

56. Let $\vec{a}$ and $\vec{b}$ be the vectors along the diagonals of a parallelogram having area $2\sqrt{2}$. Let the angle between $\vec{a}$ and $\vec{b}$ be acute $|\vec{a}|=1$, and $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$. If $c=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$, then an angle between $\vec{b}$ and $\vec{c}$ is : **[JEE(Main)June2022]**

माना एक समांतर चतुर्भुज, जिसका क्षेत्रफल $2\sqrt{2}$ है, के विकर्णों के अनुदिश सदिश $\vec{a}$ तथा $\vec{b}$ हैं। माना $\vec{a}$ तथा $\vec{b}$ के बीच कोण, न्यून कोण है, $|\vec{a}|=1$ तथा $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$ हैं। यदि $c=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$ है, तो $\vec{b}$ तथा $\vec{c}$ के बीच एक कोण है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{5\pi}{6}$ (D) $\frac{3\pi}{4}$

Ans. D

Sol. $\cos\theta=\sin\theta$

$\tan\theta=1$

$\theta=\frac{\pi}{4}$

$\frac{1}{2}|\vec{a}\times\vec{b}|=2\sqrt{2}$

$|\vec{a}||\vec{b}|\sin\theta=4\sqrt{2}$

$$\left|\vec{b}\right|.\frac{1}{\sqrt{2}}=4\sqrt{2}$$

$$\left|\vec{b}\right|=8$$

$$\left|\vec{c}\right|=\sqrt{8\left|\vec{a}\times\vec{b}\right|^2+4\left|\vec{b}\right|^2}$$

$$\sqrt{8\left(4\sqrt{2}\right)^2+4(8)^2}$$

$$=\sqrt{32\times8+32\times8}$$

$$=16\sqrt{2}$$

$$\vec{c}=2\sqrt{2}\left(\vec{a}\times\vec{b}\right)-2\vec{b}$$

$$\vec{b}.\vec{c}=0-2\vec{b}.\vec{b}$$

$$\left|\vec{c}\right|\cos\theta=-2\left|\vec{b}\right|$$

$$\cos\theta=\frac{-2(8)}{16\sqrt{2}}$$

$$\cos\theta=\frac{-1}{\sqrt{2}}$$

$$\Rightarrow\theta=\frac{3\pi}{4}$$

57. Let the plane P : $\vec{r}.\vec{a}=d$ contain the line of intersection of two planes $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=6$ and $\vec{r}.\left(-6\hat{i}+5\hat{j}-\hat{k}\right)=7$. If the plane P passes through the point $\left(2,3,\frac{1}{2}\right)$, then the value of $\frac{\left|13\vec{a}\right|^2}{d^2}$ is equal to :

[JEE(Main)June2022]

माना दो समतलतों $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=6$ तथा $\vec{r}.\left(-6\hat{i}+5\hat{j}-\hat{k}\right)=7$ की प्रतिच्छेदन रेखा समतल P : $\vec{r}.\vec{a}=d$ में है। यदि समतल P, बिन्दु $\left(2,3,\frac{1}{2}\right)$ से होकर जाता है, तो $\frac{\left|13\vec{a}\right|^2}{d^2}$ का मान है : **[JEE(Main)June2022]**

(A) 90 (B) 93 (C) 95 (D) 97

Ans. B

Sol. $P_1 : x + 3y – z = 6$

$P_2 : –6x + 5y – z = 7$

Let the required plane is $P_1 + \lambda P_2 = 0$

$\Rightarrow (1 – 6\lambda)x + (3 + 5\lambda)y – z(1 + \lambda) = 6 + 7\lambda$

$\left(2,3,\frac{1}{2}\right)$ will satisfy it

$$\lambda = 1$$

So the plane : $-5x + 8y - 2z - 13 = 0$

$$\Rightarrow \vec{r}.\left(-5\hat{i}+8\hat{j}-2\hat{k}\right)=13$$

$$\frac{|13\vec{a}|^2}{d^2}=\frac{13^2}{13^2}.(25+64+4)=93$$

58. Let $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$, where $\alpha \in R$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15\left(\alpha^2+4\right)}$, then the value of $2|\vec{a}|^2+\left(\vec{a}.\vec{b}\right)|\vec{b}|^2$ is equal to :

[JEE(Main)June2022]

माना $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$ तथा $\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$ हैं, जहाँ $\alpha \in R$ है। यदि समान्तर चतुर्भुज, जिसकी संलग्न भुजाएँ $\vec{a}$ तथा $\vec{b}$ हैं, क्षेत्रफल $\sqrt{15\left(\alpha^2+4\right)}$ हैं, तो $2|\vec{a}|^2+\left(\vec{a}.\vec{b}\right)|\vec{b}|^2$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 10 (B) 7 (C) 9 (D) 14

Ans. D

Sol. Given $\vec{a}=\alpha\hat{i}+2\hat{j}-\hat{k}$

$\vec{b}=-2\hat{i}+\alpha\hat{j}+\hat{k}$

area of parallelogram is $\sqrt{15\left(\alpha^2+4\right)}$

$$\Rightarrow |\vec{a}\times\vec{b}|=\sqrt{15\left(\alpha^2+4\right)}$$

$$\left\|\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ \alpha & 2 & -1\\ -2 & \alpha & 1\end{matrix}\right\|=\sqrt{15\left(\alpha^2+4\right)}$$

$$\Rightarrow \sqrt{(2+\alpha)^2+(\alpha-2)^2+\left(\alpha^2+4\right)^2}=\sqrt{15\alpha^2+60}$$

Squaring both sides and solving it.

$\alpha^4 - 5\alpha^2 - 36 = 0$

$\alpha^2 = 9$ and $\alpha^2 = -4$ (not possible)

$\Rightarrow \alpha = \pm 3$

Therefore

$$2|\vec{a}|^2+\left(\vec{a}.\vec{b}\right).|\vec{b}|^2$$

$= 2 \times 14 - 14 = 28 - 14 = 14$

59. Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$. If $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i}+2\hat{j}+\hat{k}$ is : **[JEE(Main)June2022]**

माना सदिश $3\hat{i}+\frac{1}{2}\hat{j}+2\hat{k}$ के लंबवत एक सदिश $\vec{a}$ है। यदि $\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$ है, तो सदिश $\vec{a}$ का सदिश $2\hat{i}+2\hat{j}+\hat{k}$ पर प्रक्षेप है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{5}{3}$ (D) $\frac{7}{3}$

Ans. C

Sol. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{a}.\left(3\hat{i}-\frac{1}{2}\hat{j}+2\hat{k}\right)=0$

$\Rightarrow 3a_1-\frac{a_2}{2}+2a_3=0$

$\Rightarrow 6a_1-a_2+4a_3=0$ ———(1)

Also

$$\vec{a}\times\left(2\hat{i}+\hat{k}\right)=2\hat{i}-13\hat{j}-4\hat{k}$$

$$-2a_2\hat{k}+2a_3\hat{j}-a_1\hat{j}+a_2\hat{i}=2\hat{i}-13\hat{j}-4\hat{k}$$

By comparison

$a_2=2$ and $a_1-2a_3=13$ ———(2)

from eq. (1) and (2)

$a_1=3$ and $a_3=-5$

$\vec{a}=3\hat{i}+2\hat{j}-5\hat{k}$

Projection of $\vec{a}$ on $2\hat{i}+2\hat{j}+\hat{k}=\frac{6+4-5}{\sqrt{2^2+2^2+1^2}}$

$=\frac{5}{3}$ units

60. Let $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ where $\alpha, \beta \in R$, be three vectors. If the projection of $\vec{a}$ on $\vec{c}$ is $\frac{10}{3}$ and $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$, then the value of $\alpha+\beta$ is equal to : **[JEE(Main)June2022]**

माना तीन सदिश $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ तथा $\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$ हैं, जहाँ $\alpha, \beta \in R$ हैं। यदि $\vec{a}$ का $\vec{c}$ पर प्रक्षेप $\frac{10}{3}$ है तथा $\vec{b}\times\vec{c}=-6\hat{i}+10\hat{j}+7\hat{k}$ है, तो $\alpha+\beta$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 3 (B) 4 (C) 5 (D) 6

Ans. A

Sol. $\vec{a}=\alpha\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}-\beta\hat{j}+4\hat{k}$ $\alpha, \beta \in R$

$\vec{c}=\hat{i}+2\hat{j}-2\hat{k}$

Projection of $\vec{a}$ on $\vec{c}=\dfrac{\vec{a}.\vec{c}}{|\vec{c}|}=\dfrac{10}{3}$

$\alpha+6+2=10$

$$\vec{b}\times\vec{c}=\begin{vmatrix} i & j & k \\ 3 & -\beta & 4 \\ 1 & 2 & -2 \end{vmatrix}=i(2\beta-8)+j(4+6)+k(6+\beta)=-6i+10j+7k$$

$2\beta-8=-6$

$\beta=1$

$\alpha+\beta=3$

61. Let A, B, C be three points whose position vectors respectively are

$\vec{a}=\hat{i}+4\hat{j}+3\hat{k}$

$\vec{b}=2\hat{i}+\alpha\hat{j}+4\hat{k}$, $\alpha \in R$

$\vec{c}=3\hat{i}-2\hat{j}+5\hat{k}$

If α is the smallest positive integer for which $\vec{a}, \vec{b}, \vec{c}$ are noncollinear, then the length of the median, in ΔABC, through A is : **[JEE(Main)June2022]**

माना तीन बिन्दुओं A, B, C के स्थिति सदिश क्रमशः

$\vec{a}=\hat{i}+4\hat{j}+3\hat{k}$

$\vec{b}=2\hat{i}+\alpha\hat{j}+4\hat{k}$, $\alpha \in R$

$\vec{c}=3\hat{i}-2\hat{j}+5\hat{k}$

है। यदि α न्यूनतम धन पूर्णांक है जिसके लिए $\vec{a}, \vec{b}, \vec{c}$ असंरेख है, तो ΔABC की A से माध्यिका की लंबाई है :

[JEE(Main)June2022]

(A) $\dfrac{\sqrt{82}}{2}$ (B) $\dfrac{\sqrt{62}}{2}$ (C) $\dfrac{\sqrt{69}}{2}$ (D) $\dfrac{\sqrt{66}}{2}$

Ans. A

Sol. Given $\overrightarrow{AB} \| \overrightarrow{AC}$ if $\dfrac{1}{2}=\dfrac{\alpha-4}{-6}=\dfrac{1}{2}$

$\Rightarrow \alpha=1$

$\therefore \vec{a}, \vec{b}, \vec{c}$ are non-collineal for $\alpha = 2$ (smallest positive integel)

Mid point of $BC = M\left(\frac{5}{2}, 0, \frac{9}{2}\right)$

$AM = \sqrt{\frac{9}{4}+16+\frac{9}{4}} = \frac{\sqrt{82}}{2}$

62. Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, where $|\vec{a}|=4, |\vec{b}|=3$ and $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$. Then $\left|\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}+\vec{b}\right)\right|^2 + 4\left(\vec{a}.\vec{b}\right)^2$ is equal to _____. **[JEE(Main)June2022]**

माना सदिशों $\vec{a}$ तथा $\vec{b}$ के बीच कोण θ है, जहाँ $|\vec{a}|=4, |\vec{b}|=3$ तथा $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$ है। तो $\left|\left(\vec{a}-\vec{b}\right)\times\left(\vec{a}+\vec{b}\right)\right|^2 + 4\left(\vec{a}.\vec{b}\right)^2$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 576

Sol. $|\vec{a}|=4, |\vec{b}|=3 \quad \theta \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$

$|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})|^2 + 4(\vec{a}\cdot\vec{b})^2$

$|\vec{a}\times\vec{b}-\vec{b}\times\vec{a}|^2 + 4a^2b^2\cos^2\theta$

$4|\vec{a}\times\vec{b}|^2 + 4a^2b^2\cos^2\theta$

$4a^2b^2\sin^2\theta + 4a^2b^2\cos^2\theta$

$4a^2b^2 = 4\times16\times9 = 576$

63. Let the lines

$L_1\ \vec{r} = \lambda\left(\hat{i}+2\hat{j}+3\hat{k}\right), \lambda \in R$

$L_2\ \vec{r} = \left(\hat{i}+3\hat{j}+\hat{k}\right)+\mu\left(\hat{i}+\hat{j}+5\hat{k}\right); \mu \in R$, intersects at the point S. If a plane $ax + by - z + d = 0$ passes through S and is parallel to both the lines L_1 and L_2, then the value of $a + b + d$ is equal to _____. **[JEE(Main)June2022]**

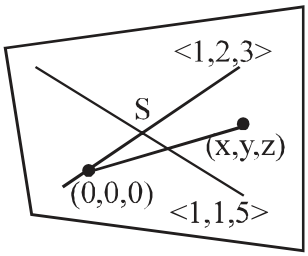
माना रेखाएँ

$L_1\ \vec{r} = \lambda\left(\hat{i}+2\hat{j}+3\hat{k}\right), \lambda \in R$

$L_2\ \vec{r} = \left(\hat{i}+3\hat{j}+\hat{k}\right)+\mu\left(\hat{i}+\hat{j}+5\hat{k}\right); \mu \in R$ बिन्दु S पर काटती हैं। यदि एक समतल $ax + by - z + d = 0$, बिन्दु S से होकर जाता है तथा दोनों रेखाओं L_1 तथा L_2 के समांत है, तो $a + b + d$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 5

Sol. Both the lines lie in the same plane



$\therefore$ equation of the plane

$$\begin{vmatrix} x & y & z \\ 1 & 2 & 3 \\ 1 & 1 & 5 \end{vmatrix} = 0$$

$\Rightarrow 7x - 2y - z = 0$

$\therefore \quad a + b + d = 5$

64. Let $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}, \lambda \in R$. If $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} \cdot \vec{b} + 21 = 0$, then $(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ is equal to _____. **[JEE(Main)June2022]**

माना $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}, \lambda \in R$ है। यदि एक सदिश $\vec{a}$ के लिए $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ तथा $\vec{a} \cdot \vec{b} + 21 = 0$ हैं, तो, $(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 14

Sol. Let $\vec{a} = xi + yj + zk$

$\vec{b} = i + j + \lambda k$

$\vec{a} \cdot \vec{b} = x + y + \lambda z = -21$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ x & y & z \\ 1 & 1 & \lambda \end{vmatrix}$$

$= (\lambda y - z)\, i - (\lambda x - z) j + (x - y) k = 13i - j - 4k$

$$\Rightarrow \left.\begin{aligned} \lambda y - z &= 13 \\ \lambda x - z &= 1 \\ x - y &= -4 \end{aligned}\right\} \Rightarrow \begin{aligned} \lambda &= 3 \\ x &= -2 \\ y &= 2 \\ z &= -7 \end{aligned}$$

$\vec{a} = -2i + 2j - 7k$

$\vec{b} = i + j + 3k$

$\vec{b}-\vec{a}=3i-j+10k$

$\vec{b}+\vec{a}=-i+3j-4k$

$\left(\vec{b}.\vec{a}\right).(k-j)+\left(\vec{b}+\vec{a}\right).(i-k)=(10+1)+(-1+4)$

$=14$

65. If $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}+3\hat{j}+\hat{k}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ are coplanar vectors and $\vec{a}.\vec{c}=5$, $\vec{b}\perp\vec{c}$ then $122(c_1+c_2+c_3)$ is equal to _____. **[JEE(Main)June2022]**

यदि $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$, $\vec{b}=3\hat{i}+3\hat{j}+\hat{k}$ तथा $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$ के सह–तलीय है $\vec{a}.\vec{c}=5$ है तथा $\vec{b}\perp\vec{c}$ के लंबवत है तब $122(c_1+c_2+c_3)$ का मान है _____ **[JEE(Main)June2022]**

Ans. 150

Sol. $\bar{a}\cdot\bar{c}=5\Rightarrow 2c_1+c_2+3c_3=5$ _____(1)

$\bar{b}\cdot\bar{c}=0\Rightarrow 3c_1+3c_2+c_3=0$ _____(2)

And $\left[\bar{a}\ \bar{b}\ \bar{c}\right]=0\Rightarrow\begin{vmatrix}c_1 & c_2 & c_3\\ 2 & 1 & 3\\ 3 & 3 & 1\end{vmatrix}=0$

$\Rightarrow 8c_1-7c_2-3c_3=0$ _____(3)

By solving (1), (2), (3) we get

$c_1=\frac{10}{122}, c_2=\frac{-85}{122}, c_3=\frac{225}{122}$

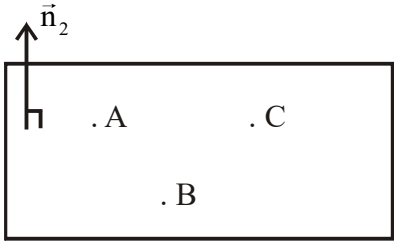
$\therefore\ 122(c_1+c_2+c_3)=150$

66. Let $P_1:\vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4$ be a plane. Let P_2 be another plane which passes through the points (2, –3, 2), (2, –2, –3) and (1, –4, 2). If the direction ratios of the line of intersection of P_1 and P_2 be 16, α, β, then the value of $\alpha+\beta$ is equal to _____. **[JEE(Main)June2022]**

माना एक समतल $P_1:\vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4$ है। माना P_2 एक अन्य समतल है जो बिन्दुओं (2, –3, 2), (2, –2, –3) तथा (1, –4, 2) से होकर जाता है। यदि P_1 तथा P_2 की प्रतिच्छेदन रेखा के दिक् अनुपात 16, α, β हैं, तो $\alpha+\beta$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 28

Sol. $P_1:\vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4\ \Rightarrow\vec{n}_1=2i+j-3k$



P_2 : A (2, –3, 2), B (2, –2, –3), C (1, –4, 2)

$$\Rightarrow \vec{n}_2 = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} i & j & k \\ 0 & 1 & -5 \\ -1 & -1 & 0 \end{vmatrix} = -5i + 5j + k(1)$$

direction vector of line of intersection of planes P_1 and P_2

$$\vec{n}_1 \times \vec{n}_2 \begin{vmatrix} i & j & k \\ 2 & 1 & -3 \\ -5 & 5 & 1 \end{vmatrix} = +16i + 13j + 15k$$

$\text{drs} \equiv 16, \alpha, \beta \equiv 16, 13, 15 \Rightarrow \alpha + \beta = 28$ Ans.

67. Let $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} + (\vec{b} \times \vec{c}) = \vec{0}$ and $\vec{b}.\vec{c} = 5$. Then the value of $3(\vec{c}.\vec{a})$ is equal to _____. (*Data error in question) **[JEE(Main)June2022]**

माना $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ है तथा एक सदिश $\vec{c}$ इस प्रकार है कि $\vec{a} + (\vec{b} \times \vec{c}) = \vec{0}$ तथा $\vec{b}.\vec{c} = 5$ हैं। तो $3(\vec{c}.\vec{a})$ का मान बराबर है _____. (*Data error in question) **[JEE(Main)June2022]**

Ans. 0

Sol. $\vec{a} \times (\vec{b} \times \vec{c}) = 0$

$\vec{c}.\vec{a} + \vec{c}.(\vec{a} \times \vec{c}) = 0$

$\vec{c}.\vec{a} + 0 = 0$

$\therefore 3(\vec{c}.\vec{a}) = 0$

3D Geometry

68. If the shortest distance between the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible values of λ is : **[JEE(Main)June2022]**

यदि रेखाओं $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ तथा $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ के बीच न्यूनतम दूरी $\frac{1}{\sqrt{3}}$ है, तो λ के सभी संभव मानों का योगफल है : **[JEE(Main)June2022]**

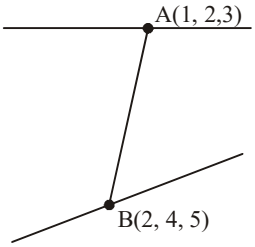
(A) 16 (B) 6 (C) 12 (D) 15

Ans. A

Sol. $\vec{p} = 2i + 3j + \lambda k$

$\vec{q} = i + 4j + 5k$

$\overrightarrow{AB} = i + 2j + 2k$



minimum distance

$$\left|\frac{\overrightarrow{AB}.(\vec{p}\times\vec{q})}{\vec{p}\times\vec{q}}\right| = \frac{1}{\sqrt{3}}$$

$$\frac{\left|\overrightarrow{AB}.(\vec{p}\times\vec{q})\right|}{|\vec{p}\times\vec{q}|} = \frac{1}{\sqrt{3}}$$

$$\overrightarrow{AB}.(\vec{p}\times\vec{q}) = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 3 & \lambda \\ 1 & 4 & 5 \end{vmatrix} = 5 - 2\lambda$$

$$|\vec{p}\times\vec{q}| = |i(15-4\lambda) - j(10-\lambda) + k(5)|$$

$= 17\lambda^2 - 140\lambda + 350$

Now

$$\frac{5-2\lambda}{\sqrt{17\lambda^2 - 140\lambda + 350}} = \frac{1}{\sqrt{3}}$$

$\lambda^2 - 16\lambda + 55 = 0$

Sum of roots = 16

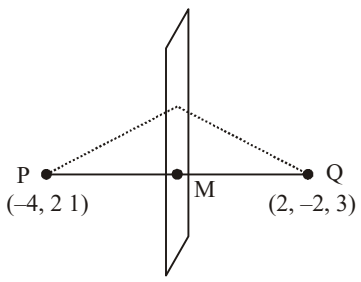
69. Let the points on the plane P be equidistant from the points (–4, 2, 1) and (2, –2, 3). Then the acute angle between the plane P and the plane $2x + y + 3z = 1$ is : **[JEE(Main)June2022]**

माना समतल P के बिन्दु, बिन्दुओं (–4, 2, 1) तथा (2, –2, 3) के बराबर दूरी पर है। तो समतल P तथा समतल $2x + y + 3z = 1$ के बीच न्यूनकोण है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol.



M (–1, 0, 2) is point on plane

Normal $\overrightarrow{PQ} = \vec{n} = 6i - 4j + 2k$

equation of required plane

$\vec{r}.\vec{n} = \vec{a}.\vec{n}$

$\left(x\hat{i} + y\hat{j} + z\hat{k}\right).\left(6\hat{i} - 4\hat{j} + 2\hat{k}\right) = -6 + 4$

$6x - 4y + 2z + 2 = 0$

angle between two plane

$$\cos\theta = \left|\frac{12 - 4 + 6}{\sqrt{36 + 16 + 4} \times \sqrt{4 + 1 + 9}}\right|$$

$$= \left|\frac{14}{\sqrt{56} \times \sqrt{14}}\right|$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

70. Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S : x + y + z = 5. If a line L passing through (1, –1, –1), parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

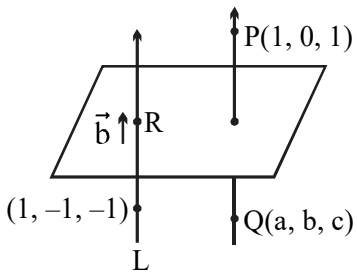
[JEE(Main)June2022]

माना समतल S : x + y + z = 5 के सापेक्ष बिन्दु P(1, 0, 1) का दर्पण प्रतिबिंब Q है। यदि बिन्दु (1, –1, –1) से होकर जाने वाली तथा रेखा PQ के समांतर रेखा L, समतल S को बिन्दु R पर मिलती है, तो QR^2 बराबर है : **[JEE(Main)June2022]**

(A) 2 (B) 5 (C) 7 (D) 11

Ans. B

Sol.



Let parallel vector of $L = \vec{b}$

mirror image of Q on given plane x + y + z = 5

$$\frac{a-1}{1} = \frac{b-0}{1} = \frac{c-1}{1} = \frac{-2(2-5)}{3}$$

$a = 3, b = 2, c = 3$

$Q \equiv (3,2,3)$

$\because \quad \vec{b} \parallel \overrightarrow{PQ}$

so, $\vec{b} = (1,1,1)$

Equation of line $L : \frac{x-1}{1} = \frac{y+1}{1} = \frac{z+1}{1}$

Let point R, $(\lambda+1, \lambda-1, \lambda-1)$

lying on plane $x + y + z = 5$

so, $3\lambda - 1 = 5$

$\Rightarrow \quad \lambda = 2$

Point R is (3, 1, 1)

$QR^2 = 5$

71. Let P be the plane passing through the intersection of the planes $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=5$ and $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=3$ and the point (2, 1, –2). Let the position vectors of the points X and Y be $\hat{i}-2\hat{j}+4\hat{k}$ and $5\hat{i}-\hat{j}+2\hat{k}$ respectively. Then the points : **[JEE(Main)June2022]**

(A) X and X + Y are on the same side of P

(B) Y and Y – X are on the opposite side of P

(C) X and Y are on the opposite side of P

(D) X + Y and X – Y are on the same side of P

माना समतलों $\vec{r}.\left(\hat{i}+3\hat{j}-\hat{k}\right)=5$ तथा $\vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=3$ के प्रतिच्छेदन तथा बिन्दु (2, 1, –2) से होकर जाने वाला समतल P है। माना बिन्दुओं X तथा Y के स्थिति सदिश क्रमशः $\hat{i}-2\hat{j}+4\hat{k}$ तथा $5\hat{i}-\hat{j}+2\hat{k}$ हैं। तो बिन्दु : **[JEE(Main)June2022]**

(A) X तथा X + Y समतल P के एक ही ओर हैं

(B) Y तथा Y – X, समतल P के भिन्न ओर हैं

(C) X तथा Y, समतल P के भिन्न ओर हैं

(D) X + Y तथा X – Y, समतल P के एक ही ओर हैं

Ans. C

Sol. $P_1 : x + 3y - z - 5 = 0$

$P_2 : 2x - y + z - 3 = 0$

$P : P_1 + \lambda P_2 = O$

$(x + 3y - z - 5) + \lambda(2x - y + z - 3) = 0$

Passes through (2, 1, –2)

$\Rightarrow \lambda = 1$

P : 3x + 2y – 8 = O

X= (1, –2, 4)

Y = (5, –1, 2)

Putting value in plane P

X $\Rightarrow$ 3 – 4 – 8 < 0

Y $\Rightarrow$ 15 – 2 – 8 > 0

X and Y are on the opposite side of P.

72. If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}$, z = 2 and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is : **[JEE(Main)June2022]**

यदि दो रेखाएँ $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ तथा $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ लंबवत है, तो रेखा l_2 तथा $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ के बीच एक कोण है : **[JEE(Main)June2022]**

(A) $\cos^{-1}\left(\frac{29}{4}\right)$ (B) $\sec^{-1}\left(\frac{29}{4}\right)$

(C) $\cos^{-1}\left(\frac{2}{29}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Ans. B

Sol. $l_1 : \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-2}{0}$

$l_2 : \frac{x-1}{1} = \frac{y+\frac{3}{2}}{\frac{\alpha}{2}} = \frac{z+5}{2}$

$l_3 : \frac{x-1}{-3} = \frac{y-\frac{1}{2}}{-2} = \frac{z-0}{4}$

$l_1 \perp l_2 \Rightarrow \frac{|3-\alpha+0|}{\sqrt{13}\sqrt{1+\frac{\alpha^2}{4}+4}} = 0 \Rightarrow \alpha = 3$

angle between l_2 & l_3

$$\cos\theta = \frac{\left|1\times(-3)+(-2)\left(\frac{\alpha}{2}\right)+2\times 4\right|}{\sqrt{1+4+\frac{\alpha^2}{4}}\sqrt{9+16+4}}$$

$$\cos\theta = \frac{\left|-3-\alpha+8\right|}{\sqrt{5+\frac{\alpha^2}{4}}\sqrt{29}}$$

put $\alpha = 3$

$$\cos\theta = \frac{2}{\sqrt{\frac{29}{4}}\sqrt{29}} = \frac{4}{29}$$

$$\theta = \cos^{-1}\left(\frac{4}{29}\right) \Rightarrow \theta = \sec^{-1}\left(\frac{29}{4}\right)$$

73. Let the plane $2x + 3y + z + 20 = 0$ be rotated through a right angle about its line of intersection with the plane $x - 3y + 5z = 8$. If the mirror image of the point $\left(2,-\frac{1}{2},2\right)$ in the rotated plane is B(a, b, c), then :

[JEE(Main)June2022]

माना समतल $2x + 3y + z + 20 = 0$ को, इसकी समतल $x - 3y + 5z = 8$ के साथ प्रतिच्छेदन रेखा के सापेक्ष एक समकोण तक घुमाया जाता है। यदि बिंदु $\left(2,-\frac{1}{2},2\right)$ का घुमे हुए समतल में दर्पण प्रतिबिंब B(a, b, c) है, तो :

[JEE(Main)June2022]

(A) $\frac{a}{8}=\frac{b}{5}=\frac{c}{-4}$ (B) $\frac{a}{4}=\frac{b}{5}=\frac{c}{-2}$

(C) $\frac{a}{8}=\frac{b}{-5}=\frac{c}{4}$ (D) $\frac{a}{4}=\frac{b}{5}=\frac{c}{2}$

Ans. A

Sol. Let equation of rotated plane be :

$(2x + 3y + z + 20) + \lambda(x - 3y + 5z - 8) = 0$

$(2 + \lambda 1)x + (3 - 3\lambda)y + (1 + 5\lambda)z + 20 - 8\lambda = 0$

Above plane is perpendicular to $2x + 3y + z + 20 = 0$

So, $(2 +\lambda).2 + (3 - 3\lambda).3 + (1 + 5\lambda).1 = 0 \Rightarrow \lambda = 7$

$\Rightarrow$ Equation of rotated plane : $x - 2y + 4z - 4 = 0$

Mirror image of $A\left(2, \frac{-1}{2}, 2\right)$ in rotated plane is B(a, b, c)

Equation of AB : $\frac{x-2}{1} = \frac{y+\frac{1}{2}}{-2} = \frac{z-2}{4} = k$

Let coordinate of B be $\left(2+k, \frac{-1}{2} - 2k, 2+4k\right)$

midpoint of AB is $\left(2+\frac{k}{2}, \frac{-1}{2} - k, 2+2k\right)$ which willm lie on the plane $x - 2y + 4z - 4 = 0$

Hence $k = \frac{-2}{3}$

Therefore B is $\left(\frac{4}{3}, \frac{5}{6}, \frac{-2}{3}\right) \equiv \left(\frac{8}{6}, \frac{5}{6}, \frac{-4}{6}\right)$

So, $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$

74. If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes throught the point : **[JEE(Main)June2022]**

समतल $2x + y - 5z = 0$ को, इसकी समतल $3x - y + 4z - 7 = 0$ से प्रतिच्छेदन रेखा के सापेक्ष $\frac{\pi}{2}$ कोण तक घुमाने पर प्राप्त समतल किस बिन्दु से होकर जाता है : **[JEE(Main)June2022]**

(A) (2, –2, 0) (B) (–2, 2, 0) (C) (1, 0, 2) (D) (–1, 0, –2)

Ans. C

Sol. $(2x + y - 5z) + \lambda(3x - y + 4z - 7) = 0$

Rotated by $\frac{\pi}{2}$

$(2+3\lambda)x + (1-\lambda)y + (-5+4\lambda)z - 7\lambda = 0$

& $2x + y - 5z = 0$

$\Rightarrow 2(2+3\lambda) + 1 - (1-\lambda) - 5(-5+4\lambda) - 7\lambda = 0$

$30 = 15\lambda$

$\lambda = 2$

Required plane : $8x - y + 32 - 14 = 0$

75. If two straight lines whose direction cosines are given by the relations $l+m-n=0, 3l^2+m^2+cnl=0$ are parallel, then the positive value of c is : **[JEE(Main)June2022]**

यदि दो सरल रेखाएँ, जिनकी दिक्कोज्याएँ $l+m-n=0, 3l^2+m^2+cnl=0$ द्वारा दी गई हैं, समांतर हैं, तो c का धनात्मक मान है: **[JEE(Main)June2022]**

(A) 6 (B) 4 (C) 3 (D) 2

Ans. A

Sol. $l + m - n = 0$

$3l^2 + m^2 + nlc = 0$ ——(1)

$m = n - l$

$m^2 = n^2 + l^2 - 2nl$ ——(2)

From (1)

$3\ell^2 + \left(n^2 + \ell^2 - 2n\ell\right) + n\ell c = 0$

divide by n^2

$3\frac{\ell^2}{n^2} + 1 + \frac{\ell^2}{n^2} - \frac{2\ell}{n} + \frac{\ell c}{n} = 0$

$4\frac{\ell^2}{n^2} + \frac{\ell}{n}(c-2) + 1 = 0$

lines are parallel so

$\frac{\ell_1}{\ell_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$

$\Rightarrow \frac{\ell_1}{n_1} = \frac{\ell_2}{n_2} \Rightarrow$ equal roots

$\Rightarrow D = 0$

$(c - 2)^2 - 4(4)(1) = 0$

$c = 6$ OR -2

$c = 6$

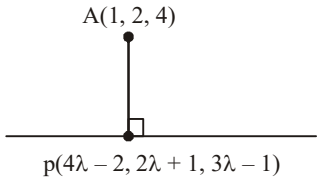
76. Let the foot of the perpendicular from the point (1, 2, 4) on the line $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ be P. Then the distance of P from the plane $3x + 4y + 12z + 23 = 0$ is : **[JEE(Main)June2022]**

माना बिन्दु (1, 2, 4) से रेखा $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ पर लंब का पाद P है। तो P की समतल $3x + 4y + 12z + 23 = 0$ से दूरी है : **[JEE(Main)June2022]**

(A) 5 (B) 50/13 (C) 4 (D) 63/13

Ans. A

Sol.



$\overrightarrow{AP}\cdot\left(4\hat{i}+2\hat{j}+3\hat{k}\right)=0$

$4(4\lambda-3)+2(2\lambda-1)+3(3\lambda-5)=0$

$29\lambda = 29$

$\lambda = 1$

P(2, 3, 2)

$d=\left|\frac{6+12+24+23}{\sqrt{3^2+4^2+12^2}}\right|=\frac{65}{13}=5$

77. The shortest distance between the line $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{-1}$ and $\frac{x+3}{2}=\frac{y-6}{1}=\frac{z-5}{3}$, is :

[JEE(Main)June2022]

रेखाओं $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{-1}$ तथा $\frac{x+3}{2}=\frac{y-6}{1}=\frac{z-5}{3}$ के बीच न्यूनतम दूरी है : **[JEE(Main)June2022]**

(A) $\frac{18}{\sqrt{5}}$ (B) $\frac{22}{3\sqrt{5}}$ (C) $\frac{46}{3\sqrt{5}}$ (D) $6\sqrt{3}$

Ans. A

Sol. $SD=\left|\frac{\left(6\hat{i}-4\hat{j}-4\hat{k}\right).\left(10\hat{i}-8\hat{j}-4\hat{k}\right)}{\sqrt{10^2+8^2+4^2}}\right|$

$\vec{n}_1\times\vec{n}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&-1\\2&1&3\end{vmatrix}$

$=\frac{60+32+16}{2\sqrt{45}}$

$=10\hat{i}-8\hat{j}-4\hat{k}$

$=\frac{54}{3\sqrt{5}}=\frac{18}{\sqrt{5}}$

78. If two distinct points Q, R lie on the line of intersection of the planes –x + 2y – z = 0 and 3x – 5y + 2z = 0 and PQ = PR = $\sqrt{18}$ where the point P is (1, –2, 3), then the area of the triangle PQR is equal to : **[JEE(Main)June2022]**

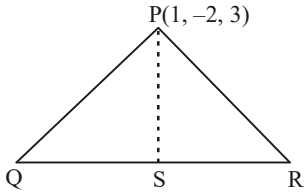
यदि समतलों –x + 2y – z = 0 तथा 3x – 5y + 2z = 0 की प्रतिच्छेदन रेखा पर दो भिन्न बिन्दु Q, R हैं तथा PQ = PR = $\sqrt{18}$ है, जहां P बिन्दु (1, –2, 3) है, तो त्रिभुज PQR का क्षेत्रफल बराबर है : **[JEE(Main)June2022]**

(A) $\frac{2}{3}\sqrt{38}$ (B) $\frac{4}{3}\sqrt{38}$ (C) $\frac{8}{3}\sqrt{38}$ (D) $\sqrt{\frac{152}{3}}$

Ans. B

Sol. Plane are $-x + 2y - z = 0$

$3x - 5y + 2z = 0$



Let R (k, k, k) Now PQ = $\sqrt{18}$

$\Rightarrow (k-2)^2 + (k+2)^2 + (k-3)^2 = 18$

$\Rightarrow 3k^2 - 4k - 4 = 0$

$\Rightarrow k = -\frac{2}{3}, 2$

So $Q\left(-\frac{2}{3}, -\frac{2}{3}, -\frac{2}{3}\right)$ and R(2, 2, 2)

mid point $s \equiv \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$

base of $\Delta PQR = QR = \sqrt{3}.\frac{8}{3}$, height $= \frac{\sqrt{114}}{3} = PS$

area $= \frac{1}{2} \times \sqrt{3} \times \frac{8}{3} \times \frac{\sqrt{114}}{3} = \frac{4}{3}\sqrt{38}$

79. The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points (2, 1, 3) and (0, 1, 2), respectively, is : **[JEE(Main)June2022]**

मान समतलों $5x + 8y + 13z - 29 = 0$, $8x - 7y + z - 20 = 0$ के प्रतिच्छेदन और बिन्दुओं (2, 1, 3) तथा (0, 1, 2) से होकर जाने वाले दो समतल क्रमश: P_1 तथा P_2 हैं। तो समतलों P_1 तथा P_2 के बीच न्यून कोण है : **[JEE(Main)June2022]**

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Ans. A

Sol. The required plane $P_1 + \lambda P_2 = 0$

$\Rightarrow (5 + 8\lambda)x + (8 - 7\lambda)y + (13 + \lambda)z = 29 + 20\lambda$

(2, 1, 3) will satisfy if $\lambda = \frac{7}{2}$

and plane : $2x - y + z = 6$ $(0, 1, 2)$ will satisfy if $\lambda = \frac{1}{5}$

So angle between then and plane : $x + y + 2z = 5$

$$\cos\theta = \left|\frac{2-1+2}{\sqrt{6}\sqrt{6}}\right| = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

80. Let the plane $ax + by + cz = d$ pass through $(2, 3, -5)$ and is perpendicular to the planes $2x + y - 5z = 10$ and $3x + 5y - 7z = 12$. If a, b, c, d are integers $d > 0$ and gcd $(|a|, |b|, |c|, d) = 1$, then the value of $a + 7b + c + 20d$ is equal to : **[JEE(Main)June2022]**

माना समतल $ax + by + cz = d$ बिन्दु $(2, 3, -5)$ से होकर जाता है तथा समतलों $2x + y - 5z = 10$ और $3x + 5y - 7z = 12$ के लंबवत है। यदि a, b, c, d पूर्णांक हैं, $d > 0$ है तथा gcd $(|a|, |b|, |c|, d) = 1$ है, तो $a + 7b + c + 20d$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 18 (B) 20 (C) 24 (D) 22

Ans. D

Sol. Equation of plane passing through $(2, 3, -5)$ and perpendicular to the palnes $2x + y - 5z = 10$ and $3x + 5y - 7z = 12$ is

$$\left(\vec{r} - \vec{a}\right).\vec{N} = 0 \Rightarrow \begin{vmatrix} x-2 & y-3 & z+5 \\ 2 & 1 & -5 \\ 3 & 5 & -7 \end{vmatrix} = 0$$

$-18x + y - 7z = 2$
$a = -18, b = 1, c = -7, d = 2$
$\because$ gcd of $(|a|, |b|, |c|, d) = 1$
$\therefore a + 7b + c + 20d = 22$

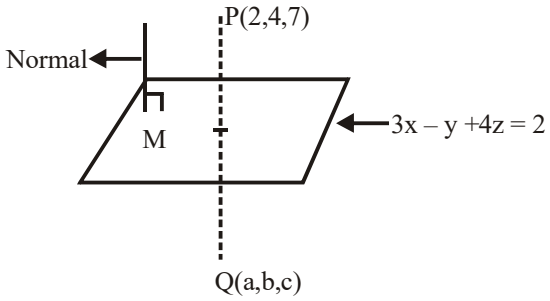
81. If the mirror image of the point $(2, 4, 7)$ in the plane $3x - y + 4z = 2$ is (a, b, c), then $2a + b + 2c$ is equal to: **[JEE(Main)June2022]**

यदि बिन्दु $(2, 4, 7)$ का समतल $3x - y + 4z = 2$ में दर्पण प्रतिबिंब (a, b, c) है, तो $2a + b + 2c$ बराबर है : **[JEE(Main)June2022]**

(A) 54 (B) 50 (C) –6 (D) –42

Ans. C

Sol.



$$M\left(\frac{a+2}{2},\frac{b+4}{2},\frac{c+7}{2}\right)$$

lies on plane

$3a - b + 4c = -26$ .....(1)

drs of PQ $a - 2, b - 4, c - 7$

drs of normal plane $3, -1, 4$

$$\frac{a-2}{3}=\frac{b-4}{-1}=\frac{c-7}{4}=k \text{ (say)}$$

$a = 3k + 2, b = 4 - k, c = 4k + 7$

Put in (1)

$13k = -28$ .....(2)

Now $2a + b + 2c = 13k + 22 = -28 + 22 = -6$ Ans.

82. Let $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$, for some $p, q \in R$. The shortest distance of the plane from the origin is : **[JEE(Main)June2022]**

माना $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}$ किसी $p, q \in R$ के लिए समतल $px - qy + z = 5$ पर स्थित है। मूल बिन्दु से इस समतल की न्यूनतम दूरी है : **[JEE(Main)June2022]**

(A) $\sqrt{\frac{3}{109}}$ (B) $\sqrt{\frac{5}{142}}$ (C) $\frac{5}{\sqrt{71}}$ (D) $\frac{1}{\sqrt{142}}$

Ans. B

Sol. Given equation of line : $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z+3}{-1}=\lambda$

$(3\lambda + 2, -2\lambda - 1, -\lambda - 3)$ lies on plane $px - qy + z = 5$

$p(3\lambda + 2) - q(-2\lambda - 1) + (-\lambda - 3) = 5$

$\lambda(3p + 2q - 1) + (2p + q - 8) = 0$

$\therefore\ 3p + 2q - 1 = 0$ _____(1)

$2p + q - 8 = 0$ _____(2)

Solve (1) and (2) we get
p = 15, q = – 22
equation of plane : 15x + 22y + z – 5 = 0

$\therefore$ Shortest distance from origin = $\left|\frac{0+0+0-5}{\sqrt{15^2+22^2+1}}\right|$

$= \frac{5}{\sqrt{710}} = \sqrt{\frac{5}{142}}$

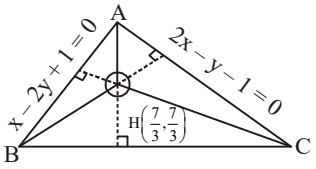
83. The distance of the origin from the centroid of the triangle whose two sides have the equations x – 2y + 1 = 0 and 2x – y – 1 = 0 and whose orthocenter is $\left(\frac{7}{3},\frac{7}{3}\right)$ is : **[JEE(Main)June2022]**

एक त्रिभुज की दो भुजाओं के समीकरण x – 2y + 1 = 0 तथा 2x – y – 1 = 0 इसका लंबकेन्द्र $\left(\frac{7}{3},\frac{7}{3}\right)$ है। इस त्रिभुज के केन्द्र से मूल बिन्दु की दूरी है : **[JEE(Main)June2022]**

(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. C

Sol. Given equation of sides of ΔABC
AB : x – 2y + 1 = 0
AC : 2x – y – 1 = 0 $\Rightarrow$ A(1,1)
Altitude from B is BH = x + 2y – 7 = 0 $\Rightarrow$ B(3, 2)
Altitude from C is CH = 2x + y – 7 = 0 $\Rightarrow$ C(2, 3)
Centroid of ΔABC = C(2, 2)



then, distance between origin and centroid of triangle = $\sqrt{(2-0)^2+(2-0)^2} = 2\sqrt{2}$

84. Let Q be the mirror image of the point P(1, 2, 1) with respect to the plane x + 2y + 2z = 16. Let T be a plane passing through the point Q and contains the line $\vec{r} = -\hat{k} + \lambda\left(\hat{i}+\hat{j}+2\hat{k}\right)$, $\lambda \in R$. Then, which of the following points lies on T? **[JEE(Main)June2022]**

माना समतल x + 2y + 2z = 16. के सापेक्ष बिन्दु P(1, 2, 1) का दर्पण प्रतिबिंब Q है। माना T एक समतल है जो बिन्दु Q से होकर जाता है तथा रेखा $\vec{r} = -\hat{k} + \lambda\left(\hat{i}+\hat{j}+2\hat{k}\right)$, $\lambda \in R$ समतल T में स्थित है। तो निम्न में से कौनसा बिन्दु T पर स्थित है? **[JEE(Main)June2022]**

(A) (2, 1, 0) (B) (1, 2, 1) (C) (1, 2, 2) (D) (1, 3, 2)

Ans. B

Sol. Image of P(1, 2, 1) in plane x + 2y + 2z = 16

$$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-1}{2}=\frac{-2\left[1+2\times2+2\times1-16\right]}{1^2+2^2+2^2}$$

$$\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-1}{2}=2$$

Q(3, 6, 5) is image of point P(1, 2, 1) in the plane x + 2y + 2z = 16.

Given $\vec{r}=-\hat{k}+\lambda\left(\hat{i}+\hat{j}+2\hat{k}\right)$ plane passing through Q.

$\dot{a}(3,8,5)$

$\dot{A}(0,0,-1)$

$\overrightarrow{AQ}=3\hat{i}+6\hat{j}+6\hat{k}=3\left(\hat{i}+2\hat{j}+2\hat{k}\right)$

$\vec{n}=\left(\hat{i}\times2\hat{j}\times2\hat{k}\right)\times\left(\hat{i}+\hat{j}+2\hat{k}\right)$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 2\\ 1 & 1 & 2\end{vmatrix}=2\hat{i}-o\hat{j}-\hat{k}$$

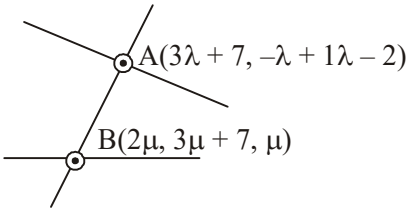
Equation of plane = 2(x – 0)+ O(y – 0) – 1(z + 1) = 0
2x – 2 = 0
So, point lying on the plane is (1, 2, 1)

85. Let a line having direction ratios 1, –4, 2 intersect the lines $\frac{x-7}{3}=\frac{y-1}{-1}=\frac{z+2}{1}$ and $\frac{x}{2}=\frac{y-7}{3}=\frac{z}{1}$ at the points. A and B. Then $(AB)^2$ is equal to _____. **[JEE(Main)June2022]**

माना एक रेखा जिसके दिक अनुपात 1, –4, 2 हैं रेखाओं $\frac{x-7}{3}=\frac{y-1}{-1}=\frac{z+2}{1}$ तथा $\frac{x}{2}=\frac{y-7}{3}=\frac{z}{1}$ को बिन्दुओं A तथा B पर काटती है। तो $(AB)^2$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 84

Sol.



DR's of AB
(3λ – 2μ + 7, –λ –3μ –6,λ – μ – 2)

$$\frac{3\lambda-2\mu+7}{1}=\frac{-\lambda-3\mu-6}{-4}=\frac{\lambda-\mu-2}{2}$$

Taking first (2) –12λ + 8μ – 28 = –λ –3μ –6
λ – μ + 2 =0
Taking second & third –2λ –6μ – 12 = –4λ + 4μ + 8

$\lambda - 5\mu - 10 = 0$

After solving above two equation $\lambda = -5$, $\mu = -3$

A = (–8, 6, – 7)

B = (–6, –2, –3)

$(AB)^2 = 4 + 64 + 16 = 84$

86. If the shortest distance between the lines $\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda\left(\hat{i} - a\hat{j}\right)$ and $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____. **[JEE(Main)June2022]**

यदि रेखाओं $\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda\left(\hat{i} - a\hat{j}\right)$ तथा $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ के बीच न्यूनतम दूरी $\sqrt{\frac{2}{3}}$ है, तो a का पूर्णांकीय मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 2

Sol. $a_1 = (-1, 0, 3)$

$a_2 = (0, -1, 2)$

$b_1 = (1, -a, 0)$ dr's of line (1)

$b_2 = (1, -1, 1)$ dr's of line (2)

$\bar{a}_2 - \bar{a}_1 = (1, -1, -1)$

$$\bar{b}_1 \times \bar{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & k \\ 1 & -a & 0 \\ 1 & -1 & 1 \end{vmatrix}$$

$\bar{b}_1 \times \bar{b}_2 = \hat{i}(-a) - \hat{j} + k(a-1)$

$\left|\bar{b}_1 \times \bar{b}_2\right| = \sqrt{a^2 + 1 + (a-1)^2}$

$a_2 - a_1 . \bar{b}_1 \times \bar{b}_2 = 2 - 2a$

$$\frac{2(1-a)}{\sqrt{a^2 + 1 + (a-1)^2}} = \sqrt{\frac{2}{3}}$$

Squaring an both the side

After solving a = $2, \frac{1}{2}$

87. Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____. **[JEE(Main)June2022]**

माना रेखा l_1 xy-समतल में है तथा इसके x तथा y अंतःखंड $\frac{1}{8}$ और $\frac{1}{4\sqrt{2}}$ है तथा रेखा l_2 zx-समतल में है तथा इसके x तथा z अंतःखंड क्रमशः $-\frac{1}{8}$ और $-\frac{1}{6\sqrt{3}}$ हैं। यदि रेखाओं l_1 तथा l_2 के बीच न्यूनतम दूरी d है, तो d^{-2} बराबर है _____

[JEE(Main)June2022]

Ans. 51

Sol. $l_1 : \frac{x}{\frac{1}{8}} + \frac{y}{\frac{1}{4\sqrt{2}}} = 1$ and $z = 0$

$l_1 : \frac{x-\frac{1}{8}}{\sqrt{2}} = \frac{y-0}{-2} = \frac{z-0}{0}$, $A = \left(\frac{1}{8}, 0, 0\right)$, $\vec{p} = \sqrt{2}i - 2j$

$l_2 : \frac{x}{-\frac{1}{8}} + \frac{z}{\frac{-1}{6\sqrt{3}}} = 1$ and $y = 0$

$l_2 : \frac{x+\frac{1}{8}}{3\sqrt{3}} = \frac{y-0}{0} = \frac{z-0}{4}$, $B = \left(\frac{-1}{8}, 0, 0\right)$, $\vec{q} = 3\sqrt{3}i + 4k$

d = Projection of $\overrightarrow{AB}$ on $\vec{p} \times \vec{q}$

$\overrightarrow{AB} = \frac{-1}{4}i$

$$\vec{p} \times \vec{q} = \begin{vmatrix} i & j & k \\ \sqrt{2} & -2 & 0 \\ 3\sqrt{3} & 0 & 4 \end{vmatrix}$$

$\vec{p} \times \vec{q} == -2\left(4i + 2\sqrt{2}j - 3\sqrt{3}\right)$

$d = \left|\frac{\overrightarrow{AB}.(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right|$

$d = \frac{1}{\sqrt{51}}$

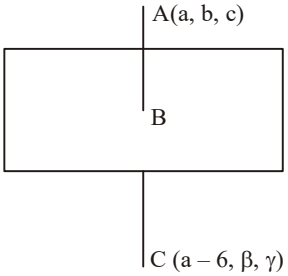
$d^{-2} = 51$

88. Let the mirror image of the point (a, b, c) with respect to the plane $3x - 4y + 12z + 19 = 0$ be $(a - 6, \beta, \gamma)$. If $a + b + c = 5$, then $7\beta - 9\gamma$ is equal to _____. **[JEE(Main)June2022]**

माना बिन्दु (a, b, c) का समतल $3x - 4y + 12z + 19 = 0$ के सापेक्ष दर्पण प्रतिबिंब $(a - 6, \beta, \gamma)$ है। यदि $a + b + c = 5$, है, तो $7\beta - 9\gamma$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 137

Sol. $3x - 4y + 12z + 19 = 0$



Direction ratio of $\overrightarrow{AC} = (-6, \beta - b, \gamma - c)$

D. R. of plane = (3, –4, 12)

So $\dfrac{-6}{3} = \dfrac{\beta - b}{-4} = \dfrac{\gamma - c}{12}$

$\beta - b = 8$ $\qquad$ $\gamma - c = -24$

$\beta = 8 + b$ $\qquad$ $\gamma = c - 24$

Now mid point of AC will lie on given plane

So $B\left(\dfrac{a + a - 6}{2}, \dfrac{b + \beta}{2}, \dfrac{c + \gamma}{2}\right)$

$= \left(a - 3, \dfrac{b + \beta}{2}, \dfrac{c + \gamma}{2}\right)$

$3(a - 3) - 4\left(\dfrac{b + \beta}{2}\right) + 12\left(\dfrac{C + \gamma}{2}\right) + 19 = 0$

$3a - 2b + 6c - 2\beta + 6\gamma + 10 = 0$

$3a - 2b + 6c - 2(8 + b) + 6(c - 24) + 10 = 0$

$3(5 - b - c) - 2b + 6c - 2b + 6c - 150 = 0$

$-7b + 9c - 135 = 0$

$-7(\beta - 8) + 9(\gamma + 24) - 135 = 0$

$7\beta - 9\gamma = 137$

89. Let the image of the point P(1, 2, 3) in the line L : $\dfrac{x - 6}{3} = \dfrac{y - 1}{2} = \dfrac{z - 2}{3}$ be Q. Let R(α, β, γ) be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of $22(\alpha + \beta + \gamma)$ is equal to _____.

[JEE(Main)June2022]

माना रेखा L : $\dfrac{x - 6}{3} = \dfrac{y - 1}{2} = \dfrac{z - 2}{3}$ में बिन्दु P(1, 2, 3) का प्रतिबिंब Q है। माना बिन्दु R(α, β, γ), रेखा खण्ड PQ को अंत 1 : 3 के अनुपात में विभाजित करता है। तो $22(\alpha + \beta + \gamma)$ का मान बराबर है _____.

[JEE(Main)June2022]

Ans. 125

Sol. From point P(1, 2, 3), P′ is foot of perpendicular on line $\frac{x-6}{3}=\frac{y-1}{2}=\frac{z-2}{3}=\lambda$

$\therefore$ P′ = $(3\lambda + 6, 2\lambda + 1, 3\lambda + 2)$

d.r's of PP′ $\equiv 3\lambda + 5, 2\lambda - 1, 3\lambda - 1$

as PP′ and line l is perpendicular

$\therefore (3\lambda + 5)3 + 2(2\lambda - 1) + 3(3\lambda - 1) = 0$

$$\Rightarrow \lambda = \frac{-5}{11}$$

$$P' = \left(\frac{51}{11}, \frac{1}{11}, \frac{7}{11}\right)$$

The point dividing PQ in the ratio 1 : 3 will be mid point of P and foot of perpendicular from P on line $\Rightarrow$ R will be mid point of PP′

$$\Rightarrow R = (\alpha, \beta, \gamma) = \left(\frac{\frac{51}{11}+1}{2}, \frac{\frac{1}{11}+2}{2}, \frac{\frac{7}{11}+3}{2}\right)$$

$$(\alpha, \beta, \gamma) = \left(\frac{62}{22}, \frac{23}{22}, \frac{40}{22}\right)$$

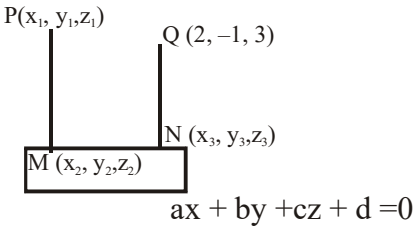
$$22(\alpha, \beta, \gamma) = 22\left(\frac{62+23+40}{22}\right) = 125$$

90. Let d be the distance between the foot of perpendiculars of the points P(1, 2, –1) and Q(2, –1, 3) on the plane $-x + y + z = 1$. Then d^2 is equal to _____. **[JEE(Main)June2022]**

माना बिन्दुओं P(1, 2, –1) तथा Q(2, –1, 3) से समतल $-x + y + z = 1$ पर डाले गए लंबों के पादों के बीच की दूरी d है, तो d^2 बराबर है _____ **[JEE(Main)June2022]**

Ans. 26

Sol.



for $M \equiv \frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{z_2 - z_1}{c} = -\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$

P(1, 2, –1)

$$M \equiv \frac{x_2 - 1}{1} = \frac{y_2 - 2}{-1} = -\frac{z_2 + 1}{-1} = -\frac{1 - 2 + 1 + 1}{3} = \frac{-1}{3}$$

plane x –y – z + 1 = 0

$M\ (x_2\, y_2\, z_2) \equiv \left(\frac{2}{3}, \frac{7}{3}, \frac{-2}{3}\right)$

Q (2, – 1, 3)

$N \to$ foot of $\perp N(x_3, y_3, z_3)$ when $\frac{x_3-2}{1} = \frac{y_3+1}{-1} = \frac{z_3-3}{-1} = -\frac{1}{3}$

$N\left(\frac{5}{3}, \frac{-2}{3}, \frac{10}{3}\right)$

$MN = d = \sqrt{1^2+3^2+4^2} = \sqrt{26}$

$d^2 = 26$

91. Let ABC be a triangle such that $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$, $\overrightarrow{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ and $\vec{b}.\vec{c} = 12$. Consider the statements :

(S1) : $\left|\left(\vec{a}\times\vec{b}\right)+\left(\vec{c}\times\vec{b}\right)\right| - |\vec{c}| = 6\left(2\sqrt{2}-1\right)$

(S2) : $\angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then : **[JEE(Main)July2022]**

(A) Both (S1) and (S2) are true (B) Only (S1) is true

(C) Only (S2) is true (D) Both (S1) and (S2) are false

माना एक त्रिभुज ABC के लिए $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$, $\overrightarrow{AB} = \vec{c}$, $|\vec{a}| = 6\sqrt{2}$, $|\vec{b}| = 2\sqrt{3}$ तथा $\vec{b}.\vec{c} = 12$ हैं :

कथनों :

(S1) : $\left|\left(\vec{a}\times\vec{b}\right)+\left(\vec{c}\times\vec{b}\right)\right| - |\vec{c}| = 6\left(2\sqrt{2}-1\right)$

(S2) : $\angle ACB = \cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

में : **[JEE(Main)July2022]**

(A) (S1) तथा (S2) दोनों सत्य हैं (B) केवल (S1) सत्य हैं

(C) केवल (S2) सत्य है (D) दोनों (S1) तथा(S2) असत्य हैं

Ans. D

Sol. $\vec{a}+\vec{b}+\vec{c} = 0$

$\vec{b}+\vec{c} = -\vec{a}$

$|\vec{b}|^2+|\vec{c}|^2+2\vec{b}\cdot\vec{c}=|\vec{a}|^2$

$|\vec{c}|^2=36$

$|\vec{c}|=6$

$S1:|\vec{a}\times\vec{b}+\vec{c}\times\vec{b}|-|\vec{c}|$

$|(\vec{a}+\vec{c})\times\vec{b}|-|\vec{c}|$

$|-\vec{b}\times\vec{b}|-|\vec{c}|$

$0-6=-6$

$S2:\vec{a}+\vec{b}+\vec{c}=0$

$\vec{b}+\vec{c}=-\vec{a}$

$|\vec{a}|^2+|\vec{b}|^2-2|\vec{a}||\vec{b}|\cos(\angle ACB)=|\vec{c}|^2$

$\cos(\angle ACB)=\sqrt{\frac{2}{3}}$

92. Let $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$ and $\vec{a}\,.\,\vec{b}=3$. Then the projection of $\vec{b}$ on the vector $\vec{a}-\vec{b}$ is : **[JEE(Main)July2022]**

माना $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$ तथा एक सदिश $\vec{b}$ के लिए $\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$ तथा $\vec{a}\,.\,\vec{b}=3$ हैं, तो सदिश $\vec{b}$ का सदिश $\vec{a}-\vec{b}$ पर प्रक्षेप है: **[JEE(Main)July2022]**

(A) $\frac{2}{\sqrt{21}}$ (B) $2\sqrt{\frac{3}{7}}$ (C) $\frac{2}{3}\sqrt{\frac{7}{3}}$ (D) $\frac{2}{3}$

Ans. A

Sol. $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$

$\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$

$\vec{a}.\vec{b}=3$

$\left|\vec{a}\times\vec{b}\right|^2+\left|\vec{a}.\vec{b}\right|^2=\left|\vec{a}\right|^2.\left|\vec{b}\right|^2$

$\Rightarrow 5+9=6\left|\vec{b}\right|^2$

$\Rightarrow |b|^2=\frac{7}{3}$

$$\left|\vec{a}-\vec{b}\right|=\sqrt{\left|\vec{a}\right|^2+\left|\vec{b}\right|^2-2\vec{a}.\vec{b}}=\sqrt{\frac{7}{3}}$$

Projection of $\vec{b}$ on $\vec{a}-\vec{b}=\dfrac{\vec{b}.\left(\vec{a}-\vec{b}\right)}{\left|\vec{a}-\vec{b}\right|}$

$$=\frac{\vec{b}.\vec{a}-|\vec{b}|^2}{\left|\vec{a}-\vec{b}\right|}=\frac{3-\frac{7}{3}}{\sqrt{\frac{7}{3}}}$$

$$=\frac{2}{\sqrt{21}}$$

93. Let $\vec{a}=\alpha\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-\alpha\hat{k}$, $\alpha>0$. If the projection of $\vec{a}\times\vec{b}$ on the vector $-\hat{i}+2\hat{j}-2\hat{k}$ is 30, then α is equal to : **[JEE(Main)July2022]**

माना $\vec{a}=\alpha\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=2\hat{i}+\hat{j}-\alpha\hat{k}$, $\alpha>0$ यदि $\vec{a}\times\vec{b}$ का सदिश $-\hat{i}+2\hat{j}-2\hat{k}$ पर प्रक्षेप 30 है, तो α बराबर है: **[JEE(Main)July2022]**

(A) $\frac{15}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 7

Ans. D

Sol. $\vec{a}\times\vec{b}=(1-\alpha)\hat{i}+\left(\alpha^2-2\right)\hat{j}+(\alpha-2)\hat{k}$

Projection of $\vec{a}\times\vec{b}$ on $-\hat{i}+2\hat{j}-2\hat{k}$

$$=\frac{(\vec{a}\times\vec{b})\cdot(-\hat{i}+2\hat{j}-2\hat{k})}{3}=30$$

$$\Rightarrow 2\alpha^2-\alpha-91=0$$

$$\Rightarrow \alpha=7,-\frac{13}{2}$$

94. A vector $\vec{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i},\hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j},\hat{i}+\hat{k}$. The obtuse angle between $\vec{a}$ and the vector $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ is : **[JEE(Main)July2022]**

सदिशों $\hat{i},\hat{i}+\hat{j}$ द्वारा प्राप्त समतल तथा सदिशों $\hat{i}-\hat{j},\hat{i}+\hat{k}$ द्वारा प्राप्त समतल की प्रतिच्छेदन रेखा के समांतर एक सदिश $\vec{a}$ है। $\vec{a}$ तथा सदिश $\vec{b}=\hat{i}-2\hat{j}+2\hat{k}$ के बीच अधिक कोण है : **[JEE(Main)July2022]**

(A) $\frac{3\pi}{4}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{5}$ (D) $\frac{5\pi}{6}$

Ans. A

Sol. $\vec{n}_1 = \hat{i}\times\left(\hat{i}+\hat{j}\right) = \hat{i}$

$\vec{n}_2 = \left(\hat{i}+\hat{j}\right)\times\left(\hat{i}-\hat{j}\right)$

$= \hat{i}+\hat{j}-\hat{k}$

Line of intersection along $\vec{n}_1\times\vec{n}_2$

$= \hat{k}\times\left(\hat{i}+\hat{j}-\hat{k}\right) = -\hat{i}+\hat{j}$

D.R of $\vec{a} = -\hat{i}+\hat{j}$

D.E of $\vec{b} = \hat{i}-2\hat{j}+2\hat{k}$

$\vec{a}.\vec{b} = -3$ and $\left(\vec{a}\wedge\vec{b}\right) = \theta$

$\cos\theta = \frac{-3}{\sqrt{2}\times 3}$

$\theta = \frac{3\pi}{4}$

95. Let $\vec{a} = \alpha\hat{i}+\hat{j}+\beta\hat{k}$ and $\vec{b} = 3\hat{i}-5\hat{j}+4\hat{k}$ be two vectors, such that $\vec{a}\times\vec{b} = -\hat{i}+9\hat{i}+12\hat{k}$. Then the projection of $\vec{b}-2\vec{a}$ on $\vec{b}+\vec{a}$ is equal to : **[JEE(Main)July2022]**

माना दो सदिशों $\vec{a} = \alpha\hat{i}+\hat{j}+\beta\hat{k}$ तथा $\vec{b} = 3\hat{i}-5\hat{j}+4\hat{k}$ के लिए $\vec{a}\times\vec{b} = -\hat{i}+9\hat{i}+12\hat{k}$ है, तो $\vec{b}-2\vec{a}$ का $\vec{b}+\vec{a}$ पर प्रक्षेप बराबर है : **[JEE(Main)July2022]**

(A) 2 (B) $\frac{39}{5}$ (C) 9 (D) $\frac{46}{5}$

Ans. D

Sol. Let $\vec{a} = a\hat{i}+\hat{j}+\beta\hat{k}, \vec{b} = 3\hat{i}-5\hat{j}+4\hat{k}$

$\vec{a}\times\vec{b} = -\hat{i}+9\hat{j}+12\hat{k}$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 1 & \beta \\ 3 & -5 & 4 \end{vmatrix}$$

$\Rightarrow (4+5\beta)\hat{i}+(3\beta-4\alpha)\hat{j}+(-5\alpha-3)\hat{k}$

$=-\hat{i}+9\hat{j}+12\hat{k}$

$\therefore 4+5\beta=-1, 3\beta-4\alpha=9, -5\alpha-3=12$

$\beta=-1,\ \alpha=-3$

$\therefore \vec{\alpha}=-3\hat{i}+\hat{j}-\hat{k}, \vec{b}=3\hat{i}-5\hat{j}+4\hat{k}$

$\therefore \vec{\alpha}+\vec{b}=-4\hat{j}+3\hat{k}$

$\left|\vec{\alpha}\right|^2=11, \left|\vec{b}\right|^2=50$

$\vec{\alpha}.\vec{b}=-9+(-5)-4=-18$

$\therefore$ Pojectile of $\left(2\vec{b}-2\vec{a}\right)$ on $\vec{a}+\vec{b}$ is

$$\frac{\left(\vec{b}-2\vec{a}\right).\left(\vec{a}+\vec{b}\right)}{\left|\vec{a}+\vec{b}\right|}$$

$$=\frac{\left|\vec{b}\right|^2-2\left|\vec{a}\right|^2-\left(\vec{a}.\vec{b}\right)}{\left|\vec{a}+\vec{b}\right|}=\frac{50-22-(-18)}{5}=\frac{46}{5}$$

Ans. $\left(\frac{46}{5}\right)$

96. Let $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ and $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$. If $\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$, then $\left|\vec{b}\times 2\hat{j}\right|$ is equal to :

[JEE(Main)July2022]

माना $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ तथा $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$ हैं। यदि $\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$ है, तो $\left|\vec{b}\times 2\hat{j}\right|$ बराबर है :

[JEE(Main)July2022]

(A) 4 (B) 5 (C) $\sqrt{21}$ (D) $\sqrt{17}$

Ans. B

Sol. $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}, \vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$

$\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$, then $\left|\vec{b}\times 2\hat{j}\right|$ is

$\left(\left(\vec{a}.\hat{i}\right)\vec{b}-\left(\vec{b}.\hat{i}\right)\vec{a}\right).\hat{k}=\frac{23}{2}$

$$\left(\vec{a}.\hat{i}\right)\left(\vec{b}.\hat{i}\right)-\left(\vec{b}.\hat{i}\right)\left(\vec{a}.\hat{k}\right)=\frac{23}{2}$$

$$2\times 2-\alpha\times 5=\frac{23}{2}\Rightarrow 5\alpha=4-\frac{23}{2}\Rightarrow \alpha=\frac{-3}{2}$$

$$\vec{b}\times 2\hat{j}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \alpha & \beta & 2\\ 0 & 2 & 0\end{vmatrix}=-4\hat{i}+2\alpha\hat{k}$$

$$\therefore\left|\vec{b}\times 2\hat{j}\right|=\sqrt{16+4\alpha^2}=\sqrt{16+4\times\frac{9}{4}}=5$$

97. Let the vectors $\vec{a}=(1+t)\hat{i}+(1-t)\hat{j}+\hat{k}$, $\vec{b}=(1-t)\hat{i}+(1+t)\hat{j}+2\hat{k}$ and $\vec{c}=t\hat{i}-t\hat{j}+\hat{k}$, $t\in R$ be such that for $\alpha, \beta, \gamma \in R$, $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\vec{0}\Rightarrow\alpha=\beta=\gamma=0$. Then, the set of all values of t is :

[JEE(Main)July2022]

(A) a non-empty finite set (B) equal to N

(C) equal to R – {0} (D) equal to R

माना सदिश $\vec{a}=(1+t)\hat{i}+(1-t)\hat{j}+\hat{k}$, $\vec{b}=(1-t)\hat{i}+(1+t)\hat{j}+2\hat{k}$ तथा $\vec{c}=t\hat{i}-t\hat{j}+\hat{k}$, $t\in R$ इस प्रकार है कि $\alpha, \beta, \gamma \in R$, के लिए $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\vec{0}\Rightarrow\alpha=\beta=\gamma=0$ तो t के सभी मानों का समुच्चय है :

[JEE(Main)July2022]

(A) एक अरिक्त परिमित समुच्चय है (B) N के बराबर है

(C) R – {0} के बराबर है (D) R के बराबर है

Ans. C

Sol. By its given condition

$:\vec{a},\vec{b},\vec{c}$ are linearly independent vectors

$$\Rightarrow\left[\vec{a}\,\vec{b}\,\vec{c}\right]\neq 0 \qquad \text{....(i)}$$

Now, $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$$=\begin{vmatrix}1+t & 1-t & 1\\ 1-t & 1+t & 2\\ t & -t & 1\end{vmatrix}$$

$C_2\to C_1+C_2$

$$\begin{vmatrix} 1+t & 2 & 1 \\ 1-t & 2 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2\begin{vmatrix} 1+t & 2 & 1 \\ 1-t & 2 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2\left[(1+t)-(1-t)+t\right]$$

$$= 2\left[3t\right] = 6t$$

$$\left[\bar{a}\ \bar{b}\ \bar{c}\right] \neq 0 \Rightarrow t \neq 0$$

98. Let a vector $\vec{a}$ has magnitude 9. Let a vector $\vec{b}$ be such that for every $(x, y) \in R \times R - \{(0, 0)\}$, the vector $\left(x\vec{a} + y\vec{b}\right)$ is perpendicular to the vector $\left(6y\vec{a} - 18x\vec{b}\right)$. Then the value of $\left|\vec{a} \times \vec{b}\right|$ is equal to :

[JEE(Main)July2022]

माना एक सदिश $\vec{a}$ का परिमाण 9 है। माना एक सदिश $\vec{b}$ इस प्रकार है कि प्रत्येक $(x, y) \in R \times R - \{(0, 0)\}$ के लिए, सदिश $\left(x\vec{a} + y\vec{b}\right)$, सदिश $\left(6y\vec{a} - 18x\vec{b}\right)$ के लंबवत है। तो $\left|\vec{a} \times \vec{b}\right|$ का मान बराबर है : **[JEE(Main)July2022]**

(A) $9\sqrt{3}$ (B) $27\sqrt{3}$ (C) 9 (D) 81

Ans. B

Sol. $|\vec{a}| = 9 \ \& \ \left(x\vec{a} + y\vec{b}\right).\left(6y\vec{a} - 18x\vec{b}\right) = 0$

$$\Rightarrow 6xy|\bar{a}|^2 - 18x^2\left(\bar{a}.\bar{b}\right) + 6y^2\left(\bar{a}.\bar{b}\right) - 18xy|\bar{b}|^2 = 0$$

$$\Rightarrow 6xy\left(|\bar{a}|^2 - 3|\bar{b}|^2\right) + \left(\bar{a}.\bar{b}\right)\left(y^2 - 3x^2\right) = 0$$

This should hold $\forall x, y \in R \times R$

$$\therefore |\bar{a}|^2 = 3|\bar{b}|^2 \ \& \ \left(\bar{a}.\bar{b}\right) = 0$$

Now $\left|\bar{a} \times \bar{b}\right|^2 = |\bar{a}|^2|\bar{b}|^2 - \left(\bar{a}.\bar{b}\right)^2$

$$= \bar{a}.\frac{|\bar{a}|^2}{3}$$

$$\therefore |\bar{a} \times \bar{b}| = \frac{|\bar{a}|^2}{\sqrt{3}} = \frac{81}{\sqrt{3}} = 27\sqrt{3}$$

99. Let S be the set of all $a \in R$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}, (b > 1)$ is acute. Then S is equal to : **[JEE(Main)July2022]**

माना सभी $a \in R$, जिनके लिए सदिशों $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ तथा $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}, (b > 1)$ के बीच का कोण न्यून कोण है, का समुच्चय S है। तो S बराबर है : **[JEE(Main)July2022]**

(A) $\left(-\infty, -\frac{4}{3}\right)$ (B) Φ (C) $\left(-\frac{4}{3}, 0\right)$ (D) $\left(\frac{12}{7}, \infty\right)$

Ans. B

Sol. For angle to be acute

$\vec{u}.\vec{v} > 0$

$\Rightarrow a(\log_e b)^2 - 12 + 6a(\log_e b) > 0$

$\forall b > 1$

let $\log_e b = t \Rightarrow t > 0$ as $b > 1$

$y = at^2 + 6at - 12$ & $y > 0, \forall t > 0$

$\Rightarrow a \in \phi$

100. Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is : **[JEE(Main)July2022]**

माना $\vec{a} = 3\hat{i} + \hat{j}$ तथा $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ है। माना एक सदिश $\vec{c}$ के लिए $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$ है। यदि $\vec{b}$ तथा $\vec{c}$ समांतर नहीं हैं, तो λ का मान है : **[JEE(Main)July2022]**

(A) –5 (B) 5 (C) 1 (D) –1

Ans. A

Sol. $\vec{a} = 3\hat{i} + \hat{j}, \vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

As $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$

$\Rightarrow \vec{a}.\vec{c}(\vec{b}) - (\vec{a}.\vec{b})\vec{c} = \vec{b} + \lambda\vec{c}$

$\Rightarrow \vec{a}.\vec{c} = 1, \vec{a}.\vec{b} = -\lambda$

$\Rightarrow (3\hat{i} + \hat{j}).(\hat{i} + 2\hat{j} + \hat{k}) = -\lambda$

$\Rightarrow \lambda = -5$

101. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $\left(\hat{a}+\hat{b}\right)$ and $\left(\hat{a}+2\hat{b}+2\left(\hat{a}\times\hat{b}\right)\right)$, then the value of $164\cos^2\theta$ is equal to : **[JEE(Main)July2022]**

माना दो इकाई सदिशों $\hat{a}$ तथा $\hat{b}$ के बीच का कोण $\frac{\pi}{4}$ है। यदि सदिशों $\left(\hat{a}+\hat{b}\right)$ तथा $\left(\hat{a}+2\hat{b}+2\left(\hat{a}\times\hat{b}\right)\right)$ के बीच का कोण θ है, तो $164\cos^2\theta$ का मान बराबर है : **[JEE(Main)July2022]**

(A) $90+27\sqrt{2}$ (B) $45+18\sqrt{2}$ (C) $90+3\sqrt{2}$ (D) $54+90\sqrt{2}$

Ans. A

Sol. $\hat{a}\hat{}\hat{b}=\frac{\pi}{4}=\phi$

$\hat{a}\cdot\hat{b}=|\hat{a}||\hat{b}|\cos\phi$

$\hat{a}\cdot\hat{b}=\cos\phi=\frac{1}{\sqrt{2}}$

$\cos\theta=\frac{(\hat{a}+\hat{b})\cdot(\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b}))}{|\hat{a}+\hat{b}||\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b})|}$

$|\hat{a}+\hat{b}|^2=(\hat{a}+\hat{b})\cdot(\hat{a}+\hat{b})$

$|\hat{a}+\hat{b}|^2=2+2\hat{a}\cdot\hat{b}$

$=2+\sqrt{2}$

$\hat{a}\times\hat{b}=|\hat{a}||\hat{b}|\sin\phi\hat{n}$

$\hat{\mathbf{a}}\times\hat{\mathbf{b}}=\frac{\hat{n}}{\sqrt{2}}$ when $\hat{n}$ is vector $\perp\hat{a}$ and $\hat{b}$

let $\vec{c}=\hat{a}\times\hat{b}$

We know.

$\vec{c}\cdot\vec{a}=0$

$\vec{c}\cdot\vec{b}=0$

$|\hat{a}+2\hat{b}+2\vec{c}|^2$

$=1+4+\frac{(4)}{2}+4\hat{a}\cdot\hat{b}+8\hat{b}\cdot\vec{c}+4\vec{c}\cdot\hat{a}$

$=7+\frac{4}{\sqrt{2}}=7+2\sqrt{2}$

Now

$(\hat{a}+\hat{b})\cdot(\hat{a}+2\hat{b}+2\vec{c})$

$=|\hat{a}|^2+2\hat{a}\cdot\hat{b}+0+\hat{b}\cdot\hat{a}+2|\hat{b}|^2+0$

$=1+\frac{2}{\sqrt{2}}+\frac{1}{\sqrt{2}}+2$

$=3+\frac{3}{\sqrt{2}}$

$\cos\theta=\frac{3+\frac{3}{\sqrt{2}}}{\sqrt{2+\sqrt{2}}\sqrt{7+2\sqrt{2}}}$

$\cos^2\theta=\frac{9(\sqrt{2}+1)^2}{2(2+\sqrt{2})(7+2\sqrt{2})}$

$\cos^2\theta=\left(\frac{9}{2\sqrt{2}}\right)\frac{(\sqrt{2}+1)}{(7+2\sqrt{2})}$

$164\cos^2\theta=\frac{(82)(9)}{\sqrt{2}}\frac{(\sqrt{2}+1)}{(7+2\sqrt{2})}\frac{(7-2\sqrt{2})}{(7-2\sqrt{2})}$

$=\frac{(82)}{\sqrt{2}}\frac{(9)[7\sqrt{2}-4+7-2\sqrt{2}]}{(41)}$

$=(9\sqrt{2})[5\sqrt{2}+3]$

$=90+27\sqrt{2}$

102. If (2, 3, 9), (5, 2, 1), (1, λ, 8) and (λ, 2, 3) are coplanar, then the product of all possible values of λ is :

[JEE(Main)July2022]

यदि बिन्दु (2, 3, 9), (5, 2, 1), (1, λ, 8) तथा (λ, 2, 3) सह–तलीय है, तो λ के सभी संभव मानों का गुणनफल है :

[JEE(Main)July2022]

(A) $\frac{21}{2}$ (B) $\frac{59}{8}$ (C) $\frac{57}{8}$ (D) $\frac{95}{8}$

Ans. D

Sol. $A(2,3,9);B(5,2,1);C(1,\lambda,8);D(\lambda,2,3)$

$\left[\overrightarrow{AB}\quad\overrightarrow{AC}\quad\overrightarrow{AD}\right]=0$

$$\begin{vmatrix} 3 & -1 & -8 \\ -1 & \lambda-3 & -1 \\ \lambda-2 & -1 & -6 \end{vmatrix} = 0$$

$$\Rightarrow [-6(\lambda-3)-1]-8(1-(\lambda-3)(\lambda-2))+(6+(\lambda-2)=0$$

$$3(-6\lambda+17)-8\left(-\lambda^2+5\lambda-5\right)+(\lambda+4)=8$$

$$8\lambda^2-57\lambda+95=0$$

$$\lambda_1\lambda_2=\frac{95}{8}$$

103. Let $\vec{a},\vec{b},\vec{c}$ be three coplanar concurrent vectors such that angles between any two of them is same. If the product of their magnitudes is 14 and $\left(\vec{a}\times\vec{b}\right).\left(\vec{b}\times\vec{c}\right)+\left(\vec{b}\times\vec{c}\right).\left(\vec{c}\times\vec{a}\right)+\left(\vec{c}\times\vec{a}\right).\left(\vec{a}\times\vec{b}\right)=168$, then $|\vec{a}|+|\vec{b}|+|\vec{c}|$ is equal to: **[JEE(Main)July2022]**

माना तीन सहतलीय संगामी सदिश $\vec{a},\vec{b},\vec{c}$ इस प्रकार हैं कि किन्हीं भी दो भी बीच के कोण बराबर हैं। यदि इन सदिशों के परिमाणों का गुणनफल 14 है तथा $\left(\vec{a}\times\vec{b}\right).\left(\vec{b}\times\vec{c}\right)+\left(\vec{b}\times\vec{c}\right).\left(\vec{c}\times\vec{a}\right)+\left(\vec{c}\times\vec{a}\right).\left(\vec{a}\times\vec{b}\right)=168$ है, तो $|\vec{a}|+|\vec{b}|+|\vec{c}|$ बराबर है : **[JEE(Main)July2022]**

(A) 10 (B) 14 (C) 16 (D) 18

Ans. C

Sol. $\vec{a}\,\|\,\vec{b}\,\|\,\vec{c}\,|=14$

$$\vec{a}\hat{}\,\vec{b}=\vec{b}\wedge\vec{c}=\vec{c}\hat{}\,\vec{a}=\theta=\frac{2\pi}{3}$$

So, $\vec{a}\cdot\vec{b}=-\frac{1}{2}ab,\vec{b}\cdot\vec{c}=-\frac{1}{2}bc,\vec{a}\cdot\vec{c}.=-\frac{1}{2}ac$

(let)

$$(\vec{a}\times\vec{b})\cdot(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{c})-(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{b})$$

$$=\frac{1}{4}ab^2c+\frac{1}{2}ab^2c=\frac{3}{4}ab^2c$$

Similarly

$$(\vec{b}\times\vec{c})\cdot(\vec{c}\times\vec{a})=\frac{3}{4}abc^2$$

$$(\vec{c}\times\vec{a})\cdot(\vec{a}\times\vec{b})=\frac{3}{4}a^2bc$$

$$168=\frac{3}{4}abc(a+b+c)$$

So, (a + b + c) = 16

104. The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$ and $\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$ from the point (2, 1, 4) is $\sqrt{3}$, is _____. **[JEE(Main)July2022]**

a का अधिकतम मान, जिसके लिए रेखाओं $\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$ तथा $\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$ को अंतर्विष्ट करने वाले समतल की बिन्दु (2, 1, 4) से लंबवत् दूरी $\sqrt{3}$ है, है_____ **[JEE(Main)July2022]**

Ans. 2

Sol. $\vec{r}=\left(\hat{i}+\hat{j}\right)+\lambda\left(\hat{i}+a\hat{j}-\hat{k}\right)$

$\vec{r}=\left(\hat{i}+\hat{j}\right)+\mu\left(-\hat{i}+\hat{j}-a\hat{k}\right)$

D.R's of plane containing these lines is

$$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & a & -1\\ -1 & 1 & -a\end{vmatrix}=\hat{i}\left(1-a^2\right)-\hat{j}(-a-1)+\hat{k}(1+a)$$

$\vec{n}=(1-a)\hat{i}+\hat{j}+\hat{k}$

One point in plane : (1, 1, 0)

$\therefore$ equation of plane is

$(1-a)(x-1)+(y-1)+(z-0)=0$

$(1-a)x+y+z+a-2=0$

$$\therefore D=\frac{|(1-a)2+1+4+a-2|}{\sqrt{(1-a)^2+1+1}}$$

$$\Rightarrow|5-a|=\sqrt{3}.\sqrt{a^2-2a+3}$$

$\Rightarrow a^2+2a-8=0$

$\Rightarrow a=2,-4$

$\therefore$ largest value of a = 2

105. Let $\vec{a},\vec{b},\vec{c}$ be three non-coplanar vectors such that $\vec{a}\times\vec{b}=4\vec{c},\ \vec{b}\times\vec{c}=9\vec{a}$ and $\vec{c}\times\vec{a}=\alpha\vec{b}$, $\alpha>0$. If $|\vec{a}|+|\vec{b}|+|\vec{c}|=36$, then α is equal to _____. **[JEE(Main)July2022]**

माना तीन सदिश $\vec{a},\vec{b},\vec{c}$ सहतलीय नहीं है तथा $\vec{a}\times\vec{b}=4\vec{c},\ \vec{b}\times\vec{c}=9\vec{a}$, $\vec{c}\times\vec{a}=\alpha\vec{b}$, $\alpha>0$ हैं। यदि $|\vec{a}|+|\vec{b}|+|\vec{c}|=36$ है, तो α बराबर है _____ **[JEE(Main)July2022]**

Ans. 36

Sol. $\vec{a}+\vec{b}=4\vec{c}\Rightarrow \vec{a}.\vec{c}=0=\vec{b}.\vec{c}$

$\vec{b}\times\vec{c}=9\vec{a}\Rightarrow \vec{a}.\vec{b}=0=\vec{a}.\vec{c}$

$\therefore \vec{a},\vec{b},\vec{c}$ are mutually $\perp$ set of vectors.

$\Rightarrow |\vec{a}||\vec{b}|=4|\vec{c}|, |\vec{b}||\vec{c}|=9|\vec{a}| \ \& \ |\vec{c}||\vec{a}|=\alpha|\vec{b}|$

$$\Rightarrow \frac{|\vec{a}|}{|\vec{c}|}=\frac{4}{9}\frac{|\vec{c}|}{|\vec{a}|}$$

$$\Rightarrow \frac{|\vec{c}|}{|\vec{a}|}=\frac{3}{2}$$

$$\therefore \text{If } |a|=\lambda, |c|=\frac{3\lambda}{2} \ \& \ |b|=6$$

Now $|a|+|b|+|c|=\frac{1}{36}$

$$\Rightarrow \frac{5}{2}\lambda+6=\frac{1}{36}, \lambda=\frac{-43}{18}=|a|$$

which gives negative value of λ or $|a|$ which is NOT possible & hence data seems to be wrong.

But if $|\vec{a}|+|\vec{b}|+|\vec{c}|=36$

$$\frac{5}{2}\lambda+6=36$$

$$\lambda=12$$

$$\alpha=\frac{|\vec{c}||\vec{a}|}{|\vec{b}|}=\frac{3\times12}{2}\times\frac{12}{6}$$

$$\alpha=36$$

106. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{a}+\vec{b}\right|^2=|\vec{a}|^2+2|\vec{b}|^2, \vec{a}.\vec{b}=3$ and $\left|\vec{a}\times\vec{b}\right|^2=75$. Then $|\vec{a}|^2$ is equal to _____. **[JEE(Main)July2022]**

माना सदिशों $\vec{a}$ तथा $\vec{b}$ के लिए $\left|\vec{a}+\vec{b}\right|^2=|\vec{a}|^2+2|\vec{b}|^2, \vec{a}.\vec{b}=3$ तथा $\left|\vec{a}\times\vec{b}\right|^2=75$ हैं, तो $|\vec{a}|^2$ बराबर है ____

[JEE(Main)July2022]

Ans. 14

Sol. $|\vec{a}+\vec{b}|^2=|\vec{a}|^2+2|\vec{b}|^2; \vec{a}\cdot\vec{b}=3$

As $|\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b}=|\vec{a}|^2+2|\vec{b}|^2$

$|\vec{b}|^2=2\vec{a}\cdot\vec{b}=6$

$|\vec{a}\times\vec{b}|^2=75$

$|\vec{a}|^2|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2=75$

$6|\vec{a}|^2-9=75\Rightarrow|\vec{a}|^2=14$

3D Geometry

107. Let P be the plane containing the straight line $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$. If d is the distance of P from the point(2, –5, 11), then d^2 is equal to :

[JEE(Main)July2022]

माना रेखा $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$, समतल P में स्थित है तथा समतल P, रेखाओं $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ तथा $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$ को अंतर्विष्ट करने वाले समतल के लंबवत है। यदि P की बिन्दु (2, –5, 11) से दूरी d है, तो d^2 बराबर है : **[JEE(Main)July2022]**

(A) $\frac{147}{2}$ (B) 96 (C) $\frac{32}{3}$ (D) 54

Ans. C

Sol. Let <a, b, c> be direction ratios of plane containing

lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$.

$\therefore$ 2a + 3b + 5c = 0 .....(i)

and 3a + 7b + 8c = 0 .....(ii)

from eq. (i) and (ii) : $\frac{a}{24-35}=\frac{b}{15-16}=\frac{c}{14-9}$

$\therefore$ D.R^S. of plane are < 11, 1, –5>

Let D.R^S of plane P be $<a_1, b_1, c_1>$ then.

$11a_1+b_1-5c_1=0$ .....(iii)

and $9a_1-b_1-5c_1=0$ .....(iv)

From eq. (iii) and (iv) :

$\frac{a_1}{-5-5}=\frac{b_1}{-45-55}=\frac{c_1}{-11-9}$

$\therefore$ D.A^5. of plane P are < 1, –1, 2>

Equation plane P is : $1(x-3)-1(y+4)+2(z-7)=0$

$\Rightarrow x-y+2z-21=0$

Distance from point (2, –5, 11) is $d=\frac{|2+5+22-2|}{\sqrt{6}}$

$\therefore \quad d^2=\frac{32}{3}$

108. A plane E is perpendicular to the two planes $2x-2y+z=0$ and $x-y+2z=4$, and passes through the point P(1, –1, 1). If the distance of the plane E from the point Q(a, a, 2) is $3\sqrt{2}$, then $(PQ)^2$ is equal to :

[JEE(Main)July2022]

एक समतल E दो समतलों $2x-2y+z=0$ तथा $x-y+2z=4$ के लंबवत है, तथा बिन्दु P(1, –1, 1) से होकर जाता है। समतल E की बिन्दु Q(a, a, 2) से दूरी $3\sqrt{2}$ है, तो $(PQ)^2$ बराबर है : **[JEE(Main)July2022]**

(A) 9 (B) 12 (C) 21 (D) 33

Ans. C

Sol. First plane, $P_1 = 2x-2y+z=0$, normal vector $\equiv \bar{n}_1=\langle 2,-2,1\rangle$

Second plane, $P_2 \equiv x-y+2z=4$, normal vector $\equiv \bar{n}_2=(1,-1,2)$

Plane perpendicular to P_1 and P_2 will have normal vector $\bar{n}_3$

Where $\bar{n}_3=(\bar{n}_1\times\bar{n}_2)$

Hence, $\bar{n}_3=(-3,-3,0)$

Equation of plane E through P(1, –1, 1) and $\bar{n}_3$ as normal vector

$(x-1,y+1,z-1)\cdot(-3,-3,0)=0$

$\Rightarrow x+y=0\equiv E$

Distance of PQ(a, a, 2) from $E=\left|\frac{2a}{\sqrt{2}}\right|$

as given, $\left|\frac{2a}{\sqrt{2}}\right|=3\sqrt{2}\Rightarrow a=\pm 3$

Hence, $Q\equiv(\pm 3,\pm 3,2)$

Distance $7Q=\sqrt{21}\Rightarrow (PQ)^2=21$

109. The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is :

[JEE(Main)July2022]

रेखाओं $\frac{x+7}{-6}=\frac{y-6}{7}=z$ तथा $\frac{7-x}{2}=y-2=z-6$ के बीच न्यूनतम दूरी है : **[JEE(Main)July2022]**

(A) $2\sqrt{29}$ (B) 1 (C) $\sqrt{\frac{37}{29}}$ (D) $\frac{\sqrt{29}}{2}$

Ans. A

Sol. $L_1 : \frac{x+7}{-6}=\frac{y-6}{7}=\frac{z-0}{1}$

Any point on it $\vec{a}_1$ (–7, 6, 0)

and L_1 is parallel to $\vec{b}_1$ (–6, 7, 1)

$L_2 : \frac{x-7}{-2}=\frac{y-2}{1}=\frac{z-6}{1}$

Any point on it, $\vec{a}_2$ (7, 2, 6)

and L_2 is parallel to $\vec{b}_2$ (–2, 1, 1)

Shortest distance between L_1 and L_2

$$=\left|\frac{(\vec{a}_1-\vec{a}_2)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|=\left|\frac{(-14,4,-6)\cdot(3,2,4)}{\sqrt{9+4+16}}\right|$$

$$=2\sqrt{29}$$

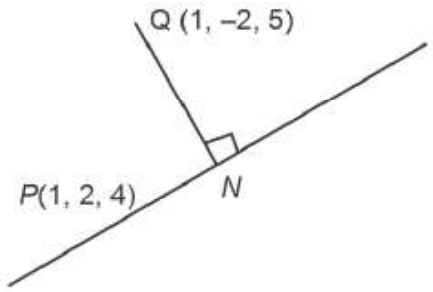
110. The length of the perpendicular from the point (1, –2, 5) on the line passing through (1, 2, 4) and parallel to the line $x+y-z=0=x-2y+3z-5$ is : **[JEE(Main)July2022]**

बिन्दु (1, 2, 4) से होकर जाने वाली तथा रेखा $x+y-z=0=x-2y+3z-5$ के समांतर रेखा की, बिन्दु (1, – 2, 5) से लंब की लंबाई है : **[JEE(Main)July2022]**

(A) $\sqrt{\frac{21}{2}}$ (B) $\sqrt{\frac{9}{2}}$ (C) $\sqrt{\frac{73}{2}}$ (D) 1

Ans. A

Sol.



The line $x + y - z = 0 = x - 2y + 3z - 5$ is parallel to the vector

$$\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -2 & 3 \end{vmatrix} = (1, 4, -3)$$

Equation of line though P(1, 2, 3) and parallel to $\vec{b}$

$$\frac{x-1}{1} = \frac{y-2}{-4} = \frac{z-4}{-3}$$

Let $N \equiv (\lambda + 1, -4\lambda + 2, -3\lambda + 4)$

$\overrightarrow{QN} = (\lambda, -4\lambda + 4, -3\lambda - 1)$

$\overrightarrow{QN}$ is perpendicular to $\vec{b}$

$\Rightarrow (\lambda, -4\lambda + 4, -3\lambda - 1).(1, 4, -3) = 0$

$\Rightarrow \lambda = \frac{1}{2}$

Hence $\overrightarrow{QN} = \left(\frac{1}{2}, 2, \frac{-5}{2}\right)$ and $\left|\overrightarrow{QN}\right| = \sqrt{\frac{21}{2}}$

111. If the plane P passes through the intersection of two mutually perpendicular planes $2x + ky - 5z = 1$ and $3kx - ky + z = 5$, $k < 3$ and intercepts a unit length on positive x-axis, then the intercept made by the plane P on the y-axis is : **[JEE(Main)July2022]**

यदि दो परस्पर लंब समतलों $2x + ky - 5z = 1$ तथा $3kx - ky + z = 5$, $k < 3$ के प्रतिच्छेदन से होकर जाने वाले समतल P धनात्मक x-अक्ष पर अंतःखण्ड लंबाई एक है, तो P का y-अक्ष पर अंतः खण्ड है : **[JEE(Main)July2022]**

(A) $\frac{1}{11}$ (B) $\frac{5}{11}$ (C) 6 (D) 7

Ans. D

Sol. Two given planes mutually perpendicular

$2(3k) + k(-k) + (-5)\,1 = 0$

$k = 1, 5$

but $k < 3$ So $k = 1$

Plane passing through these planes is

$2x + y - 5z - 1 + \lambda(3x - y + z - 5) = 0$

$$\frac{x}{\frac{5\lambda+1}{2+3\lambda}}+\frac{y}{\frac{5\lambda+1}{1-\lambda}}+\frac{z}{\frac{5\lambda+1}{\lambda-5}}=1$$

Given $\frac{5\lambda+1}{2+3\lambda}=1\Rightarrow\lambda=\frac{1}{2}$

So intecept on y-axis is $=\frac{5\lambda+1}{1-\lambda}=7$

112. If the length of the perpendicular drawn from the point P(a, 4, 2), a > 0 on the line $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ is $2\sqrt{6}$ units and Q(α_1, α_2, α_3) is the image of the point P in this line, then $a+\sum_{i=1}^{3}\alpha_1$ is equal to :

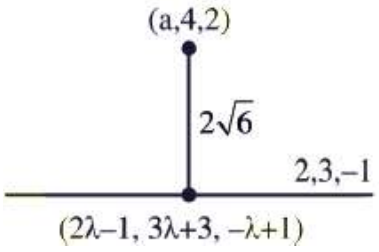
[JEE(Main)July2022]

यदि बिन्दु P(a, 4, 2), a > 0, से रेखा $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ पर डाले गए लंब की लंबाई $2\sqrt{6}$ इकाई है तथा इस रेखा में बिन्दु P का प्रतिबिम्ब Q(α_1, α_2, α_3) है, तो $a+\sum_{i=1}^{3}\alpha_1$ बराबर है : **[JEE(Main)July2022]**

(A) 7 (B) 8 (C) 12 (D) 14

Ans. B

Sol.



$$(2\lambda-1-a)2+(3\lambda-1)3+(-\lambda-1)(-1)=0$$

$$\Rightarrow 4\lambda-2-2a+9\lambda-3+\lambda+1=0$$

$$\Rightarrow 14\lambda-4-2a=0$$

$$\Rightarrow 7\lambda-2-a=0$$

and,

$$(2\lambda-1-a)^2+(3\lambda-1)^2+(\lambda+1)^2=24$$

$$\Rightarrow (5\lambda-1)^2+(3\lambda-1)^2+(\lambda+1)^2=24$$

$$\Rightarrow 35\lambda^2-14\lambda-21=0$$

$$\Rightarrow (\lambda-1)(35\lambda+21)=0$$

For, $\lambda = 1$ $\Rightarrow a = 5$

Let $(\alpha_1, \alpha_2, \alpha_3)$ be reflection of point P

$\alpha_1 + 5 = 2$ $\alpha_2 + 4 = 12$ $\alpha_3 + 2 = 0$

$\alpha_1 = -3$ $\alpha_2 = 8$ $\alpha_3 = -2$

$a + \alpha_1 + \alpha_2 + \alpha_3 = 8$

113. If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0$, $a > 0$ makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are : **[JEE(Main)July2022]**

यदि समतलों $ax + by = 3$ तथा $ax + by + cz = 0$, $a > 0$ की प्रतिच्छेदन रेखा L, समतल $y - z + 2 = 0$ से 30º का कोण बनाती है, तो रेखा L की दिक् कोज्या हैं : **[JEE(Main)July2022]**

(A) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ (B) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$ (C) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$ (D) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Ans. B

Sol. $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & 0 \\ a & b & c \end{vmatrix}$

$= bc\hat{i} - ac\hat{j}$

Direction ratios of line are **b, –a, 0**

Direction ratios of normal of the plane are **0, 1, – 1**

$$\cos 60^\circ = \left| \frac{-a}{\left|\sqrt{2}\right| \sqrt{b^2 + a^2}} \right| = \frac{1}{2}$$

$$\Rightarrow \left| \frac{a}{\sqrt{a^2 + b^2}} \right| = \frac{1}{\sqrt{2}}$$

$\Rightarrow b = \pm a$

So, D.R.'s can be $(\pm a, -a, 0)$

$\therefore$ D.C.'s can be $\pm\left(\frac{\pm 1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right)$

114. The foot of the perpendicular from a point on the circle $x^2 + y^2 = 1, z = 0$ to the plane $2x + 3y + z = 6$ lies on which one of the following curves? **[JEE(Main)July2022]**

वृत्त $x^2 + y^2 = 1, z = 0$ के एक बिन्दु से समतल $2x + 3y + z = 6$ पर डाले गए लंब का पाद निम्न में से किस वक्र पर है ? **[JEE(Main)July2022]**

(A) $(6x + 5y - 12)^2 + 4(3x + 7y - 8)^2 = 1, z = 6 - 2x - 3y$

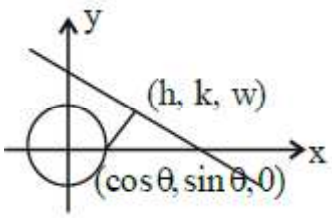
(B) $(5x + 6y - 12)^2 + 4(3x + 5y - 9)^2 = 1, z = 6 - 2x - 3y$

(C) $(6x + 5y - 14)^2 + 9(3x + 5y - 7)^2 = 1, z = 6 - 2x - 3y$

(D) $(5x + 6y - 14)^2 + 9(3x + 7y - 8)^2 = 1, z = 6 - 2x - 3y$

Ans. B

Sol.



$$\frac{h-\cos\theta}{2}=\frac{k-\sin\theta}{3}=\frac{w-0}{1}$$

$$=\frac{-1\left(2\cos\theta+3\sin\theta-6\right)}{14}$$

$$h=\cos\frac{-2\left(2\cos\theta+3\sin\theta-6\right)}{14}$$

$$=\frac{10\cos\theta-6\sin\theta+12}{14}$$

$$k=\sin\theta-\frac{3}{14}\left(2\cos\theta+3\sin\theta-6\right)$$

$$k=\frac{5\sin\theta-6\cos\theta+18}{14}$$

Elementaary $\sin\theta$ and $\cos\theta$

$$\left(5h+6k-12\right)^2+4\left(3h+5k-9\right)^2=1$$

115. Let the line $\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ be coplanar and P be the plane containing these two lines. Then which of the following points does NOT lie no P? **[JEE(Main)July2022]**

माना रेखाएँ $\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ तथा $\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ सह–तलीय हैं तथा P वह समतल है जिस में यह दोनों रेखाएँ स्थित हैं। तो निम्न में से कौनसा बिन्दु, समतल P पर नहीं है ? **[JEE(Main)July2022]**

(A) (0, –2, –2) (B) (–5, 0, –1) (C) (3, –1, 0) (D) (0, 4, 5)

Ans. D

Sol. Given, $L_1 : \frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$

and $L_2 : \frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$

are coplanar

$$\Rightarrow \begin{vmatrix} 27 & 20 & 31 \\ \lambda & 1 & 2 \\ -2 & 3 & \lambda \end{vmatrix} = 0$$

$\Rightarrow \lambda = 3$

Now, normal of plane P, which contains L_1 and L_2

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ -2 & 3 & 3 \end{vmatrix}$$

$= -3\hat{i} - 13\hat{j} + 11\hat{k}$

$\Rightarrow$ Equation of required plane P :

$3x + 13y - 11z + 4 = 0$

(0, 4, 5) does not lie on plane P.

116. A plane P is parallel to two lines whose direction ratios are –2, 1, –3 and –1, 2, –2 and it contains the point(2, 2, –2). Let P intersect the co-ordinate axes at the points A, B, C making the intercepts α, β, γ. If V is the volume of the tetrahedron OABC, where O is the origin, and $p = \alpha + \beta + \gamma$, then the ordered pair (V, p) is equal to : **[JEE(Main)July2022]**

एक समतल P, दो रेखाओं, जिनके दिक् अनुपात –2, 1, –3 तथा –1, 2, –2 हैं, के समान्तर है तथा बिन्दु (2, 2, –2) समतल P पर है। माना P निर्देशांक अक्षों को बिन्दुओं A, B, C पर काटता है तथा अंतःखण्ड α, β, γ बनाता है। यदि चतुष्फलक OABC का आयतन V है, जहाँ O मूल बिन्दु है, तथा $p = \alpha + \beta + \gamma$ है, तो क्रमित युग्म (V, p) बराबर है : **[JEE(Main)July2022]**

(A) (48, –13) (B) (24, –13) (C) (48, 11) (D) (24, –5)

Ans. B

Sol. Normal of plane P :

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & -3 \\ -1 & 2 & -2 \end{vmatrix} = 4\hat{i} - \hat{j} - 3\hat{k}$$

Equation of plane P which passes through (2, 2, –2) is $4x - 7 - 3z - 12 = 0$

Now, A (3, 0, 0), B (0, – 12, 0), C (0, 0, – 4)

$\Rightarrow \alpha = 3, \beta = -12, \gamma = -4$

$\Rightarrow p = \alpha + \beta + \gamma = -13$

Now, volume of tetrahedron OABC

$$V = \left| \frac{1}{6} \overrightarrow{OA}.\left(\overrightarrow{OB} \times \overrightarrow{OC} \right) \right| = 24$$

(V, p) = (24, –13)

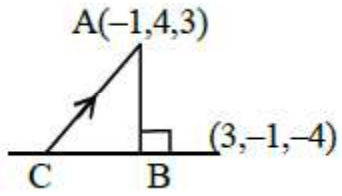
117. If the foot of the perpendicular from the point A(–1, 4, 3) on the plane P : $2x + my + nz = 4$, is $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios 3, –1, –4, is equal to : **[JEE(Main)July2022]**

यदि बिन्दु A(–1, 4, 3) से समतल P : $2x + my + nz = 4$, पर लंब का पाद $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$ है, तो दिक् अनुपात 3, –1, –4 की एक रेखा के समांतर नापी गई बिन्दु A की समतल P से दूरी बराबर है : **[JEE(Main)July2022]**

(A) 1 (B) $\sqrt{26}$ (C) $2\sqrt{2}$ (D) $\sqrt{14}$

Ans. B

Sol.



Let B be foot of $\perp$ coordinates of $B = \left(-2, \frac{7}{2}, \frac{3}{2}\right)$

Direction ratio of line AB is <2,1,3> so

m = 1, n = 3

So equation of AC is $\frac{x+1}{3}=\frac{y-4}{-1}=\frac{z-3}{-4}=\lambda$

So point C is $(3\lambda-1,-\lambda+4,-4\lambda+3)$. But C lies on the plane, so

$6\lambda-2-\lambda+4-12\lambda+9=4$

$\Rightarrow \lambda=1 \Rightarrow C(2,3,-1)$

$\Rightarrow AC=\sqrt{26}$

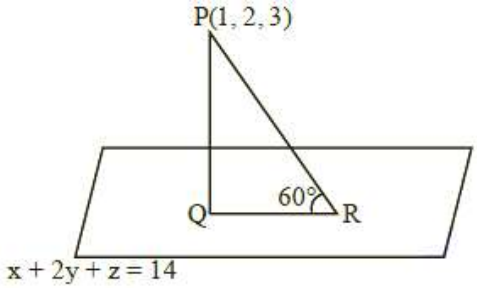
118. Let Q be the foot of perpendicular drawn from the point P(1, 2, 3) to the plane $x+2y+z=14$. If R is a point on the plane such that $\angle OPRQ = 60°$, then the area of ΔPQR is equal to : **[JEE(Main)July2022]**

माना बिन्दु P(1, 2, 3) से समतल $x+2y+z=14$ पर डाले गए लंब का पाद Q है। माना समतल पर बिन्दु R के लिए $\angle OPRQ = 60°$ है, तो ΔPQR का क्षेत्रफल बराबर : **[JEE(Main)July2022]**

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) 3

Ans. B

Sol.



Length of perpendicular

$PQ=\left|\frac{1+4+3-14}{\sqrt{6}}\right|=\sqrt{6}$

$QR=(PQ)\cot 60^\circ=\sqrt{2}$

$\therefore$ Area of $\Delta PQR=\frac{1}{2}(PQ)(QR)=\sqrt{3}$

119. The line of shortest distance between the lines $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ and $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane P : $ax-y-z=0$, $(a>0)$. If the image of the point (1, 1, –5) in the plane P is (α, β, γ), then $\alpha+\beta-\gamma$ is equal to _____. **[JEE(Main)July2022]**

रेखाओं $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ तथा $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ के बीच न्यूनतम दूरी की रेखा, समतल $P : ax - y - z = 0, (a > 0)$ से $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ का कोण बनाती है। यदि बिन्दु (1, 1, –5) P में प्रतिबिंब (α, β, γ) है, तो $\alpha + \beta - \gamma$ बराबर है _____

[JEE(Main)July2022]

Ans. 3

Sol. DR's of line of shortest distance

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix} = -\hat{i}+2\hat{j}-2\hat{k}$$

angle between line and plane is $\cos^{-1}\sqrt{\frac{2}{27}}=\alpha$

$$\cos\alpha=\left|\frac{-a-2+2}{\sqrt{4+4+1}\sqrt{a^2+1+1}}\right|=\frac{5}{3\sqrt{3}}$$

$\sqrt{3}\,|a|=5\sqrt{a^2+2}$

$3a^2=25a^2+50$

No value of (a)

120. Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point P(4, 2, 7). Then the square of the area of the triangle PQR is _____. **[JEE(Main)July2022]**

माना रेखा $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ पर बिन्दु P (4, 2, 7) से $\sqrt{26}$ दूरी पर दो बिन्दु Q तथा R है। तो त्रिभुज PQR के क्षेत्रफल का वर्ग बराबर है _____ **[JEE(Main)July2022]**

Ans. 153

Sol. Let $(2\lambda - 1, 3\lambda - 2, 2\lambda + 1)$ be any point on the line

$(2\lambda - 5)^2 + (3\lambda - 4)^2 + (2\lambda - 6)^2 = 26$

$\lambda = 1, 3$

Q (1, 1, 3) ; R (5, 7, 7) ; P (4, 2, 7)

Area of triangle $PQR=\frac{1}{2}\left|\overrightarrow{PQ}\times\overrightarrow{PR}\right|$

$= \sqrt{153}$

121. The plane passing through the line L : lx – y + 3(1 – l) z = 1, x + 2y – z = 2 and perpendicular to the plane 3x + 2y + z = 6 is 3x – 8y + 7z = 4. If θ is the acute angle between the line L and the y-axis, then 415 $\cos^2\theta$ is equal to _____. **[JEE(Main)July2022]**

रेखा L : lx – y + 3(1 – l) z = 1, x + 2y – z = 2 से होकर जाने वाले तथा समतल 3x + 2y + z = 6 के लम्बवत् समतल 3x – 8y + 7z = 4 है। यदि रेखा L तथा y-अक्ष के बीच न्यून कोण θ है, तो 415 $\cos^2\theta$ बराबर है _____

[JEE(Main)July2022]

Ans. 125

Sol. $\vec{n}_1 = \ell\hat{i} - \hat{j} + 3(1-\ell)\hat{k}$

$\vec{n}_2 = \hat{i} + 2\hat{j} - \hat{k}$

$$\text{Direction ratio of line} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \ell & -1 & 3(1-\ell) \\ 1 & 2 & -1 \end{vmatrix}$$

$= (6\ell - 5)\hat{i} + (3 - 2\ell)\hat{j} + (2\ell + 1)\hat{k}$

3x – 8y + 7z = 4 will contain the line

$(6\ell - 5)\hat{i} + (3 - 2\ell)\hat{j} + (2\ell + 1)\hat{k}$

Normal of 3x – 8y + 7z = 4 will be perpendicular to the line

$= 3(6\ell - 5) + (3 - 2\ell)(-8) + 7(2\ell + 1) = 0$

$\Rightarrow \ell = \frac{2}{3}$

$\therefore$ direction ratio of line $\left(-1, \frac{5}{3}, \frac{7}{3}\right)$

Angle with y axis

$$\cos\theta = \frac{5/3}{\sqrt{1 + \frac{25}{9} + \frac{49}{9}}}$$

$$\cos\theta = \frac{5}{\sqrt{83}}$$

$$\therefore 415\cos^2\theta = \frac{25}{83} \times 415 = 125$$

122. Let the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ and $4ax - y + 5z - 7a = 0 = 2x - 5y - z - 3$, $a \in R$ at the point $P(\alpha, \beta, \gamma)$. Then the value of $\alpha + \beta + \gamma$ equals _____. **[JEE(Main)July2022]**

माना रेखा $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ एक समतल, जिसमें रेखाऐं $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ तथा $4ax - y + 5z - 7a = 0 = 2x - 5y - z - 3$, $a \in R$ स्थित हैं, तो बिन्दु $P(\alpha, \beta, \gamma)$ पर काटती है। तो $\alpha + \beta + \gamma$ का मान है _____. **[JEE(Main)July2022]**

Ans. 12

Sol. Equation of plane

$4ax - y + 5z - 7a + \lambda (2x - 5y - z - 3) = 0$

this satisfy $(4, -1, 0)$

$16a + 1 - 7a + \lambda (8 + 5 - 3) = 0$

$9a + 1 + 10\lambda = 0$ ...(1)

Normal vector of the plane A is $(4a + 2\lambda, -1 - 5\lambda, 5 - \lambda)$ vector along the line which contained the plane A is $i - 2j + k$

$\therefore 4a + 2\lambda + 2 + 10\lambda + 5 - \lambda = 0$

$11\lambda + 4a + 7 = 0$ ..(2)

Solve (1) and (2) to get $a = 1$, $\lambda = -1$

Now equation of plane

$x + 2y + 3z - 2 = 0$

Let the point in the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}=t$

is $(7t + 3, -t + 2, -4t + 3)$ satisfy the equation of plane A

$7t + 3 - 2t + 4 + 9 - 12t - 2 = 0$

$t = 2$

So $\alpha + \beta + \gamma = 2t + 8 = 12$

123. Let $P(-2, -1, 1)$ and $Q\left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagnonal RS are $\alpha, -1, \beta$, where both α and β are integers of minimum absolute values, then $\alpha^2 + \beta^2$ is equal to _____. **[JEE(Main)July2022]**

माना P(–2, –1, 1) तथा $Q\left(\frac{56}{17},\frac{43}{17},\frac{111}{17}\right)$ एक समचतुर्भुज PRQS के शीर्ष हैं। यदि विकर्ण RS के दिक्–अनुपात α, –1, β हैं, जहाँ α तथा β दोनों निम्नतम निरपेक्ष मान के पूर्णांक हैं, तो $\alpha^2 + \beta^2$ बराबर है _____

[JEE(Main)July2022]

Ans. 450

Sol. $RS \equiv (\alpha, -1, \beta)$

DR of $PQ \equiv \left(\frac{56}{17}+2,\frac{43}{17}+1,\frac{111}{17}-1\right)$

$\equiv \left(\frac{90}{17},\frac{60}{17},\frac{94}{17}\right)$

$\frac{90}{17}\alpha + \frac{60}{17}(-1) + \frac{94}{17}\beta = 0$

$90\alpha + 94\beta = 60$

$\beta = \frac{60-90\alpha}{94}$

$\beta = \frac{30(2-3\alpha)}{94}$

$\beta = -30\frac{(3\alpha-2)}{94}$

$\beta = \frac{\beta}{-15} = \frac{3\alpha-2}{47}$

$\Rightarrow \beta = -15, \alpha = -15$

$\alpha^2 + \beta^2 = 225 + 225$

$= 450$

124. Let a line with direction ratio a, –4a, –7 be perpendicular to the lines with direction ratios 3, –1, 2b and b, a, –2. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane x – y + z = 0 is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____. **[JEE(Main)July2022]**

माना दिक् अनुपास a, –4a, –7 की एक रेखा, अनुपात 3, –1, 2b तथा b, a, –2 की रेखाओं के लंबवत है। यदि रेखा $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ तथा समतल x – y + z = 0 का प्रतिच्छेदन बिन्दु (α, β, γ) है तो $\alpha + \beta + \gamma$ बराबर है _____

[JEE(Main)July2022]

Ans. 10

Sol. $(a, -4a, -7) \perp$ to $(3, -1, 2b)$

$a = 2b$ .....(i)

$(a, -4a, -7) \perp$ to $(b, a, -2)$

$3a + 4a - 14b = 0$

$ab - 4a^2 + 14 = 0$ .....(ii)

From Equations (i) and (ii)

$2b^2 - 16b^2 + 14 = 0$

$b^2 = 1$

$a^2 = 4b^2 = 4$

$$\frac{x+1}{5} = \frac{y-2}{3} = \frac{z}{1} = k$$

$\alpha = 5k - 1, \beta = 3k + 2, \gamma = k$

As (α, β, γ) satisfies $x - y + z = 0$

$5k - 1 - (3k + 2) + k = 0$

$k = 1$

$\therefore \alpha + \beta + \gamma = 9k + 1 = 10$